

Learning what we never observe

Maximum likelihood, expectation–maximization and hidden-variable models

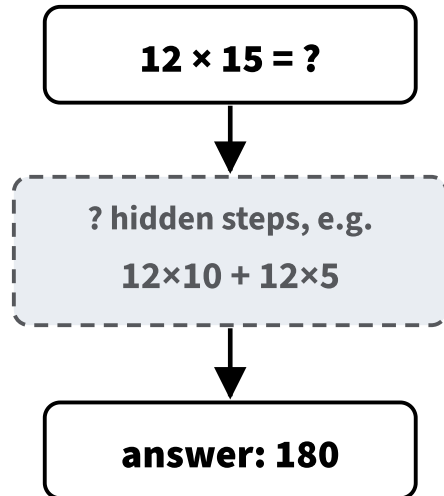
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In real data, the key variable is often never written down

 dashed box: never recorded (hidden)  solid box: in the data

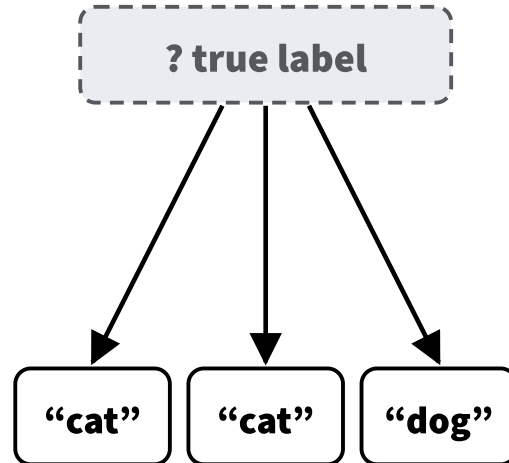
Learning to reason



data: question + answer

to learn from the data:
**good reasoning steps
from answers alone**

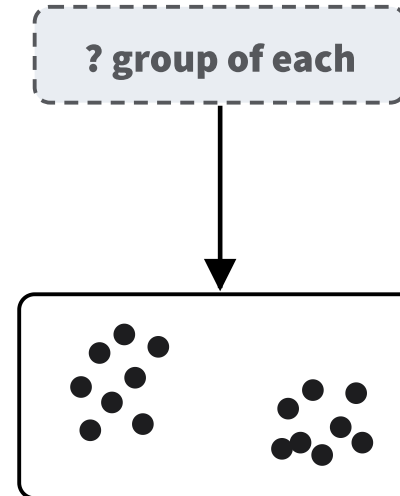
Cleaning labels



data: three annotators' labels
they disagree

to learn from the data:
**the true labels and
each annotator's skill**

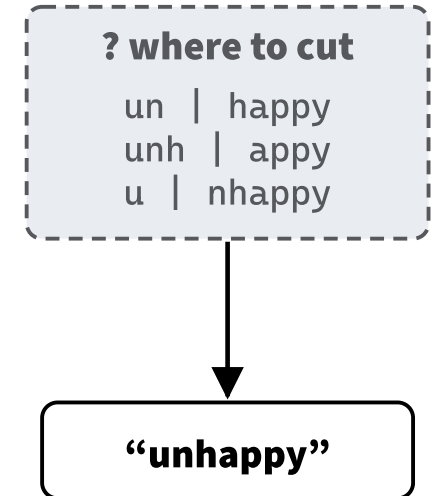
Fast vector search



data: the vectors

to learn from the data:
**the groups, so a query
checks its own group only**

Splitting words



data: the words of a text

to learn from the data:
**how often each piece
occurs, so where to cut**

All four use expectation-maximization (EM) or its hard-label form, k-means.

Two facts from earlier lectures that EM reuses

A coin shows 7 heads in 10 flips. What is our best guess of its heads chance?

1/2

7/10

3/10

Answer. 7/10: count heads, divide by flips. Today we prove this is the **maximum-likelihood** answer.



7 heads (H, blue) and 3 tails (T) in 10 flips

heads ÷ flips = 7/10

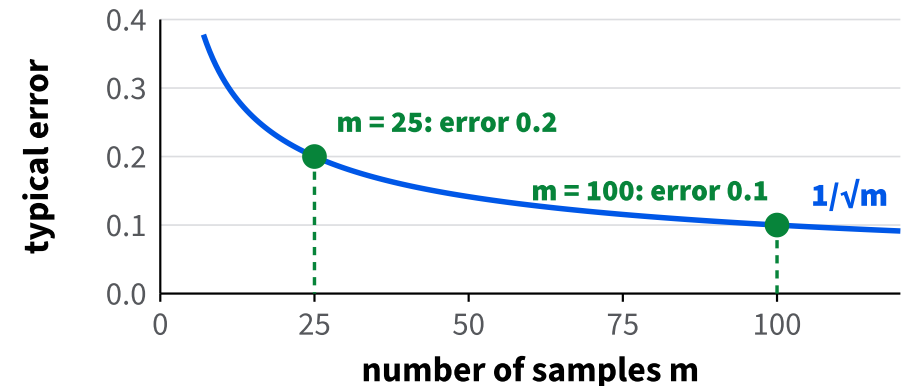
Monte Carlo (Lecture 4): with 4 times as many samples, the error shrinks by a factor of ...

4

1: no change

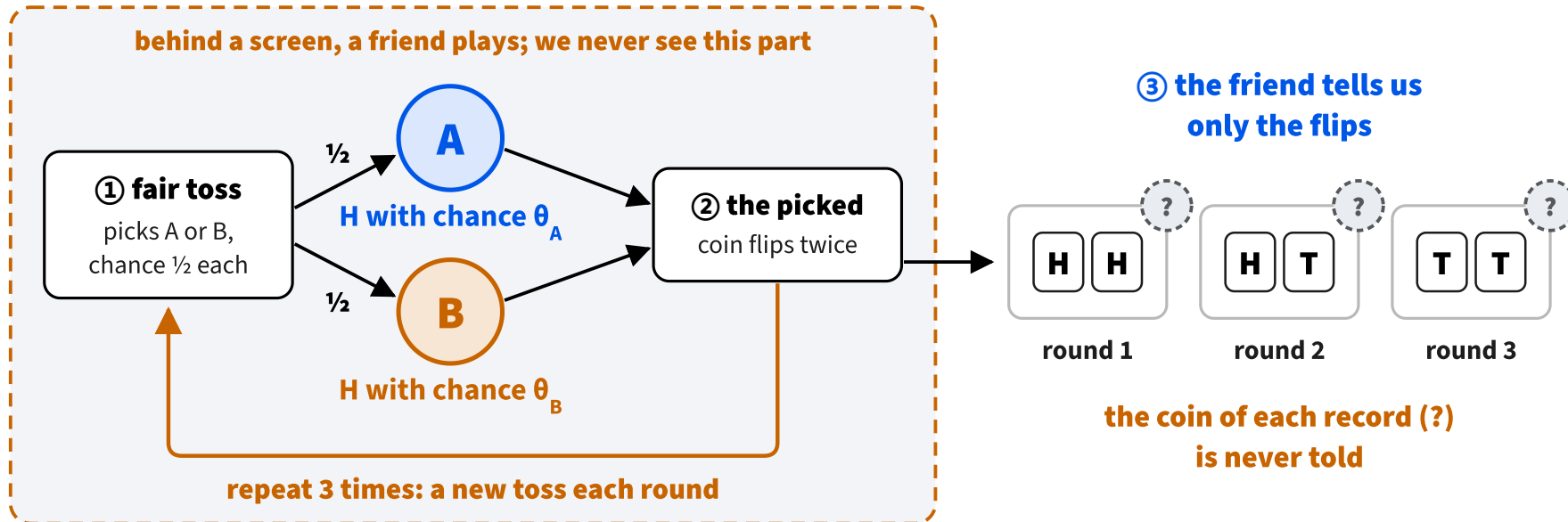
2

Answer. 2: with m samples the typical error is $\sigma/\sqrt{m}$. With spread $\sigma = 1$: $1/\sqrt{25} = 0.2$, $1/\sqrt{100} = 0.1$; so 4 times the samples halves the error, $0.2/0.1 = 2 = \sqrt{4}$.



Both return: **counting** in EM's maximization step, **sampling** for too many hidden cases.

The two-coin game: we see the flips, never the coin



We see

three records, **HH**, **HT**, **TT** (H = heads, T = tails); both flips of a record come from one coin.

Hidden

which coin made each record; a new fair toss each round, so the coins can differ.

Goal

estimate the heads chances θ_A, θ_B from the three records alone.

Every idea today is first worked out on these three records, by hand.

Eight questions; each answer raises the next

Section	Question on the coins	Answer
1 Counting	If we are told each coin, what are θ_A, θ_B ?	count heads, divide
2 Hidden variables	Not told: how likely is a record, and which coin made it?	a sum; Bayes' rule
3 EM	Can we still count?	yes, with soft labels: split each record by its posterior
4 Why EM climbs	Does every round improve the fit?	yes: each round raises a lower bound that touches the log-likelihood
5 Mixtures	What if coins become bell curves?	the same EM, now clustering points
6 Hidden chains	What if the coin switches between flips?	sum-product on a chain
7 Monte Carlo EM	Too many hidden cases to add up?	sample them
8 Trusting the fit	Is the answer right, and the only one?	restart from several guesses; check the data can pin down every unknown



Each section runs the same five steps, so you always know where you are.

By the end of today you can do six things

1 • Count

fit a model when nothing is hidden, and say why counting is the **maximum-likelihood** answer.

You can answer: why is 7 heads in 10 flips fitted by 7/10?

2 • Infer

compute a record's probability with a hidden variable, and the **posterior** of that variable.

You can answer: which coin probably made HH?

3 • Run EM

by hand: E-step (expectation: **soft labels**) and M-step (maximization: count and divide).

You can answer: how do soft labels become new heads chances?

4 • Prove

that no EM round lowers the log-likelihood (how well θ fits the data), with **one inequality**.

You can answer: can an EM round ever make the fit worse?

5 • Recognize

EM inside clustering for vector search, tokenizers, crowd labelling and **reasoning models**.

You can answer: where is the hidden coin in a tokenizer?

6 • Judge

a fit: a start that gets stuck, the names A and B swapped, or **more unknowns than the data can fix**.

You can answer: why may EM swap the names A and B?

Six skills, one example: three records of two coin flips.

We see x , never z , and want to learn θ_A, θ_B

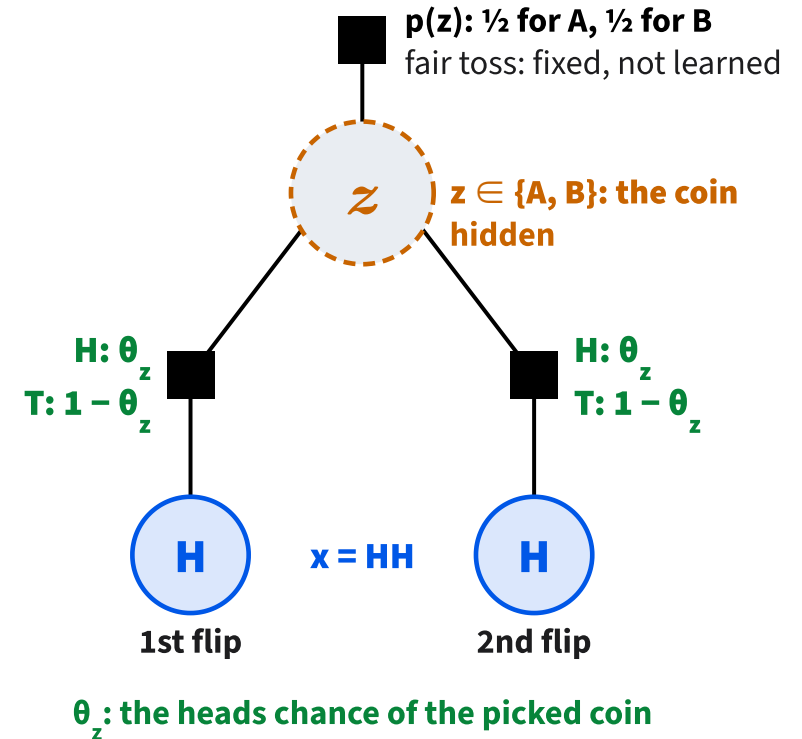
Symbol	Meaning	On the coins
x	observed record	the two flips, e.g. $x = \text{HH}$
z	hidden variable	the coin, $z \in \{A, B\}$
$\theta = (\theta_A, \theta_B)$	parameters	heads chances of A, B (unknown)
$p_\theta(x, z)$	joint probability of record and coin	$p_\theta(\text{HH}, A) = \frac{1}{2} \theta_A \theta_A$

Take a starting guess (EM will improve it later): A leans heads, $\theta_A = \frac{3}{4}$; B leans tails, $\theta_B = \frac{1}{4}$.

$$p_\theta(\text{HH}, A) = \underbrace{\frac{1}{2}}_{\text{pick } A} \cdot \underbrace{\frac{3}{4}}_{\text{1st H by } A} \cdot \underbrace{\frac{3}{4}}_{\text{2nd H by } A} = \frac{9}{32}.$$

Lecture 2's factor graph of one record

■ square: a factor (a chance) · ○ circle: a variable



The subscript θ in p_θ means "computed with these parameters"; the toss chance $\frac{1}{2}$ is fixed.

HH has chance 5/16 in all; Bayes' rule gives coin A 9/10 of it

$$p_{\theta}(x) = \sum_z p_{\theta}(x, z)$$

$$p_{\theta}(\text{HH}) = \frac{9}{32} + \frac{1}{32} = \frac{10}{32} = \frac{5}{16}$$

$$p_{\theta}(z | x) = \frac{p_{\theta}(x, z)}{p_{\theta}(x)}$$

$$p_{\theta}(A | \text{HH}) = \frac{9/32}{10/32} = \frac{9}{10}$$

Likelihood of a record: add the joint over the hidden coin.

9 HH squares of A plus 1 of B, each worth 1/32.

Posterior (Bayes' rule): which coin, given the flips.

9 of the 10 HH squares are A's.

coin A picked
chance $\frac{1}{2} \cdot \text{heads } \frac{3}{4}$

	H	H	H	T
H				
H				
H				
T				

9 squares are HH

coin B picked
chance $\frac{1}{2} \cdot \text{heads } \frac{1}{4}$

	H	T	T	T
H				
T				
T				
T				

1 square is HH

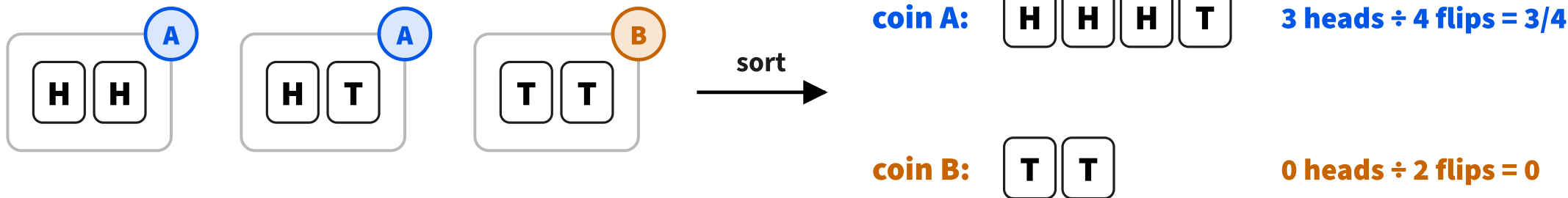
each row (1st flip) or column (2nd flip) is $\frac{1}{4}$,
so a square = $\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$

Log-likelihood: each round has its own fair toss, so the records are independent given θ , and $p_{\theta}(\text{all three}) = p_{\theta}(\text{HH}) p_{\theta}(\text{HT}) p_{\theta}(\text{TT})$; the log turns the product into a sum: $\ell(\theta) = \ln p_{\theta}(\text{HH}) + \ln p_{\theta}(\text{HT}) + \ln p_{\theta}(\text{TT})$.

Maximum likelihood: pick the θ with the largest $\ell(\theta)$. First, the easy case: the coin is told.

If the friend tells us the coin, we just count

Goal: learn θ_A, θ_B , the heads chances of coins A and B, from the records HH, HT, TT. First the easy case: the friend also tells us which coin made each record.



the friend now says which coin made each record

estimate = heads ÷ flips, coin by coin

Recall the game

A fair toss picks coin A or B, flipped twice. Normally we see **only the flips**; for now the friend also names the coin.

Estimate

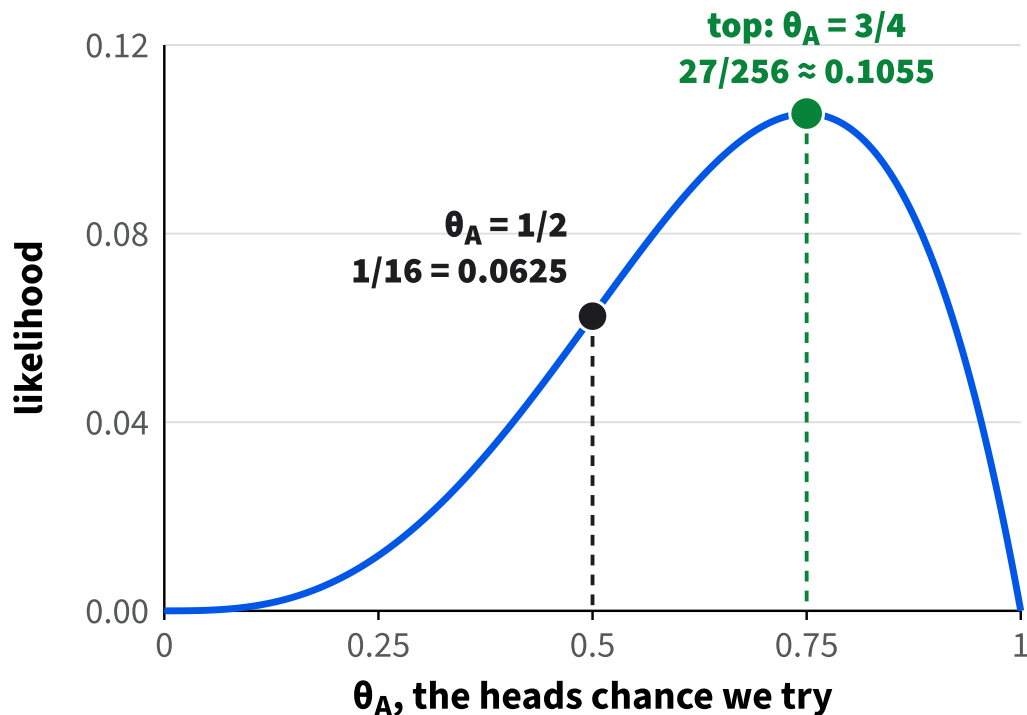
heads ÷ flips: $\theta_A = \frac{3}{4}$ and $\theta_B = \frac{0}{2} = 0$.

Two questions

Why is counting the **best** guess? Can coin B really **never** land heads?

With the coin told, learning is counting: heads ÷ flips, coin by coin.

Coin A's flips are most likely when $\theta_A = 3/4$

likelihood $L(\theta_A)$ of H, H, H, T

Likelihood: the probability of the flips we saw, read as a function of the unknown θ_A .

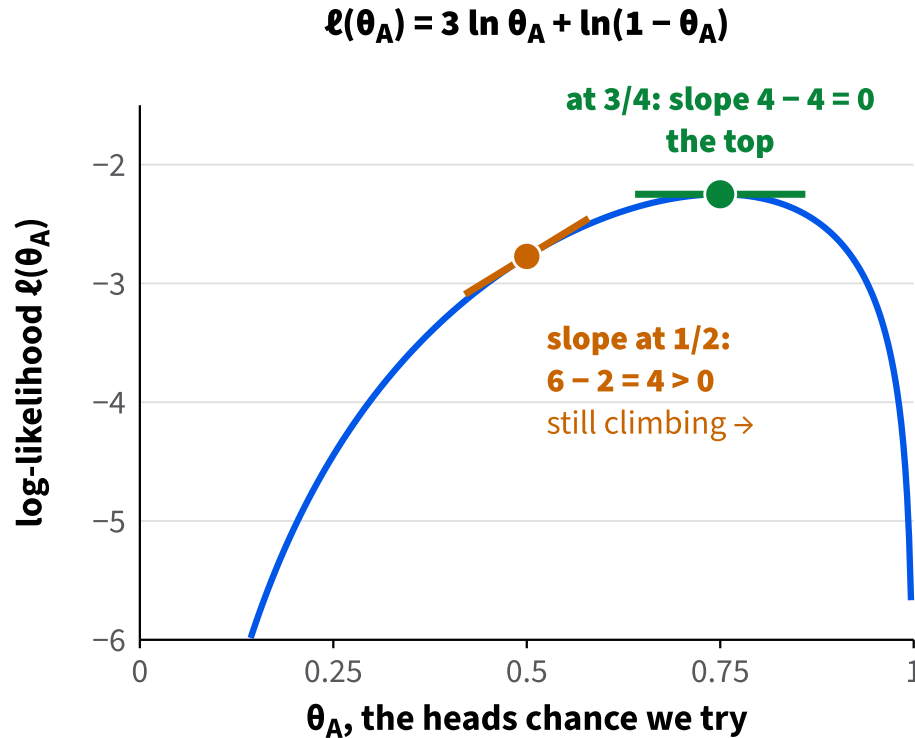
Coin A gave H, H, H, T. The flips are **independent**, so their chances multiply:

$$L(\theta_A) = \underbrace{\theta_A}_H \cdot \underbrace{\theta_A}_H \cdot \underbrace{\theta_A}_H \cdot \underbrace{(1 - \theta_A)}_T = \theta_A^3(1 - \theta_A).$$

try θ_A	likelihood $\theta_A^3(1 - \theta_A)$
$\frac{1}{2}$	$\frac{1}{8} \cdot \frac{1}{2} = \frac{1}{16} = 0.0625$
$\frac{3}{4}$	$\frac{27}{64} \cdot \frac{1}{4} = \frac{27}{256} \approx 0.1055$

The curve's top sits at $\theta_A = \frac{3}{4}$: exactly the counted fraction, 3 of 4.

Taking logs and setting the slope to zero gives $\theta_A = 3/4$



The **maximum-likelihood estimate** is the θ_A with the largest likelihood. To find it:

$$\ell(\theta_A) = \ln L = 3 \ln \theta_A + \ln(1 - \theta_A)$$

$$\ell'(\theta_A) = \frac{3}{\theta_A} - \frac{1}{1 - \theta_A}$$

$$\ell'\left(\frac{1}{2}\right) = \frac{3}{1/2} - \frac{1}{1/2} = 6 - 2 = 4 > 0$$

$$\ell' = 0 \Rightarrow 3(1 - \theta_A) = \theta_A$$

$$\Rightarrow 3 = 4\theta_A \Rightarrow \theta_A = \frac{3}{4}$$

$$\ell'\left(\frac{3}{4}\right) = \frac{3}{3/4} - \frac{1}{1/4} = 4 - 4 = 0$$

Why logs? A sum is easier to differentiate than a product, and $\ln$ only rises, so ℓ and L peak at the same θ_A .

Log: product $\rightarrow$ sum.

Slope: $(\ln u)' = \frac{u'}{u}$.

Positive: still climbing.

Times $\theta_A(1 - \theta_A)$.

Zero: the top.

Check at $\frac{3}{4}$.

Maximum likelihood gives $\theta_A = \frac{3}{4}$, the counted fraction.

For any coin, the counted fraction maximizes the likelihood

Theorem 1 (maximum likelihood by counting). Independent flips of a coin with heads chance θ show h heads and t tails. Then $\ell(\theta) = h \ln \theta + t \ln(1 - \theta)$ is largest at $\hat{\theta} = \frac{h}{h+t}$.

In words: the best heads chance is the fraction of heads seen.

$$\ell'(\theta) = \frac{h}{\theta} - \frac{t}{1-\theta} = 0$$

$$h(1-\theta) = t\theta \Rightarrow \theta = \frac{h}{h+t}$$

$$\ell''(\theta) = -\frac{h}{\theta^2} - \frac{t}{(1-\theta)^2} < 0$$

$h = 0$: $\ell = t \ln(1 - \theta)$, falling

$t = 0$: $\ell = h \ln \theta$, rising

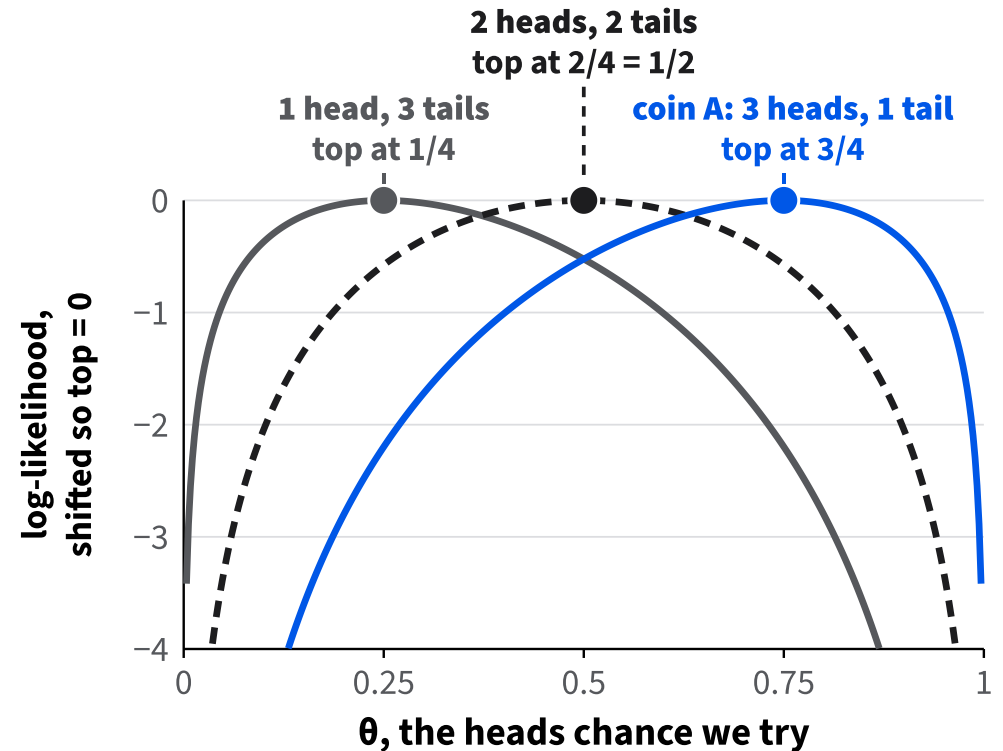
Slope zero. Coin A:
 $h = 3, t = 1$.

Times $\theta(1 - \theta)$. Coin A:
 $\frac{3}{3+1} = \frac{3}{4}$.

Curves down: one top.

Top at $\theta = 0$. Coin B: $\frac{0}{2} = 0$.

Top at $\theta = 1 = \frac{h}{h+0}$. ■



each curve peaks at its own heads $\div$ flips

Whatever the counts, the likelihood peaks at heads $\div$ flips.

Counting says coin B can never land heads

counting only

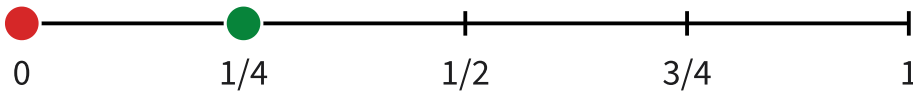
$$\boxed{T} \boxed{T} \longrightarrow 0 \text{ heads} \div 2 \text{ flips} = 0$$

with one imagined head and one imagined tail

$$\boxed{T} \boxed{T} + \boxed{H} \boxed{T} \longrightarrow 1 \text{ head} \div 4 \text{ flips} = 1/4$$

0: heads impossible

1/4: heads rare, still possible



estimate of θ_B , coin B's heads chance

Coin B showed T, T. Theorem 1 gives $\theta_B = \frac{0}{0+2} = 0$.

Too sure. Suppose B is flipped again and shows HT:

$$p(\text{HT} \mid B) = \theta_B(1 - \theta_B) = 0 \cdot 1 = 0.$$

Its log-likelihood is $\ln 0 = -\infty$: the model calls a record we could easily see **impossible**.

Weak evidence: a fair coin shows TT with chance $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

Fix: add one imagined head and one imagined tail (dashed):

$$\theta_B = \frac{0+1}{2+2} = \frac{1}{4}, \quad p(\text{HT} \mid B) = \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{16}.$$

Two tails cannot prove heads impossible; imagined flips keep every outcome possible.

Averaging over every possible θ_B gives $1/4$, not 0

↳ Why exactly one imagined head and one tail? Let θ_B be unknown, every value in $[0, 1]$ equally likely before any flip: the **uniform prior**.

$$p(\theta_B \mid \text{TT}) \propto 1 \cdot (1 - \theta_B)^2$$

$$\int_0^1 (1 - \theta)^2 d\theta = \left[-\frac{(1-\theta)^3}{3} \right]_0^1 = \frac{1}{3}$$

$$p(\theta_B \mid \text{TT}) = 3(1 - \theta_B)^2$$

$$\mathbb{E}[\theta_B \mid \text{TT}] = \int_0^1 \theta \cdot 3(1 - \theta)^2 d\theta$$

$$= 3 \int_0^1 (\theta - 2\theta^2 + \theta^3) d\theta$$

$$= 3 \left(\frac{1}{2} - 2 \cdot \frac{1}{3} + \frac{1}{4} \right)$$

$$= 3 \cdot \frac{6-8+3}{12} = 3 \cdot \frac{1}{12} = \frac{1}{4}$$

$p(\text{next H} \mid \text{TT}) = \int_0^1 \theta p(\theta \mid \text{TT}) d\theta = \mathbb{E}[\theta_B \mid \text{TT}]$: the heads chance averaged over what we **still believe**.

Bayes: prior $1 \times$ likelihood.

Area under it; θ is the integration variable for θ_B .

Divide by the area $\frac{1}{3}$.

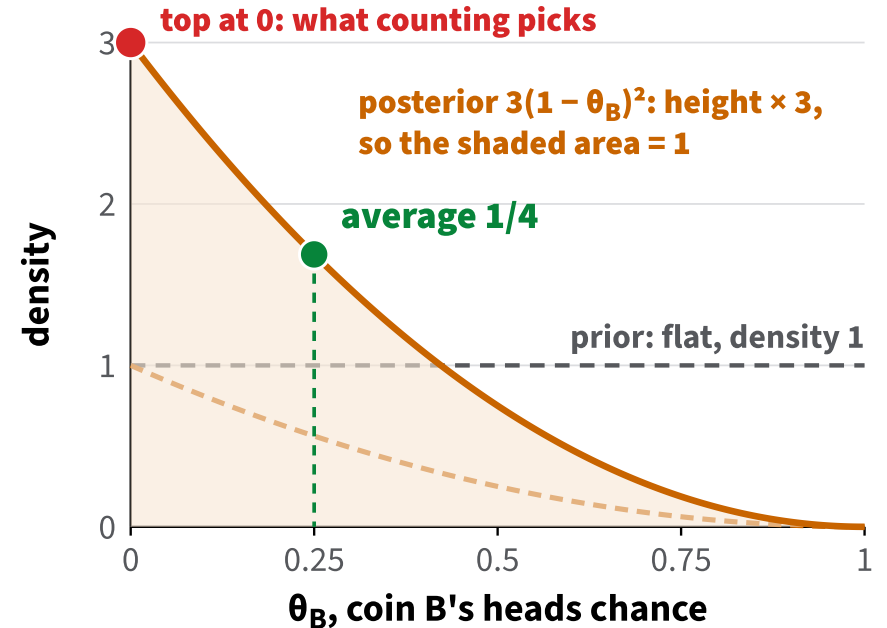
Posterior mean of θ_B .

Expand $\theta(1 - \theta)^2$.

Each term: $\int_0^1 \theta^k d\theta = \frac{1}{k+1}$.

Common denominator 12.

--- posterior $\propto (1 - \theta_B)^2$ (prior $1 \times$ likelihood)
its balance point, the average, is $1/4$



The posterior's top is 0, as counting says; its average is $\frac{1}{4}$, the imagined-flips answer.

Laplace's rule: the posterior mean adds one head and one tail

Theorem 2 (Laplace's rule). Put a uniform prior on θ , then see h heads and t tails. The posterior mean is $\mathbb{E}[\theta \mid \text{flips}] = \frac{h+1}{h+t+2}$.

In words: add one imagined head and one imagined tail, then count.

$$\mathbb{E}[\theta] = \frac{\int_0^1 \theta \cdot \theta^h (1-\theta)^t d\theta}{\int_0^1 \theta^h (1-\theta)^t d\theta}$$

$$\int_0^1 \theta^a (1-\theta)^b d\theta = \frac{a! b!}{(a+b+1)!}$$

$$= \frac{(h+1)! t! / (h+t+2)!}{h! t! / (h+t+1)!}$$

$$= \frac{h+1}{h+t+2}$$

Bayes (prior 1), then average θ .

Beta integral. Check $a = 0, b = 2$: $\frac{0! 2!}{3!} = \frac{1}{3}$, matching $\int_0^1 (1-\theta)^2 d\theta = \frac{1}{3}$ computed directly. Coin A: $\frac{3! 1!}{5!} = \frac{1}{20}$, so its density is $20\theta^3(1-\theta)$.

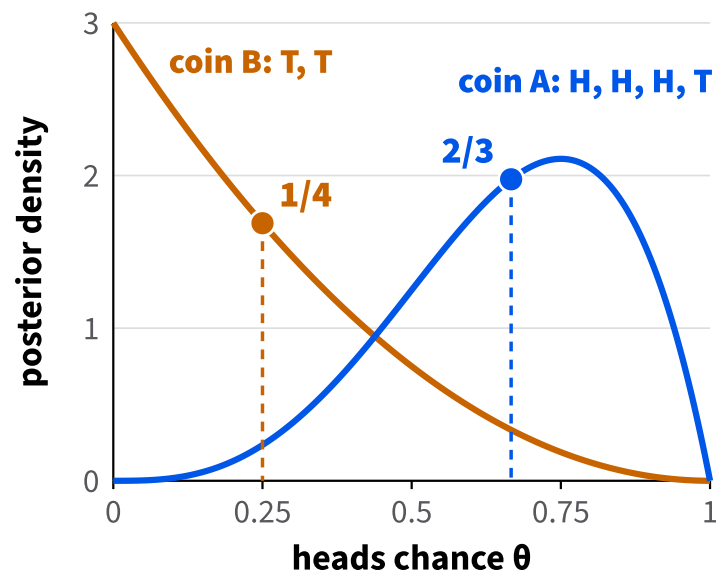
$(a, b) = (h+1, t)$ on top, (h, t) below.

$$\frac{(h+1)!}{h!} = h+1, \quad \frac{(h+t+1)!}{(h+t+2)!} = \frac{1}{h+t+2}. \blacksquare$$

Why +1 and +1? With no flips ($h = t = 0$) the rule gives $\frac{0+1}{0+2} = \frac{1}{2}$, the flat prior's average: **the imagined head and tail are the uniform prior written as counts.**

Coin B: $\frac{0+1}{2+2} = \frac{1}{4}$. **Coin A:** $\frac{3+1}{4+2} = \frac{2}{3}$, pulled from $\frac{3}{4}$ toward $\frac{1}{2}$.

posterior densities, uniform prior:
coin B: $3(1-\theta)^2$, coin A: $20\theta^3(1-\theta)$



Laplace's rule pulls the count toward $1/2$ by a weight ε

Laplace's rule mixes the **counted fraction** with the even guess $\frac{1}{2}$, over $n = h + t$ flips:

$$\frac{h+1}{n+2} = (1 - \varepsilon) \frac{h}{n} + \varepsilon \cdot \frac{1}{2}, \quad \varepsilon = \frac{2}{n+2}.$$

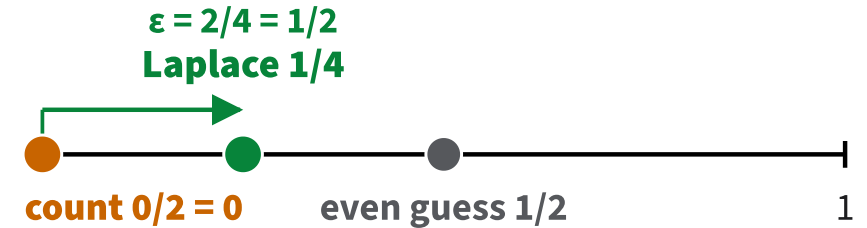
Check: $1 - \varepsilon = \frac{n}{n+2}$, so $(1 - \varepsilon) \frac{h}{n} = \frac{h}{n+2}$, and $\varepsilon \cdot \frac{1}{2} = \frac{1}{n+2}$; together $\frac{h+1}{n+2}$.

Coin B ($n = 2, \varepsilon = \frac{1}{2}$): $\frac{1}{2} \cdot 0 + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = \frac{0+1}{2+2}$.

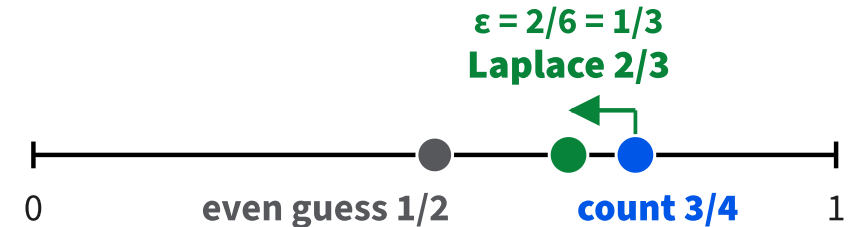
Coin A ($n = 4, \varepsilon = \frac{1}{3}$): $\frac{2}{3} \cdot \frac{3}{4} + \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{2} + \frac{1}{6} = \frac{2}{3} = \frac{3+1}{4+2}$.

More flips shrink ε , so the data soon **outweigh** the two imagined flips.

coin B: T, T ($n = 2$ flips)



coin A: H, H, H, T ($n = 4$ flips)



Laplace moves the count ε of the way toward the even guess $1/2$

Laplace moves the count $\varepsilon = \frac{2}{n+2}$ of the way to $\frac{1}{2}$; more flips, smaller pull.

Label smoothing gives every class an imagined count

An image classifier with $K = 3$ classes (cat, dog, fox) learns from labelled photos. Recall Laplace: $(1 - \varepsilon) \frac{h}{n} + \varepsilon \cdot \frac{1}{K}$.

COIN GAME

counted fraction of one flip ($H: \frac{1}{1} = 1$)

$\varepsilon \cdot \frac{1}{2}$ added to heads and to tails ($K = 2$ outcomes)

$\varepsilon = \frac{2}{n+2}$, shrinks with more flips

even guess $\frac{1}{2}$

IMAGE CLASSIFIER

$\Leftrightarrow$ hard target $(1, 0, 0)$ of one cat photo

$\Leftrightarrow \varepsilon/K$ added to every class

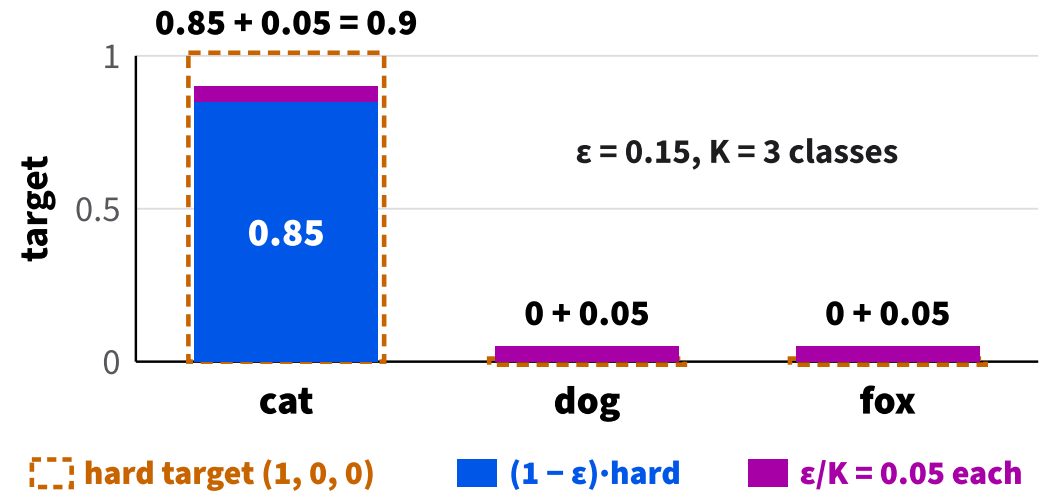
$\Leftrightarrow \varepsilon$ fixed by hand (0.1 in practice)

$\Leftrightarrow$ even guess $\frac{1}{K}$

$\text{target}_k = (1 - \varepsilon) [k = \text{cat}] + \varepsilon/K$, where $[k = \text{cat}]$ is 1 for cat, else 0.

$\varepsilon = 0.15$, $K = 3$: $1 - \varepsilon = 0.85$, $\varepsilon/K = 0.05$ (we take 0.15 so this is round). Cat $0.85 \cdot 1 + 0.05 = 0.9$, dog and fox $0.85 \cdot 0 + 0.05 = 0.05$; sum $0.9 + 0.05 + 0.05 = 1$.

In practice (Szegedy et al., CVPR = Conference on Computer Vision and Pattern Recognition, 2016): $\varepsilon = 0.1$, $K = 1000$, so each wrong class gets $0.1/1000 = 0.0001$ and the true class $0.9 + 0.0001 = 0.9001$.



Label smoothing is Laplace's rule for classes: no class gets target 0.

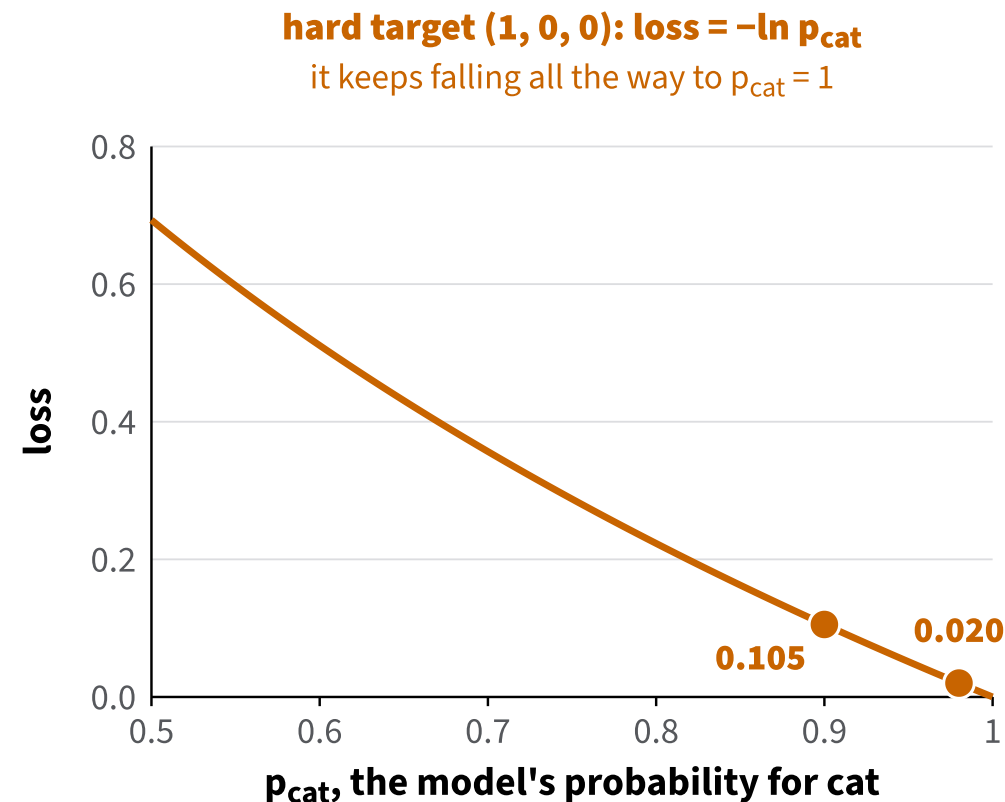
With the hard target, the over-sure prediction wins

Cross-entropy loss: $-\sum_k \text{target}_k \ln p_k$, where p_k is the model's probability for class k . Lower is better.

For a cat photo the hard target is $(1, 0, 0)$: only the cat term survives, so the loss is $-\ln p_{\text{cat}}$.

prediction (cat, dog, fox)	p_{cat}	loss $-\ln p_{\text{cat}}$
(0.9, 0.05, 0.05)	0.9	0.105
(0.98, 0.01, 0.01), over-sure	0.98	0.020

The over-sure prediction gets the **lower loss**, so training pushes p_{cat} toward 1, as counting pushed θ_B to 0 after two tails.



Under the hard target, more certainty always means a lower loss.

With the smoothed target, the over-sure prediction costs more

Smoothed target $(0.9, 0.05, 0.05)$, from $\varepsilon = 0.15$, $K = 3$. Loss

– $\sum_k \text{target}_k \ln p_k$, with logs $-\ln 0.9 = 0.105$, $-\ln 0.98 = 0.020$,
 – $-\ln 0.05 = 2.996$, $-\ln 0.01 = 4.605$.

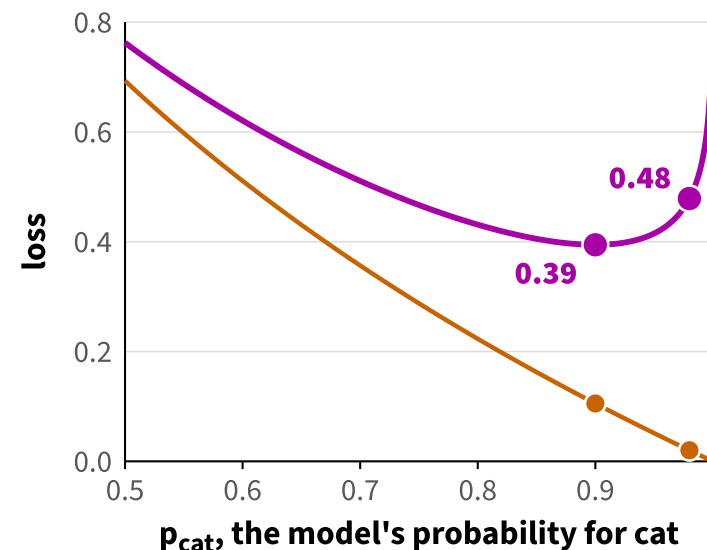
↳ Dog and fox have the same target 0.05 and, in each prediction, the same p_k , so their two terms **merge** into one with weight $0.05 + 0.05 = 0.1$.

prediction (cat, dog, fox)	$(0.9, 0.05, 0.05)$	$(0.98, 0.01, 0.01)$
cat: $0.9 (-\ln p_{\text{cat}})$	$0.9 \cdot 0.105 = 0.0945$	$0.9 \cdot 0.020 = 0.0180$
dog, fox: $0.1 (-\ln p_{\text{dog}})$	$0.1 \cdot 2.996 = 0.2996$	$0.1 \cdot 4.605 = 0.4605$
loss (sum)	$0.3941 \approx 0.39$	$0.4785 \approx 0.48$

The over-sure prediction now has the **higher loss**: $0.48 > 0.39$. The hard target ranked them the other way: $0.020 < 0.105$.

smoothed target $(0.9, 0.05, 0.05)$

hard target $(1, 0, 0)$



Under the smoothed target, more certainty is no longer always better.

The smoothed loss is lowest at $p_{\text{cat}} = 0.9$, not at 1

Let $p = p_{\text{cat}}$; dog and fox get $\frac{1-p}{2}$ each (dog and fox have equal targets, so **splitting** $1 - p$ **equally** between them is best). Under target $(0.9, 0.05, 0.05)$:

$$f(p) = -0.9 \ln p - 0.1 \ln \frac{1-p}{2}$$

$$f'(p) = -\frac{0.9}{p} + \frac{0.1}{1-p} = 0$$

$$\Rightarrow 0.9(1-p) = 0.1p$$

$$\Rightarrow p = \frac{0.9}{0.9+0.1} = 0.9$$

$$f''(p) = \frac{0.9}{p^2} + \frac{0.1}{(1-p)^2} > 0$$

Cat weight 0.9; dog and fox merged, $0.05 + 0.05 = 0.1$.

Slope zero: $(\ln u)' = \frac{u'}{u}$,
 $u = \frac{1-p}{2}$, $u' = -\frac{1}{2}$.

Times $p(1-p)$.

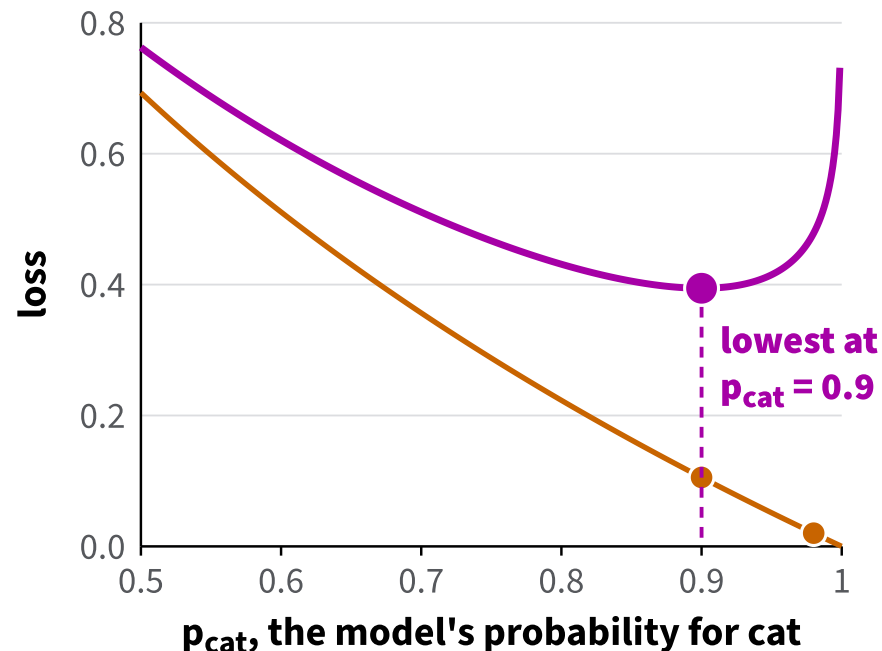
$$0.9 = (0.9 + 0.1)p.$$

Curves up: one lowest point.

Theorem 1 again: $-f(p) = 0.9 \ln p + 0.1 \ln(1-p) - 0.1 \ln 2$ is a log-likelihood with counts $h = 0.9, t = 0.1$, so its top is $\frac{h}{h+t} = 0.9$.

smoothed target (0.9, 0.05, 0.05)

hard target (1, 0, 0)



Smoothed loss is lowest at $p_{\text{cat}} = 0.9$, not 1: like imagined flips, it stops the push to certainty.

Next: the friend stops telling us the coin

we are here

Counting

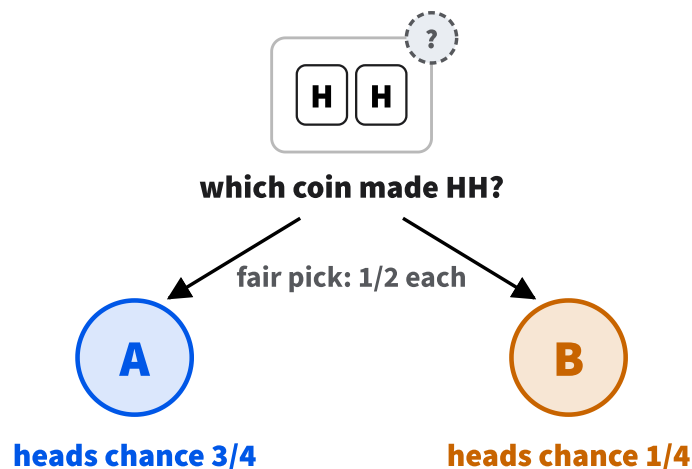
Hidden
variables

next

EM

Why EM
climbs

Mixtures

Hidden
chainsMonte Carlo
EMTrusting the
fit

Counting needs to know which coin made each flip. Now the friend reports only "HH".

Working guess for the rest of the lecture: $\theta_A = \frac{3}{4}$ (A's counted fraction), $\theta_B = \frac{1}{4}$ (B's Laplace estimate, which keeps heads possible), fair pick. How likely is HH?

9/16

5/16

1/16

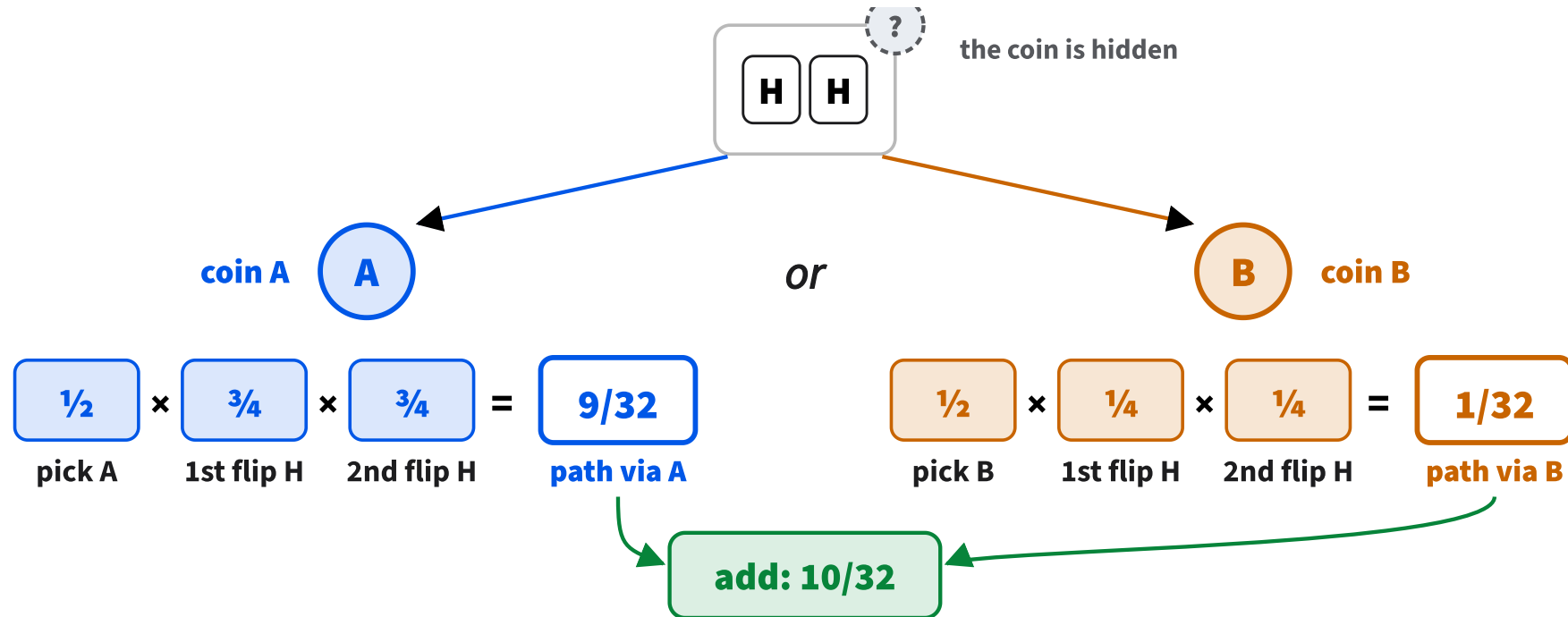
Answer. $\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{9}{32} + \frac{1}{32} = \frac{10}{32} = \frac{5}{16}$: a sum over the two coins, each weighted by its pick chance $\frac{1}{2}$.

A hidden coin turns one probability into a sum over the coins.

With the coin hidden, the chance of HH adds two paths

Goal: learn θ_A, θ_B from HH, HT, TT, now that the friend no longer says which coin was used.

Recall the game: a fair toss picks coin A or coin B; θ_A, θ_B are their heads chances. Working guess: $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$.



Why add? HH came from A **or** from B, never both, so the chances of the two paths add.

$$p(\text{HH}) = \frac{9}{32} + \frac{1}{32} = \frac{10}{32} = \frac{5}{16} : \text{a sum over the hidden coin.}$$

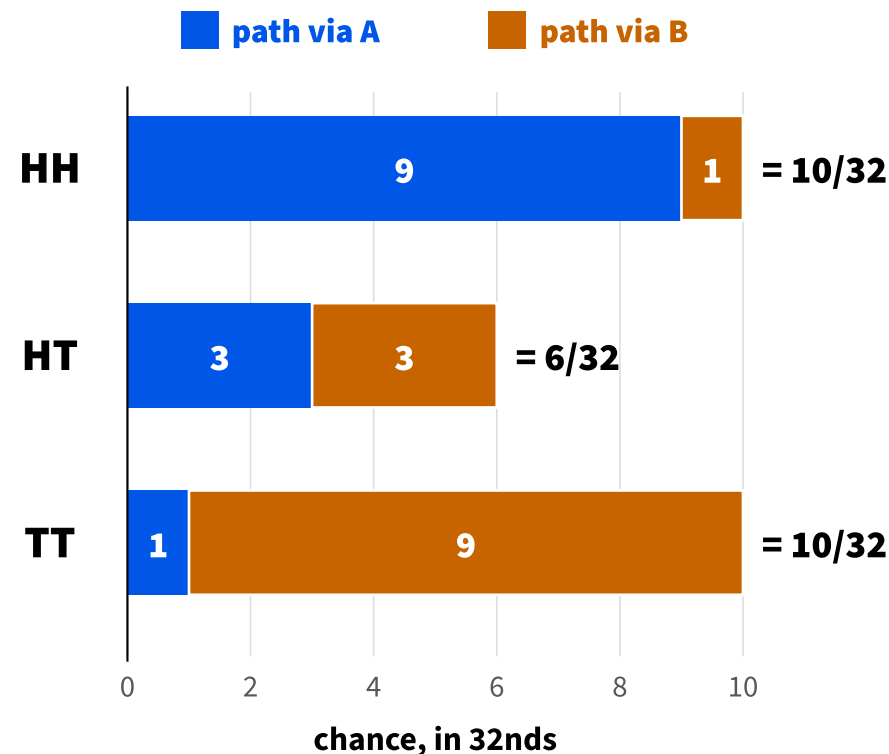
Every record adds a path via A and a path via B

We write x for a record. A path = $\frac{1}{2}$ (pick the coin) $\times$ the chance of each flip. Coin A shows H with $\frac{3}{4}$, T with $\frac{1}{4}$; coin B the reverse.

record	path via A	path via B	$p(x)$
HH	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{10}{32}$
HT	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{32}$	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{32}$	$\frac{6}{32}$
TT	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{10}{32}$

HT is a tie: A's $\frac{3}{4} \cdot \frac{1}{4}$ equals B's $\frac{1}{4} \cdot \frac{3}{4}$, so neither coin explains it better. HH favours A 9 to 1; TT favours B 9 to 1.

Sanity check: TH has the same paths as HT, so the four possible records add to $\frac{10+6+6+10}{32} = 1$.



Each record's chance is a sum of two paths: 10/32, 6/32, 10/32.

The log-likelihood scores our guess: $\ell = -4.000$

The **log-likelihood** ℓ adds the log-chance of every record.
The records are independent, so their chances multiply
and their logs add.

$$\begin{aligned}\ell &= \ln p(\text{HH}) + \ln p(\text{HT}) + \ln p(\text{TT}) \\ &= \ln \frac{10}{32} + \ln \frac{6}{32} + \ln \frac{10}{32} \\ &= (-1.163) + (-1.674) + (-1.163) \\ &= -4.000\end{aligned}$$

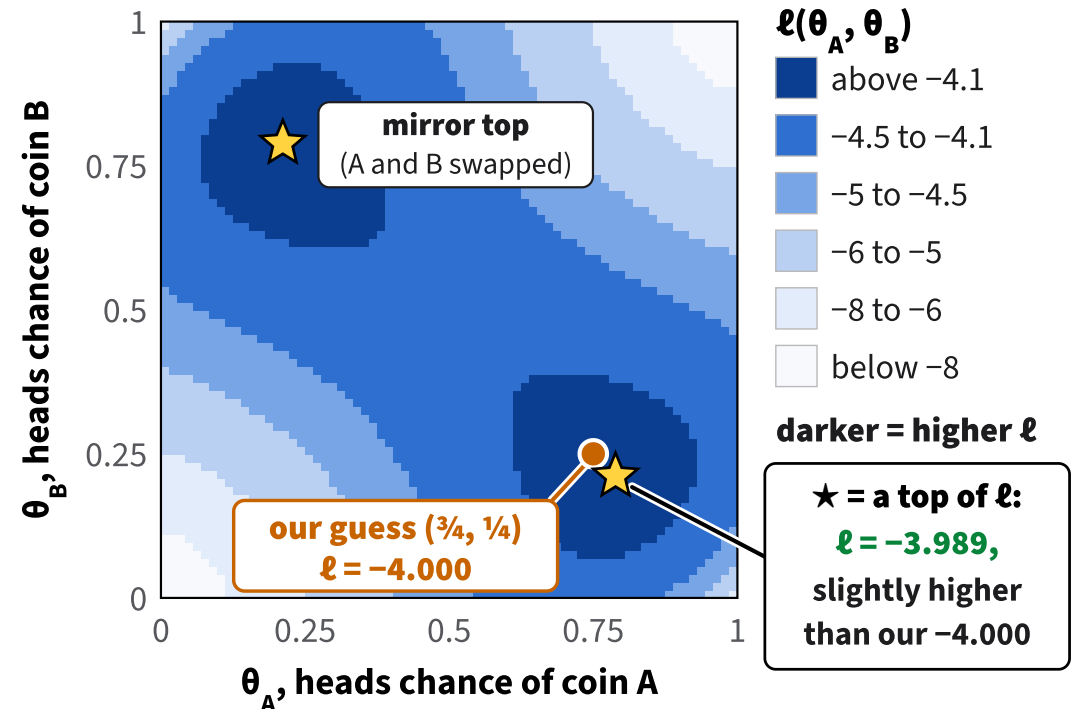
log of a product
= sum of logs

recall the totals

rarest HT ($\frac{6}{32}$)
costs most:
-1.674

add: -4.000

A higher ℓ means the data are more likely. **Learning** =
finding the (θ_A, θ_B) at a top of the map (★).



two mirror-image hills: swap the names A and B

Every guess gets one score ℓ ; ours scores -4.000, close to a top but not on it. How do we climb?

A log of a sum does not split, so counting no longer works

↳ Recall Theorem 1 (counting): a coin with h heads and t tails has best heads chance $\frac{h}{h+t}$. It needs each coin's flips **separately**.

Coin told: the log splits

Say HH and HT came from A, so A's flips are H, H, H, T:

$$\ln p = 2 \ln \frac{1}{2} + 3 \ln \theta_A + \ln(1 - \theta_A).$$

θ_B is absent, and **counting** gives $\theta_A = \frac{3}{3+1} = \frac{3}{4}$.

Coin hidden: one log, two coins

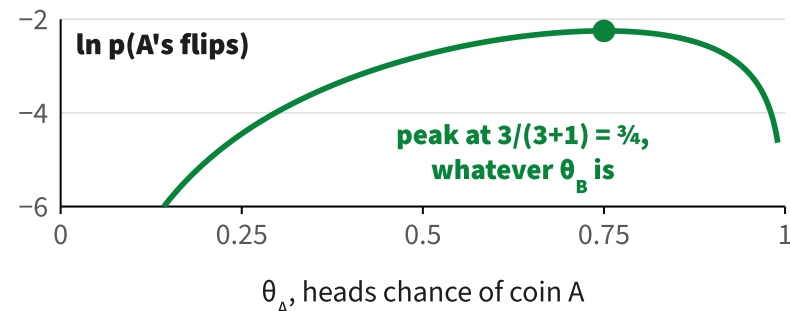
$$\ln p(\text{HH}) = \ln \left(\underbrace{\frac{1}{2}\theta_A^2}_{\text{path A}} + \underbrace{\frac{1}{2}\theta_B^2}_{\text{path B}} \right)$$

Both θ_A and θ_B sit inside **one** log, and a log of a sum is not a sum of logs:

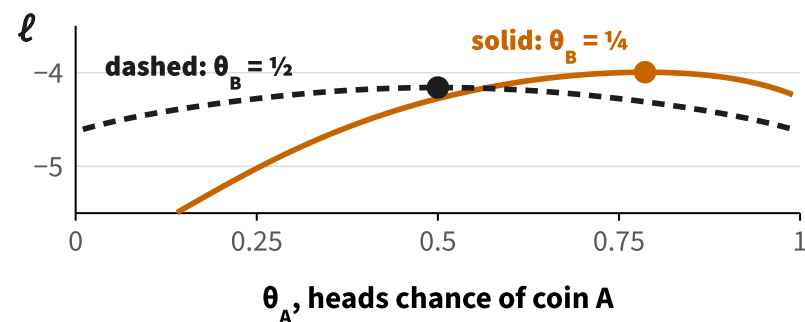
$$\ln \left(\frac{9}{32} + \frac{1}{32} \right) = \ln \frac{10}{32} = -1.163$$

$$\ln \frac{9}{32} + \ln \frac{1}{32} = \ln \left(\frac{9}{32} \cdot \frac{1}{32} \right) = \ln \frac{9}{1024} = -4.734$$

Coin told (say HH and HT were from A):
A's flips are H, H, H, T



Coin hidden: the best θ_A (dot) moves with θ_B



Hidden coin $\Rightarrow$ a log of a sum $\Rightarrow$ the best θ_A depends on θ_B : no counting formula.

Which coin made HH? Bayes' rule says A, with chance 9/10

before the flips: the fair toss



after the flips: posterior $p(A | x)$, $p(B | x)$



each bar splits one record between the coins

What we *can* compute: the **posterior** $p(A | x)$, the chance of coin A *after* seeing record x . **Bayes' rule:** a path's share of the total.

$$p(A | HH) = \frac{\overbrace{\frac{9}{32}}^{\text{path via A}}}{\underbrace{\frac{9}{32} + \frac{1}{32}}_{p(HH)}} = \frac{9}{10}.$$

record	via A	via B	$p(A x)$
HT	$\frac{3}{32}$	$\frac{3}{32}$	$\frac{3}{3+3} = \frac{1}{2}$
TT	$\frac{1}{32}$	$\frac{9}{32}$	$\frac{1}{1+9} = \frac{1}{10}$

Why it matters: seeing HH moved the chance of A from the prior $\frac{1}{2}$ to $\frac{9}{10}$.

The posterior is a soft label: HH is 0.9 coin A and 0.1 coin B.

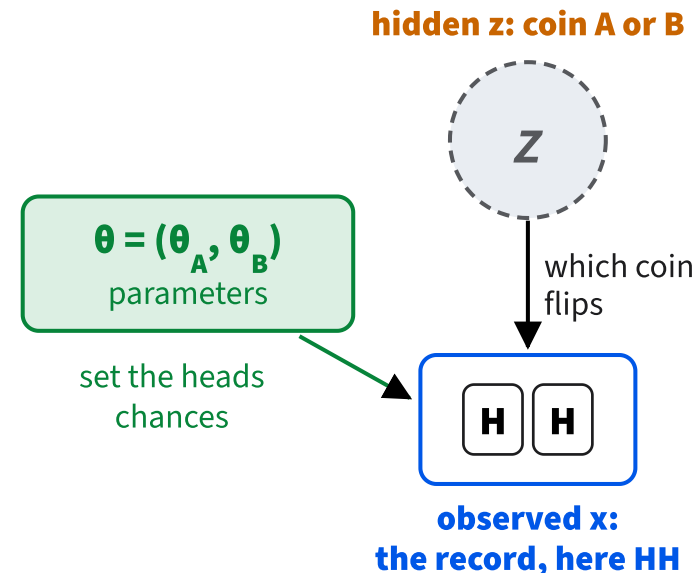
In general: an observed x , a hidden z , parameters θ

Definition (hidden-variable model). A model $p_\theta(x, z)$ gives the chance of an observed record x **together with** a hidden variable z ; θ are its parameters. The data are N records $x^{(1)}, \dots, x^{(N)}$.

name	formula	on the coins
marginal likelihood	$p_\theta(x) = \sum_z p_\theta(x, z)$	$p(\text{HH}) = \frac{9}{32} + \frac{1}{32} = \frac{10}{32}$
log-likelihood	$\ell(\theta) = \sum_{n=1}^N \ln p_\theta(x^{(n)})$	$N = 3$: recall $\ell = -4.000$
posterior	$p_\theta(z x) = \frac{p_\theta(x, z)}{p_\theta(x)}$	$p(A \text{HH}) = \frac{9/32}{10/32} = \frac{9}{10}$

Why "marginal"? Adding over z is called marginalizing z out: what is left is the chance of x alone.

Goal: the θ with the largest $\ell(\theta)$, the **maximum-likelihood estimate**.



$$p_\theta(x, z) = p(z) \cdot p_\theta(x | z)$$

$p(z) = \frac{1}{2}$: the fair toss

HH and A: $p(A) \cdot \theta_A \cdot \theta_A$

$$= \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$$

Three formulas: add the paths, take logs, divide for the posterior.

The slope of a log of a sum is a posterior-weighted slope

Why a slope? To climb ℓ toward the maximum-likelihood θ , we need its slope in each parameter.

Theorem 3 (Fisher's identity). Let θ_j be any one entry of $\theta = (\theta_A, \theta_B)$. Then

$$\frac{\partial}{\partial \theta_j} \ln p_\theta(x) = \sum_z p_\theta(z | x) \frac{\partial}{\partial \theta_j} \ln p_\theta(x, z).$$

In words: the slope of the log of a sum is each path's slope, weighted by its posterior.

$$\begin{aligned} \frac{\partial}{\partial \theta_j} \ln p_\theta(x) &= \frac{1}{p_\theta(x)} \sum_z \frac{\partial}{\partial \theta_j} p_\theta(x, z) \\ &= \sum_z \frac{p_\theta(x, z)}{p_\theta(x)} \frac{\partial}{\partial \theta_j} \ln p_\theta(x, z) \\ &= \sum_z p_\theta(z | x) \frac{\partial}{\partial \theta_j} \ln p_\theta(x, z) \end{aligned}$$

Chain rule $(\ln f)' = \frac{f'}{f}$ with $f = p_\theta(x) = \sum_z p_\theta(x, z)$.

For each path $g = p_\theta(x, z)$: $g' = g \cdot (\ln g)'$.

Bayes: path $\div$ total = posterior. ■

slope in θ_A of $\ln p_\theta(HH)$, at $(\frac{3}{4}, \frac{1}{4})$
a log of a sum: hard to handle directly

=

$p_\theta(A | HH) = 9/10$
× slope of
 $\ln p_\theta(HH, A)$
one path, no sum
on HH: $9/10 \times 2/\theta_A$

+

$p_\theta(B | HH) = 1/10$
× slope of
 $\ln p_\theta(HH, B)$
no θ_A in it
on HH: $1/10 \times 0$

one path splits into separate logs:

$\ln p_\theta(HH, A) = \ln \frac{1}{2} + \ln \theta_A + \ln \theta_A$
slope in θ_A : $2/\theta_A$, as in counting
weights: recall $p(A | HH) = 9/10$

Slope of a log of a sum = posterior-weighted single-path slopes; those split like counting.

Theorem 3 on HH: both sides give 2.4

↳ Put $x = \text{HH}$, $z \in \{A, B\}$, $\theta_j = \theta_A$ into Theorem 3, at $\theta = (\frac{3}{4}, \frac{1}{4})$. Recall the paths $\frac{9}{32}, \frac{1}{32}$ and $p(\text{HH}) = \frac{10}{32}$.

$$\underbrace{\frac{\partial}{\partial \theta_A} \ln p(\text{HH})}_{\text{left side} = 2.4 \text{ (directly)}} = \underbrace{p(A \mid \text{HH})}_{\frac{9/32}{10/32} = \frac{9}{10}} \cdot \underbrace{\frac{\partial}{\partial \theta_A} \ln p(\text{HH}, A)}_{\text{path A slope} = 2/\theta_A = 8/3} + \underbrace{p(B \mid \text{HH})}_{\frac{1/32}{10/32} = \frac{1}{10}} \cdot \underbrace{\frac{\partial}{\partial \theta_A} \ln p(\text{HH}, B)}_{\text{path B slope} = 0 \text{ (no } \theta_A \text{)}}$$

$$2.4 = \frac{9}{10} \cdot \frac{8}{3} + \frac{1}{10} \cdot 0 = \frac{24}{10} \checkmark \text{ both sides agree.}$$

Left side, directly (chain rule, no posterior):

$\ln p(\text{HH}) = \ln \left(\frac{1}{2} \theta_A^2 + \frac{1}{2} \theta_B^2 \right)$, so

$$\begin{aligned} \frac{\partial}{\partial \theta_A} \ln p(\text{HH}) &= \frac{\frac{1}{2} \cdot 2\theta_A}{p(\text{HH})} = \frac{\theta_A}{p(\text{HH})} = \frac{3/4}{10/32} \\ &= \frac{3}{4} \cdot \frac{32}{10} = \frac{12}{5} = 2.4 \end{aligned}$$

Right side, path by path:

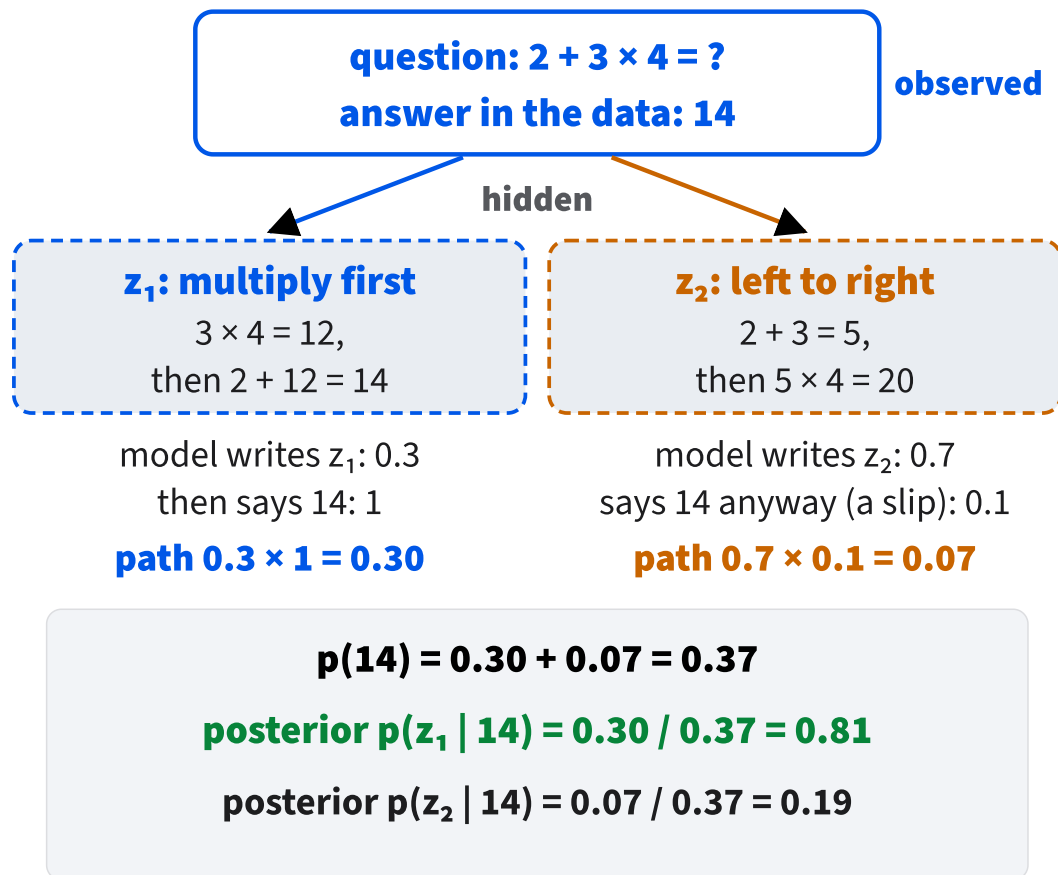
$\ln p(\text{HH}, A) = \ln \frac{1}{2} + 2 \ln \theta_A$: slope $\frac{2}{\theta_A} = \frac{8}{3}$

$\ln p(\text{HH}, B) = \ln \frac{1}{2} + 2 \ln \theta_B$: slope 0

Why check? The right side used only one-path logs, which split like counting; **no log of a sum** was differentiated. EM builds on exactly this.

2.4 > 0 both ways: raising θ_A makes HH more likely, and posterior-weighted path slopes find it.

A reasoning trace is a hidden coin



A language model is trained on **question** $\rightarrow$ **answer** pairs. The reasoning written in between, the **trace** z , is never in the data.

COIN GAME

record HH (seen)

coin z (hidden)

pick A with $\frac{1}{2}$

path via A $\frac{9}{32}, p(\text{HH}) = \frac{10}{32}$

$p(A | \text{HH}) = \frac{9}{10}$

REASONING MODEL

$\leftrightarrow$ question + answer 14 (seen)

$\leftrightarrow$ trace z (hidden)

$\leftrightarrow$ writes z_1 with 0.3 (toy)

$\leftrightarrow$ path via z_1 0.30, $p(14) = 0.37$

$\leftrightarrow$ $p(z_1 | 14) = 0.81$

What 0.81 means: given the answer 14, the trace was "multiply first" with chance 0.81. So in training, the slope of $\ln p(z_1, 14)$ gets weight 0.81 and that of $\ln p(z_2, 14)$ gets 0.19 (Theorem 3).

Zelikman et al., "STaR: Bootstrapping Reasoning With Reasoning," Neural Information Processing Systems (NeurIPS) 2022. Phan et al., "Training Chain-of-Thought via Latent-Variable Inference," NeurIPS 2023.

Same sum over paths, same Bayes' rule; only the hidden variable is a text.

Training on the answer raises the trace that explains it

Toy model: one parameter s , the chance of writing z_1 ; now $s = 0.3$.

Training climbs $\ln p(14)$, with

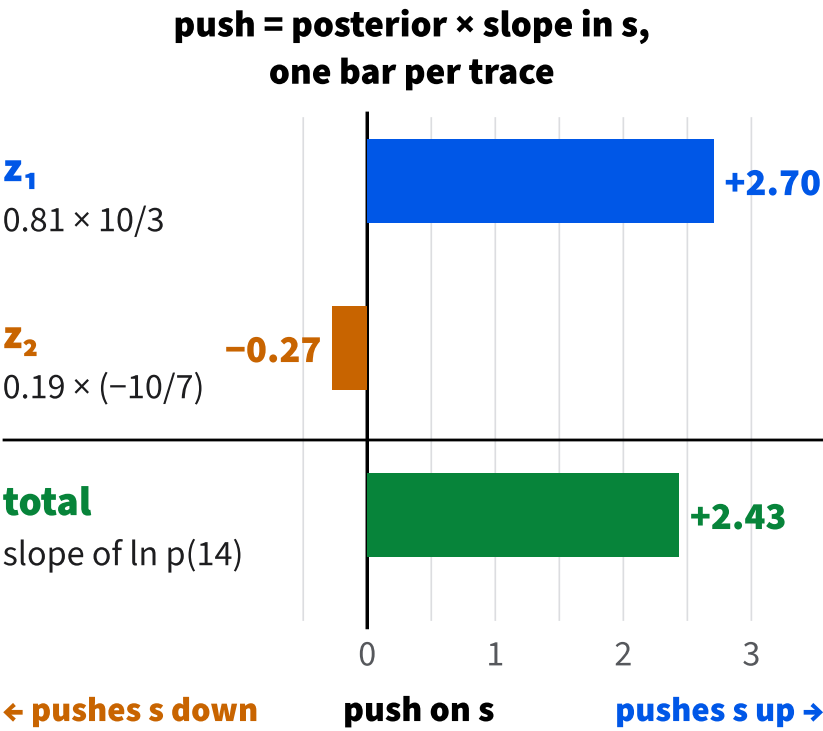
$$p(14) = s \cdot \underbrace{1}_{\text{says 14 after } z_1} + (1 - s) \cdot \underbrace{0.1}_{\text{says 14 after } z_2 \text{ (a slip)}} = 0.30 + 0.07 = 0.37.$$

Theorem 3 with $\theta_j = s$:

trace	$\ln p(z, 14)$	slope in s	posterior $p(z \mid 14)$	posterior $\times$ slope
z_1	$\ln s + \ln 1$	$\frac{1}{s} = \frac{10}{3}$	$\frac{0.30}{0.37} = 0.81$	+2.70
z_2	$\ln(1 - s) + \ln 0.1$	$\frac{-1}{1-s} = -\frac{10}{7}$	$\frac{0.07}{0.37} = 0.19$	-0.27
total				+2.43

Check: $\frac{d}{ds} \ln p(14) = \frac{1}{0.37} \frac{d}{ds} [s + 0.1(1 - s)] = \frac{1-0.1}{0.37} = 2.43 \checkmark$

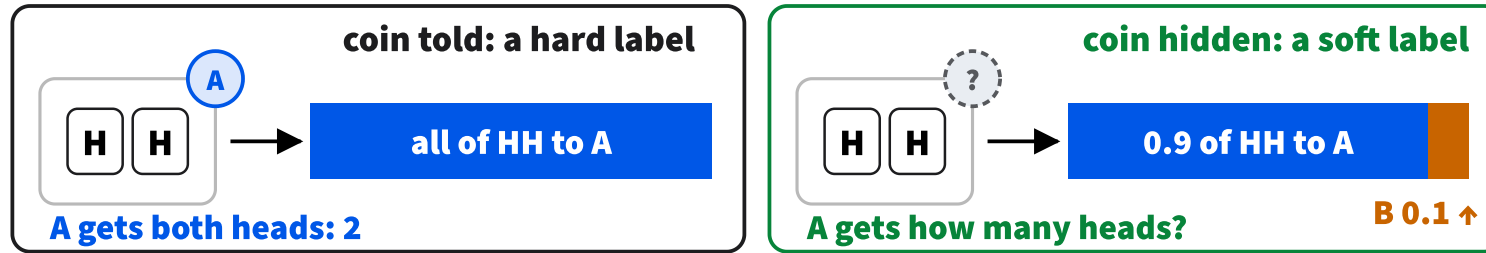
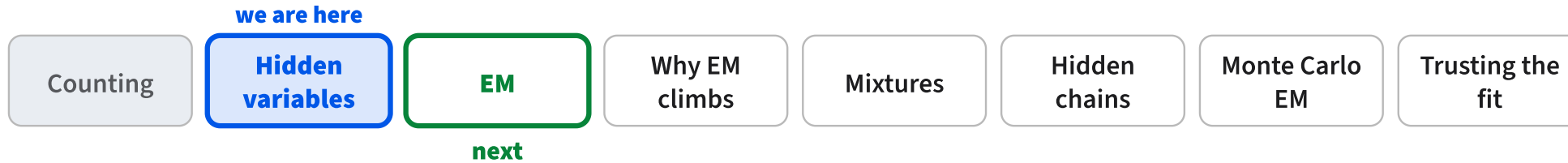
Then raise s , recompute $p(z \mid 14)$, and climb again: $p(14)$ rises toward 1.



Phan et al., Training Chain-of-Thought via Latent-Variable Inference, Neural Information Processing Systems (NeurIPS) 2023.

Slope +2.43 > 0: training on answers alone raises the chance of the trace that explains them.

Next: count with soft labels



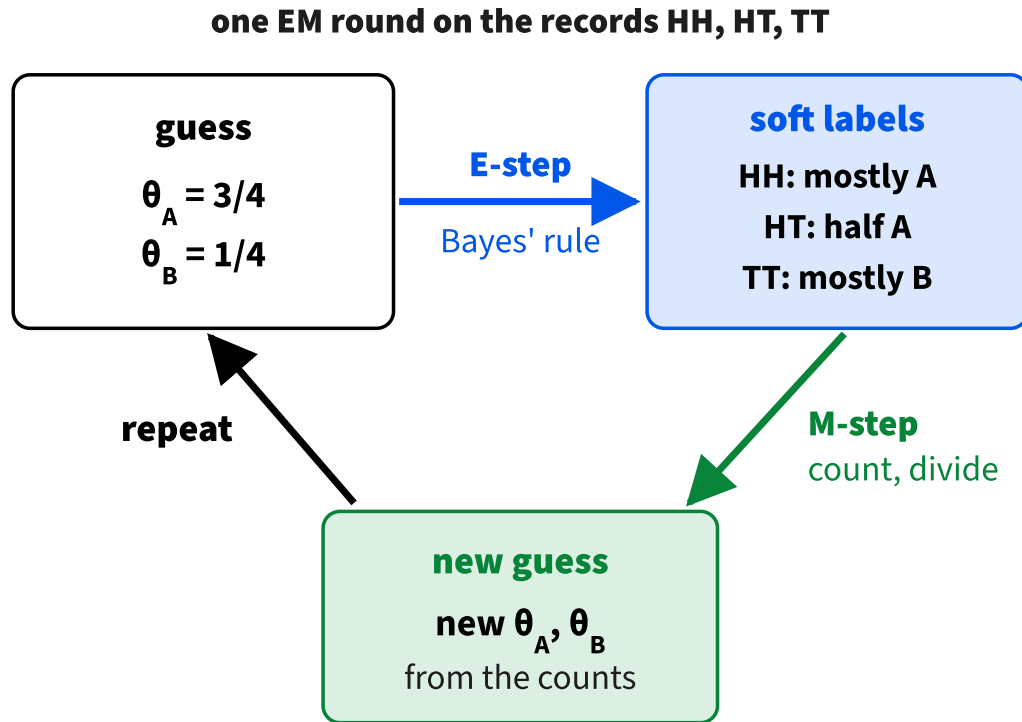
The posterior says HH is 0.9 coin A. How many heads should HH give to coin A?

1.8 2 0.9

Answer. $1.8 = 0.9 \times 2$ heads. Computing the 0.9 is the E-step (expectation); counting fractional heads and dividing is the M-step (maximization) of expectation–maximization (EM).

Soft-label every record, count fractions, divide: expectation–maximization.

EM: guess the coins, soft-label the records, count, repeat



the flips never change; only the guess does

Goal: the θ_A, θ_B with the largest **log-likelihood** $\ell = \ln p(\text{HH}) + \ln p(\text{HT}) + \ln p(\text{TT})$, where $p(\text{HH})$ is the model's chance of record HH.

A circle: with θ we could label the records; with labels we could count θ . **Break the circle with a guess.**

E-step (expectation): Bayes' rule turns the guess into **soft labels**, each record's chance of coin A.

M-step (maximization): **count heads ÷ flips** for each coin, a record counting only by its share.

Why a guess is fine: no round ever lowers ℓ , so each round's guess fits at least as well as the last.

Each round needs only the flips and the last guess, never the true coin.

E-step by hand: Bayes' rule splits each record between the coins

↳ Guess $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$; the friend picks each coin with chance $\frac{1}{2}$. A path multiplies its steps, e.g. HH via A:

$$\underbrace{\frac{1}{2}}_{\text{pick A}} \cdot \underbrace{\frac{3}{4}}_{\text{H by A}} \cdot \underbrace{\frac{3}{4}}_{\text{H by A}} = \frac{9}{32}$$

record	path via A	path via B	share to A
HH	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{9}{9+1} = 0.9$
HT	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{32}$	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{32}$	$\frac{3}{3+3} = 0.5$
TT	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{1}{1+9} = 0.1$

$$\underbrace{p(A \mid \text{record})}_{\text{share to A, the posterior}} = \frac{\text{path via A}}{\text{path via A} + \text{path via B}}$$

both paths, in 32nds

■ via A

■ via B



share to A: $9/10 = 0.9$



share to A: $3/6 = 0.5$



share to A: $1/10 = 0.1$

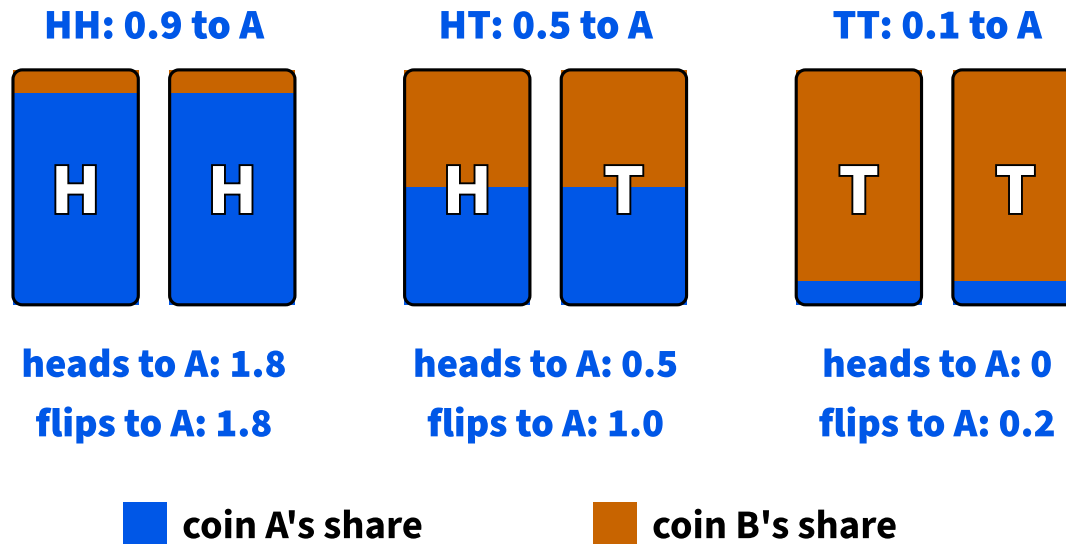
share = via A ÷ both

E-step output: soft labels 0.9, 0.5, 0.1 for coin A; coin B gets the rest.

M-step by hand: credit coin A with its share of every flip

↳ Recall Theorem 1: with the coin known, $\theta_A = \text{heads by A} \div \text{flips by A}$. Now each record is only partly A's work, so A is **credited** its share of the record's heads and flips.

blue part of each flip = coin A's share of it



blue totals: heads 2.3, flips 3

record	heads to A share × heads	flips to A share × 2
HH	$0.9 \cdot 2 = 1.8$	$0.9 \cdot 2 = 1.8$
HT	$0.5 \cdot 1 = 0.5$	$0.5 \cdot 2 = 1.0$
TT	$0.1 \cdot 0 = 0$	$0.1 \cdot 2 = 0.2$
total	2.3	3.0

$$\theta_A = \frac{\text{heads credited to A}}{\text{flips credited to A}} = \frac{2.3}{3} = 0.767$$

Credited = share × count. Then divide, exactly as in Theorem 1.

Coin B is credited the other share of every flip: $\theta_B = 0.7/3$

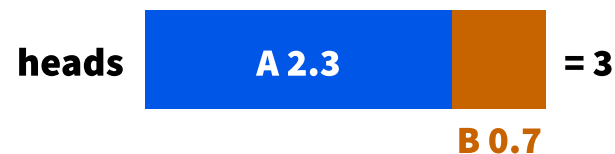
record	share to B 1 – share to A	heads to B	flips to B
HH	$1 - 0.9 = 0.1$	$0.1 \cdot 2 = 0.2$	$0.1 \cdot 2 = 0.2$
HT	$1 - 0.5 = 0.5$	$0.5 \cdot 1 = 0.5$	$0.5 \cdot 2 = 1.0$
TT	$1 - 0.1 = 0.9$	$0.9 \cdot 0 = 0$	$0.9 \cdot 2 = 1.8$
total		0.7	3.0

$$\theta_B = \frac{0.7}{3} = 0.233$$

Check: nothing lost, nothing counted twice

Heads $2.3 + 0.7 = 3$ and flips $3 + 3 = 6$: exactly the data's 3 heads in 6 flips.

the data's 3 heads and 6 flips, split between the coins



$$\theta_A = 2.3 / 3 = 0.767$$

$$\theta_B = 0.7 / 3 = 0.233$$

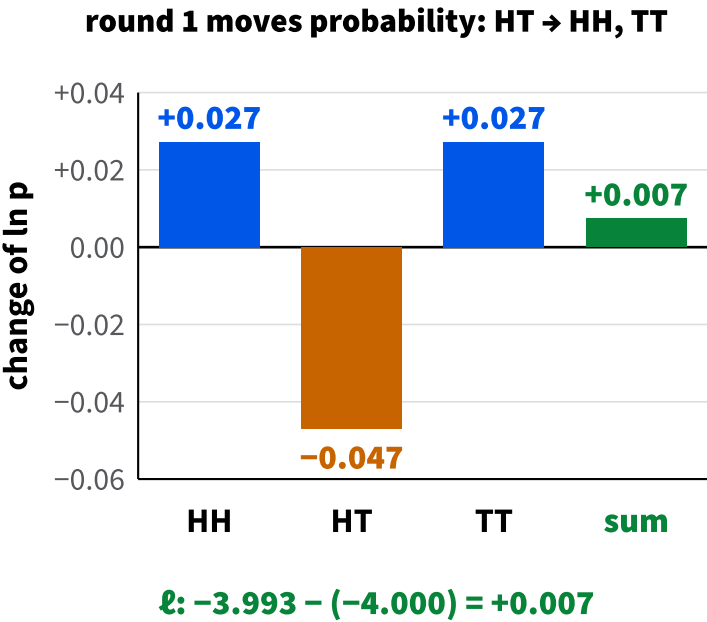
One round moved the guess from $(\frac{3}{4}, \frac{1}{4})$ to $(0.767, 0.233)$.

The new guess fits better: ℓ rises from -4.000 to -3.993

EM's target is the **log-likelihood** ℓ , the sum of $\ln p(\text{record})$, where $p(\text{record})$ adds the two paths.

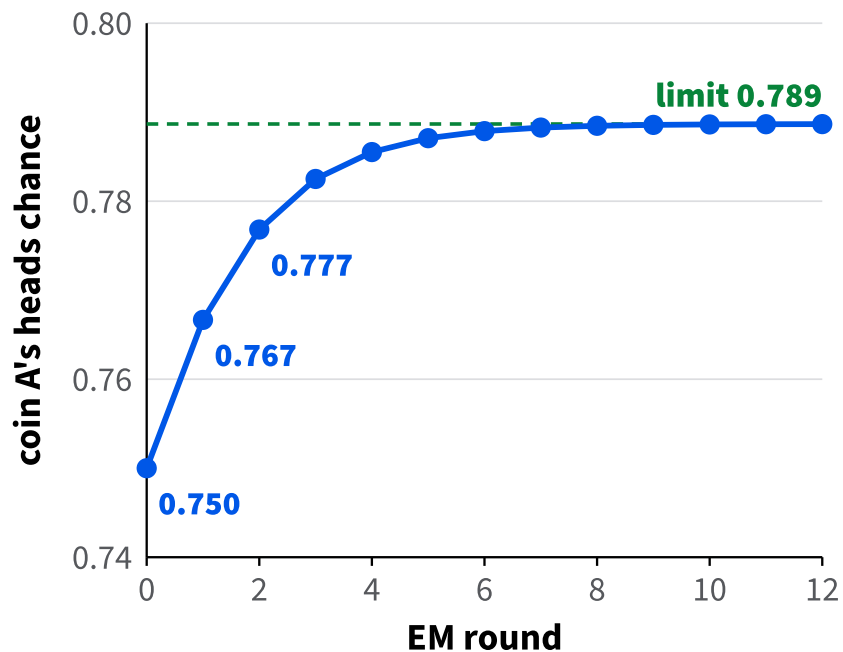
record	start ($\frac{3}{4}, \frac{1}{4}$)	after round 1 (0.767, 0.233)
HH	$\frac{9}{32} + \frac{1}{32} = 0.3125$	$\frac{1}{2} 0.767^2 + \frac{1}{2} 0.233^2 = 0.321$
HT	$\frac{3}{32} + \frac{3}{32} = 0.1875$	$\frac{1}{2}(0.767 \cdot 0.233) + \frac{1}{2}(0.233 \cdot 0.767) = 0.179$
TT	$\frac{1}{32} + \frac{9}{32} = 0.3125$	$\frac{1}{2} 0.233^2 + \frac{1}{2} 0.767^2 = 0.321$
ℓ	$2 \ln 0.3125 + \ln 0.1875$ $= -4.000$	$2 \ln 0.321 + \ln 0.179$ $= -3.993$

The coins moved apart, so mixed HT got **less likely** and HH, TT **more likely**; the net change is up.



Round 1 raised ℓ by 0.007: the soft-label count improved the fit.

Repeating the round: θ_A climbs 0.750, 0.767, 0.777, ..., 0.789



The start $(\frac{3}{4}, \frac{1}{4})$ is **mirror-symmetric** (swap H $\leftrightarrow$ T and A $\leftrightarrow$ B), so every round keeps $\theta_B = 1 - \theta_A$ and HT's share stays 0.5.

A's flips: $2q_{HH} + 2 \cdot 0.5 + 2(1 - q_{HH}) = 3$ every round. One round:

$$\theta_A^{\text{new}} = \frac{\overbrace{2 \cdot q_{HH}}^{\text{HH's heads}} + \overbrace{1 \cdot 0.5}^{\text{HT's}}}{3}, \quad q_{HH} = \frac{\theta_A^2}{\theta_A^2 + \theta_B^2}$$

round	θ_A	q_{HH} , share to A	new θ_A
0	$\frac{3}{4}$	$\frac{9}{9+1} = 0.900$	$\frac{2(0.900) + 0.5}{3} = 0.767$
1	$\frac{2.3}{3}$	$\frac{2.3^2}{2.3^2 + 0.7^2} = 0.915$ (the /3 cancels)	$\frac{2(0.915) + 0.5}{3} = 0.777$
limit	0.789	$\frac{0.789^2}{0.789^2 + 0.211^2} = 0.933$	$\frac{2(0.933) + 0.5}{3} = 0.789$

round 12 · $\theta_A = 0.789$ · $\theta_B = 0.211$

At 0.789 a round gives back 0.789: EM has reached a fixed point.

Why 0.789: there the model gives the data's own shares

The data have 2, 1 and 0 heads **once each**: shares $\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$.

The model gives HT and TH the same chance, so **1 head has chance** $2p(\text{HT})$. Matching the data: $p(\text{HH}) = p(\text{TT}) = \frac{1}{3}$ and $p(\text{HT}) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$.

At the start: $p(\text{HH}) = p(\text{TT}) = \frac{9}{32} + \frac{1}{32} = \frac{5}{16}$; 1 head = $2(\frac{3}{32} + \frac{3}{32}) = \frac{6}{16}$.

$$\frac{1}{2}(\theta_A^2 + (1 - \theta_A)^2) = \frac{1}{3}$$

$$1 - 2\theta_A(1 - \theta_A) = \frac{2}{3}$$

$$\Rightarrow \theta_A(1 - \theta_A) = \frac{1}{6}$$

$$\theta_A = \frac{3 + \sqrt{3}}{6} = 0.789$$

$$\ell = \ln \frac{1}{3} + \ln \frac{1}{6} + \ln \frac{1}{3} = -3.989$$

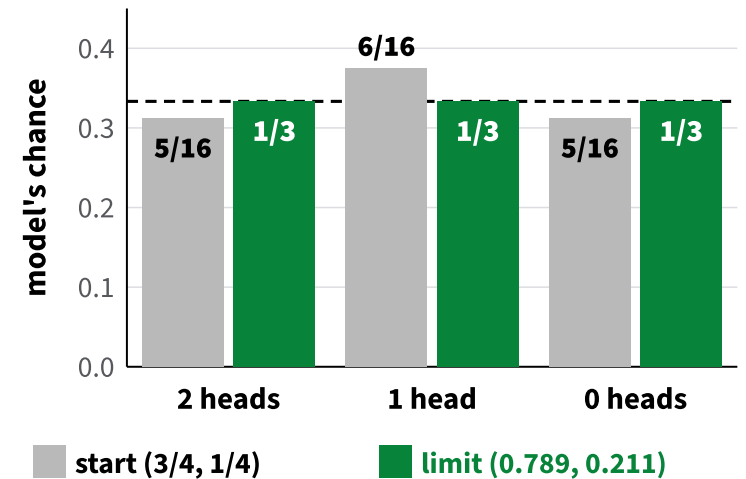
Model $p(\text{HH}) = \frac{1}{3}, \theta_B = 1 - \theta_A$.

Expand, multiply by 2.

Larger root of $\theta^2 - \theta + \frac{1}{6} = 0$; the smaller root, 0.211, is θ_B .

Uses $p(\text{HT}) = \frac{1}{2}\theta_A\theta_B + \frac{1}{2}\theta_B\theta_A = \theta_A\theta_B = \frac{1}{6}$.

each head count: model vs data



--- data: each head count in 1 of 3 records

$\ell = 3 \cdot [\text{average } \ln(\text{model share})] - \ln 2$ **peaks at the data's shares (Gibbs' inequality): -3.989 is the maximum.**

EM for any hidden-variable model: posterior, then weighted fit

Definition (expectation-maximization; Dempster, Laird & Rubin, 1977). Records $x^{(1)}, \dots, x^{(N)}$, hidden z , model $p_\theta(x, z)$. Start from a guess θ_0 ; for $t = 0, 1, 2, \dots$:

$$q_t^{(n)}(z) = p_{\theta_t}(z \mid x^{(n)})$$

$$\theta_{t+1} = \arg \max_{\theta} \sum_n \sum_z q_t^{(n)}(z) \ln p_{\theta}(x^{(n)}, z)$$

E-step: posterior soft labels. Coins: 0.9, 0.5, 0.1.

M-step: maximize each path's log, weighted by its soft label. Coins: $\theta_A = 2.3/3$.

SYMBOL	ON THE COINS
$x^{(n)}$	$\Leftrightarrow$ record n: HH, HT, TT
z	$\Leftrightarrow$ the coin, A or B
θ_t	$\Leftrightarrow$ guess after t rounds; $\theta_0 = (\frac{3}{4}, \frac{1}{4})$
$q_t^{(n)}(z)$	$\Leftrightarrow$ soft label of record n
$p_{\theta}(x, z)$	$\Leftrightarrow$ one path, e.g. $\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$ for (HH, A)

Why "expectation": the M-step maximizes $\sum_n \mathbb{E}_{z \sim q_t^{(n)}} [\ln p_{\theta}(x^{(n)}, z)]$, the **expected complete-data log-likelihood** (record and coin together). It is not ℓ itself but an easy stand-in: every term is the log of one path, with no log of a sum, so it splits into counts. Raising it raises ℓ ; Section 4 proves this.

E: infer the hidden part. M: fit as if it were seen, with weights.

Written out, the M-step objective counts expected heads

↳ Which θ maximizes $\sum_n \sum_z q_t^{(n)}(z) \ln p_\theta(x^{(n)}, z)$? Take the A-paths first, with soft labels 0.9, 0.5, 0.1.

$$\ln p_\theta(x, A) = \ln \frac{1}{2} + h \ln \theta_A + (2 - h) \ln(1 - \theta_A)$$

$$\begin{aligned} & \underbrace{0.9 [2 \ln \theta_A]}_{\text{HH}} + \underbrace{0.5 [\ln \theta_A + \ln(1 - \theta_A)]}_{\text{HT}} + \underbrace{0.1 [2 \ln(1 - \theta_A)]}_{\text{TT}} + \text{const} \\ &= \underbrace{(0.9 \cdot 2 + 0.5) \ln \theta_A}_{2.3 \text{ expected heads}} + \underbrace{(0.5 + 0.1 \cdot 2) \ln(1 - \theta_A)}_{0.7 \text{ expected tails}} + \text{const} \end{aligned}$$

$$\text{B-paths : } (0.1 \cdot 2 + 0.5) \ln \theta_B + (0.5 + 0.9 \cdot 2) \ln(1 - \theta_B) + \text{const}$$

One path: pick A, then h heads and $2 - h$ tails. A product, so its log splits.

Weight each A-path by its soft label; const = the $\ln \frac{1}{2}$ terms.

Collect the $\ln \theta_A$ and $\ln(1 - \theta_A)$ terms.

Same with B's soft labels 0.1, 0.5, 0.9: 0.7 heads, 2.3 tails. θ_A appears only in A-paths, θ_B only in B-paths, so **each is maximized on its own**.

The objective splits into Theorem 1's form, heads $\cdot \ln \theta$ + tails $\cdot \ln(1 - \theta)$, once per coin.

Slope zero gives count-and-divide: $\theta_A = 2.3 / (2.3 + 0.7)$

$$f = 2.3 \ln \theta_A + 0.7 \ln(1 - \theta_A)$$

$$f' = \frac{2.3}{\theta_A} - \frac{0.7}{1 - \theta_A}$$

$$= 0 \Rightarrow 2.3(1 - \theta_A) = 0.7 \theta_A$$

$$\theta_A = \frac{2.3}{2.3 + 0.7} = 0.767$$

$$\theta_B = \frac{0.7}{0.7 + 2.3} = 0.233$$

Coin A's part of the M-step objective.

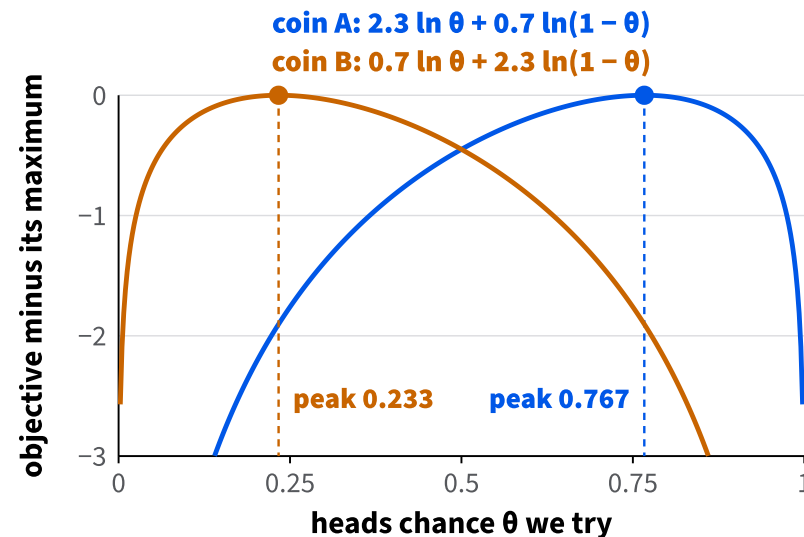
Slope of each log term: $\frac{d}{d\theta} \ln \theta = \frac{1}{\theta}$,
 $\frac{d}{d\theta} \ln(1 - \theta) = -\frac{1}{1 - \theta}$.

Slope zero; multiply by $\theta_A(1 - \theta_A)$.
 f is concave (each $\ln$ curves down),
 so this is the peak.

Expected heads ÷ expected flips.

Same steps for coin B's part.

Why count-and-divide: Theorem 1's peak, heads ÷ (heads + tails), holds for any counts, whole or fractional.



The arg max needs no search: weighted counts, then divide.

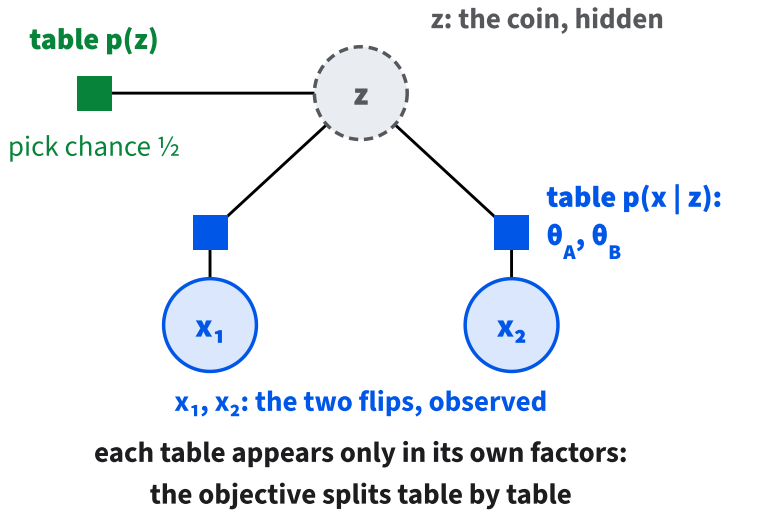
On a tree model, the M-step is count-and-divide with fractions

Corollary (of Theorem 1). If $p_{\theta}(x, z)$ is a product of tables on a tree factor graph, the M-step sets each table entry to

$$\theta_{\text{entry}} = \frac{\text{expected count of its case}}{\text{expected count of its condition}}.$$

In words: each record adds soft label $\times$ (times the case occurs in it); for example HH adds $0.9 \times 2 = 1.8$ to flips by A showing H, and $0.9 \times 2 = 1.8$ to flips by A.

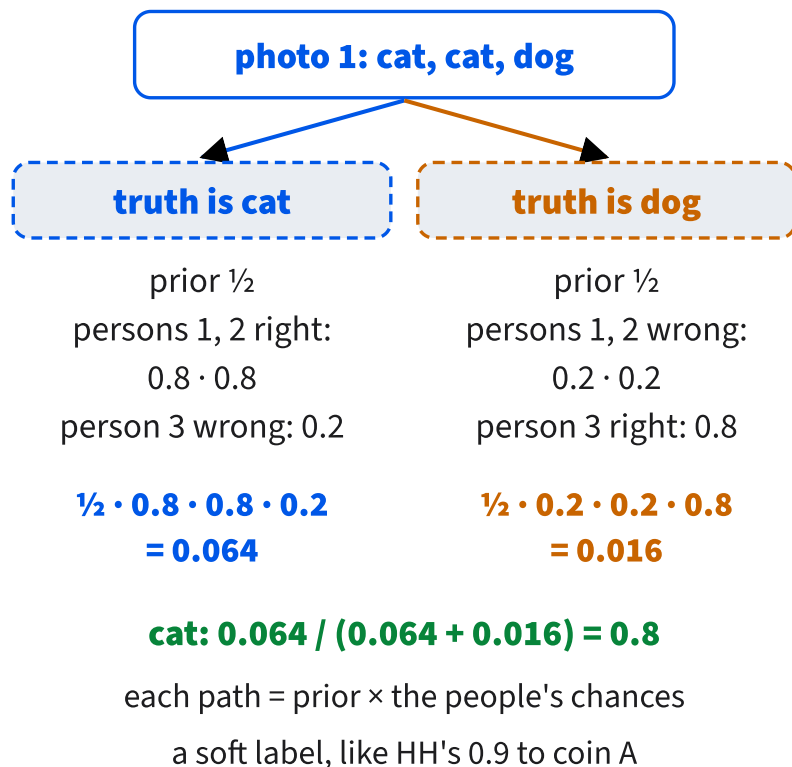
entry	case ÷ condition	value
$\theta_A = p(H \mid A)$	flips by A showing H: 2.3 ÷ flips by A: 3	0.767
$\theta_B = p(H \mid B)$	flips by B showing H: 0.7 ÷ flips by B: 3	0.233
$\pi = p(A)$, pick chance	records by A: $0.9 + 0.5 + 0.1 = 1.5$ ÷ records: 3	0.5



Why: each table is a **conditional distribution** $p(\text{child} \mid \text{parent})$ that appears only in its own factors, and each table sums to 1 on its own, so the objective splits table by table.

Every table of a tree model is refilled by counting, with soft labels as fractions.

Crowd labelling: a photo's true label is the hidden coin



Three people label photo 1: **cat, cat, dog**. The truth is hidden: cat or dog, chance $\frac{1}{2}$ each. Person k says the truth with chance a_k , their **accuracy**; start guess $a_k = 0.8$.

COIN GAME

coin A or B

two flips

θ_A : chance a flip shows H,
given the coin

soft label 0.9

CROWD LABELLING

$\Leftrightarrow$ truth cat or dog

$\Leftrightarrow$ three people's labels

$\Leftrightarrow a_k$: chance person k is right,
given the truth

$\Leftrightarrow$ posterior of cat 0.8

Photo 2, labels dog, dog, dog: dog path $\frac{1}{2} \cdot 0.8^3 = 0.256$ (all right), cat path $\frac{1}{2} \cdot 0.2^3 = 0.004$ (all wrong).

Posterior of dog = $\frac{0.256}{0.256 + 0.004} = 0.9846$.

E-step: the votes give soft labels (cat 0.8, dog 0.9846), not hard majority votes.

Dawid & Skene, Maximum likelihood estimation of observer error-rates using the EM algorithm, *J. Royal Statistical Society C* 28(1):20–28, 1979; used today to merge annotators' noisy labels.

Crowd M-step: each accuracy is counted like θ_A

↳ Recall the E-step: photo 1 (cat, cat, dog) has posterior of cat 0.8; photo 2 (dog, dog, dog) has posterior of dog 0.9846. Start accuracies $a_k = 0.8$.

M-step. Each cell: the label given, and the posterior chance it is right.

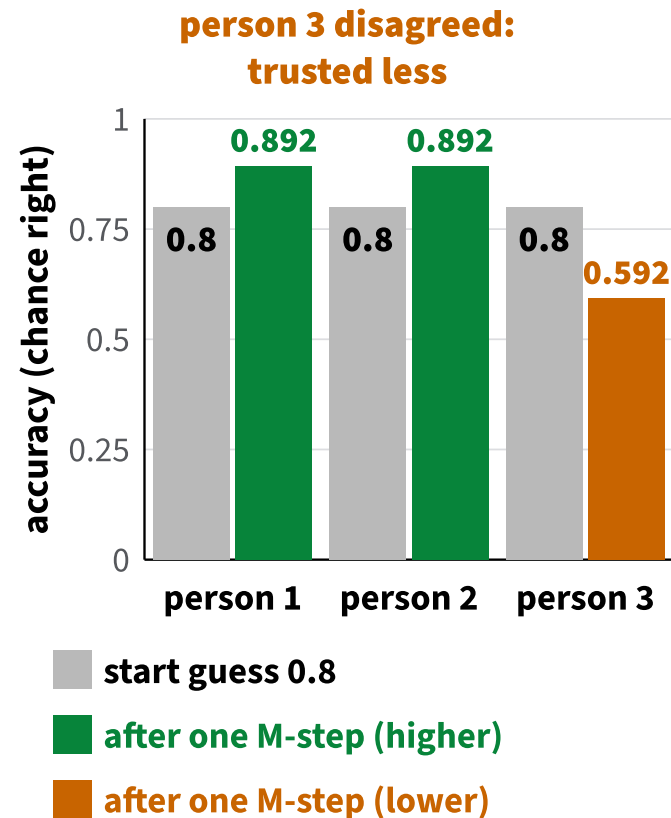
person	photo 1	photo 2	new accuracy a_k
1	cat: 0.8	dog: 0.9846	$(0.8 + 0.9846)/2 = 0.892$
2	cat: 0.8	dog: 0.9846	$(0.8 + 0.9846)/2 = 0.892$
3	dog: $1 - 0.8 = 0.2$	dog: 0.9846	$(0.2 + 0.9846)/2 = 0.592$

Like $\theta_A = \text{heads} \div \text{flips}$: **expected right answers $\div$ photos labelled.**

Next E-step on photo 1 (cat, cat, dog), with the new accuracies:

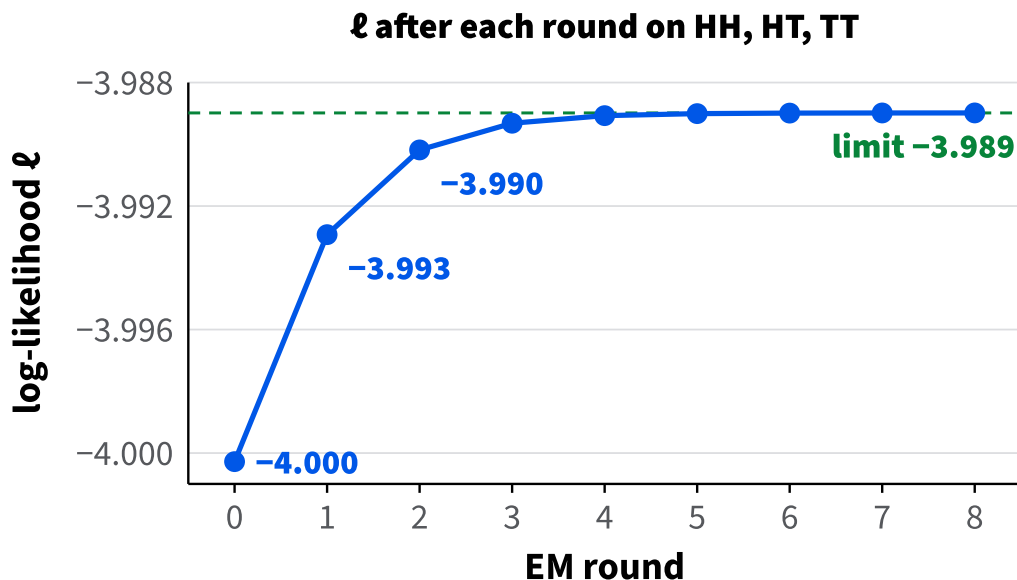
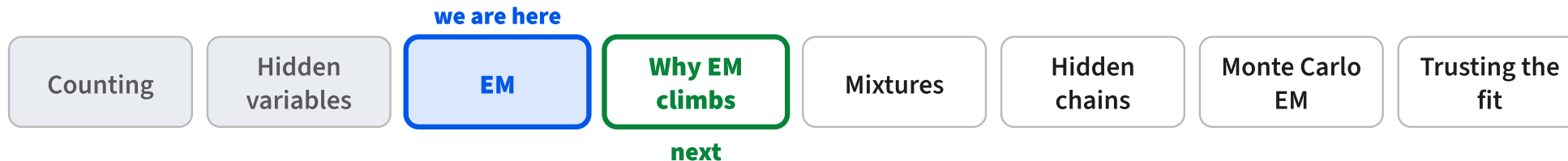
$$\text{cat: } \frac{1}{2} \cdot 0.892^2 \cdot (1 - 0.592) = 0.162, \quad \text{dog: } \frac{1}{2}(1 - 0.892)^2 \cdot 0.592 = 0.0035$$

Posterior of cat = $\frac{0.162}{0.162+0.0035} = 0.979$, up from 0.8: person 3's dog vote now counts for less.



With no true labels, EM learns whom to trust: person 3 drops to 0.592.

Next: does every EM round improve the fit?



On the coins, ℓ rose in every round: $-4.000 \rightarrow -3.993 \rightarrow -3.990 \rightarrow \dots \rightarrow -3.989$.

Round 2, $\theta_A = 0.777$, $\theta_B = 0.223$: $p(\text{HH}) = p(\text{TT}) = \frac{1}{2}(0.777^2 + 0.223^2) = 0.327$, $p(\text{HT}) = 0.777 \cdot 0.223 = 0.173$, so $\ell = 2 \ln 0.327 + \ln 0.173 = -3.990$.

For any model and any start, can an EM round lower ℓ ?

yes, sometimes

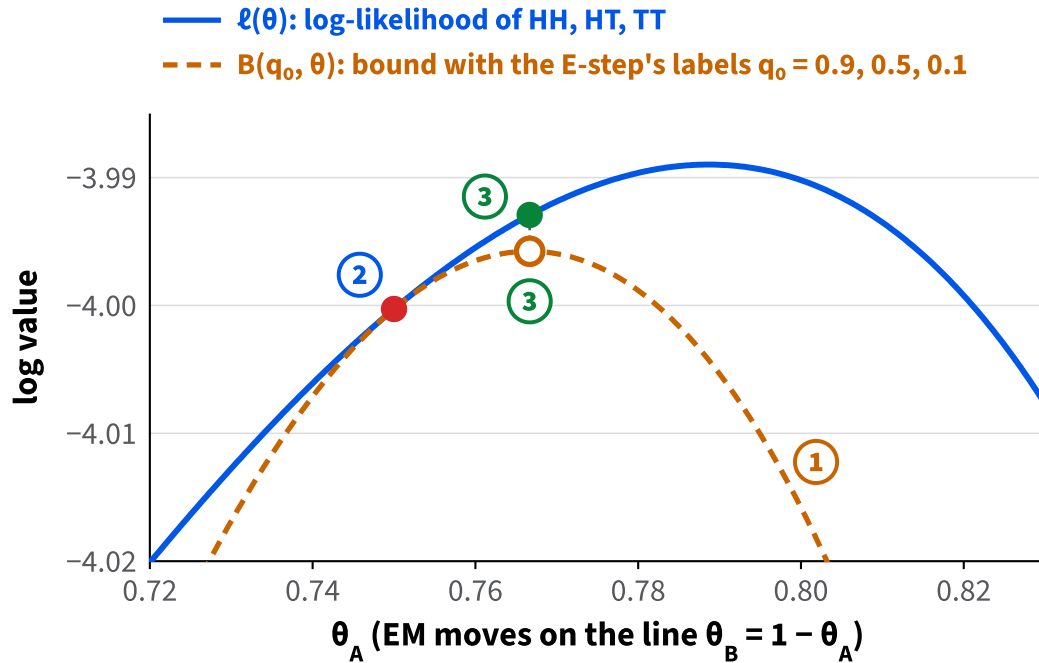
never

only at the start

Answer. Never. Section 4 proves it with one picture and one inequality.

A guarantee, not luck: EM climbs.

EM builds a floor under ℓ , then jumps to the floor's top



round **1**: θ_A moves from **0.750** (red dot) to **0.767** (circle)

Goal: show that EM's log-likelihood ℓ never goes down.

1 · The dashed bound $B(q_0, \theta)$

For each record, **average $\ln(\text{path} \div \text{label})$** over coins A and B; add the records. Labels $q_0 = 0.9, 0.5, 0.1$: the E-step's chance of A for HH, HT, TT. Path: pick the coin, see the record.

2 · Touch (red dot)

At $\theta_0 = (\frac{3}{4}, \frac{1}{4})$ the labels are the posteriors, so $\text{path} \div \text{label} = p(\text{record})$ for **both** coins: $B = \ell$.

3 · Below, then top

Elsewhere $\ln$ bends down, so the average of the logs is **below** the log of the average, $\ln p(\text{record})$: $B \leq \ell$. The M-step jumps to B 's top (circle); ℓ is **higher still** (green dot).

Touch at the current estimate, below everywhere else, jump to the top: ℓ can only go up.

On HH, any guess of the coin gives a floor under $\ln p(\text{HH})$

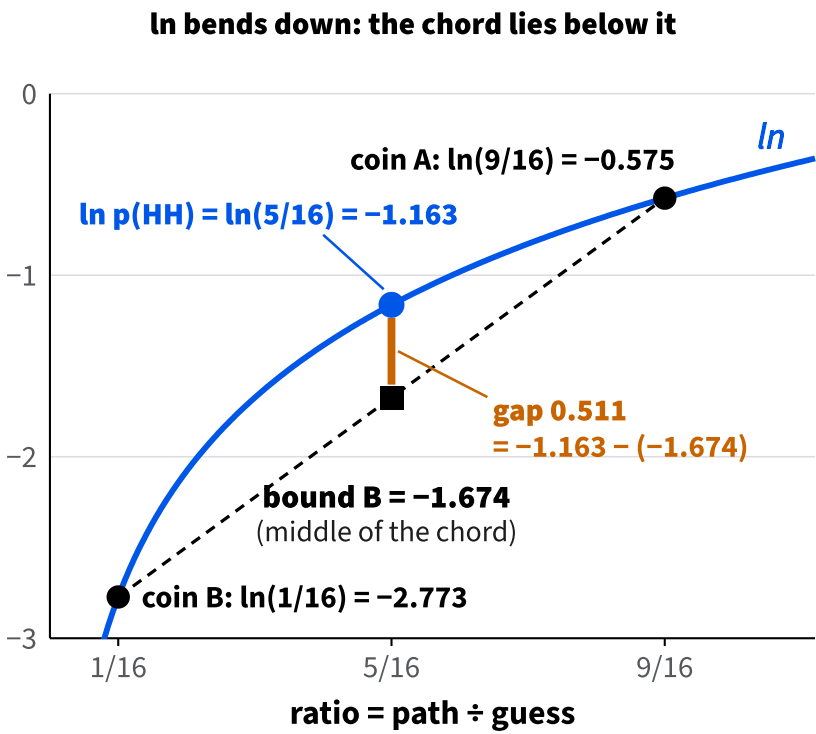
HH at $\theta = (\theta_A, \theta_B) = (\frac{3}{4}, \frac{1}{4})$	pick · H · H	path
via coin A	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4}$	$\frac{9}{32}$
via coin B	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4}$	$\frac{1}{32}$
$p(\text{HH})$	sum	$\frac{10}{32} = \frac{5}{16}$

A **path** is the chance the coin is picked and then shows H, H. A **guess** $q = (q(A), q(B))$ splits HH between the coins; its **bound** averages $\ln(\text{path} \div \text{guess})$:

$$B(q) = q(A) \ln \frac{9/32}{q(A)} + q(B) \ln \frac{1/32}{q(B)}$$

Guess $(\frac{1}{2}, \frac{1}{2})$: ratios $\frac{9}{32} \div \frac{1}{2} = \frac{9}{16}$ and $\frac{1}{32} \div \frac{1}{2} = \frac{1}{16}$, so
 $B = \frac{1}{2} \ln \frac{9}{16} + \frac{1}{2} \ln \frac{1}{16} = \frac{1}{2}(-0.575) + \frac{1}{2}(-2.773) = -1.674$.

For **any guess** the q 's cancel: $q(A) \frac{9/32}{q(A)} + q(B) \frac{1/32}{q(B)} = \frac{10}{32} = p(\text{HH})$. Since $\ln$ bends down, the average of the logs is at most the log of the average
(**Jensen's inequality**): **gap** $= \ln \frac{5}{16} - B = -1.163 - (-1.674) = 0.511$.



Because $\ln$ bends down, every guess gives a floor below $\ln p(\text{HH})$.

Only one guess makes the floor touch $\ln p(\text{HH})$

Predict: which guess $q(A)$ gives HH the highest floor?

Answer. 0.9, HH's posterior: its floor reaches $\ln p(\text{HH})$ itself.

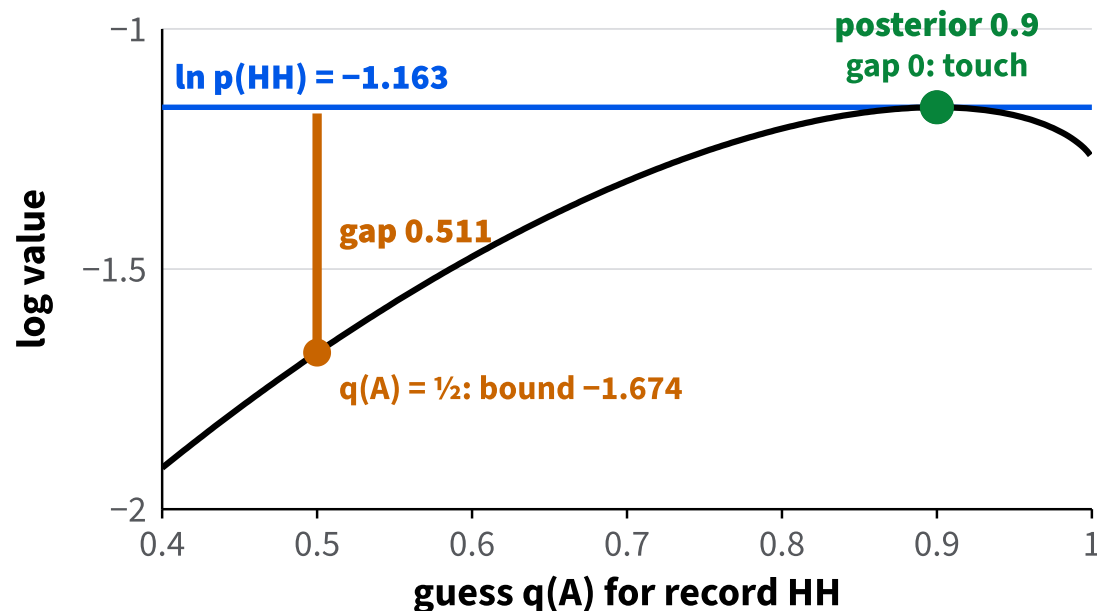
Recall HH's posterior, path $\div$ total: $\frac{9/32}{10/32} = 0.9$ for A, 0.1 for B. Guess $q = (0.9, 0.1)$; the ratios:

$$\frac{9/32}{0.9} = \frac{5}{16}, \quad \frac{1/32}{0.1} = \frac{5}{16}.$$

Why equal: posterior = path $\div p(\text{HH})$, so path $\div$ posterior = $p(\text{HH})$ for **each** coin.

Both logs are $\ln \frac{5}{16}$, so their average is too: $B = -1.163 = \ln p(\text{HH})$, **gap 0**. Drag the slider to try other guesses.

black: bound $B(q)$ for every guess $q = (q(A), 1 - q(A))$



guess $q(A) = 0.90$: bound -1.163 + gap $0.000 = -1.163$

Every guess is a floor; only the posterior touches, and that is the E-step's guess.

Lemma 1: for every guess, $\ln p(x) = \text{bound} + \text{gap}$

↳ Symbols: record x ; hidden z (the coin); guess $q(z)$, any chances over z adding to 1; posterior $p_\theta(z | x) = p_\theta(x, z)/p_\theta(x)$; log-likelihood $\ell(\theta) = \sum_n \ln p_\theta(x^{(n)})$; bound $B(q, \theta) = \sum_z q(z) \ln \frac{p_\theta(x, z)}{q(z)}$.

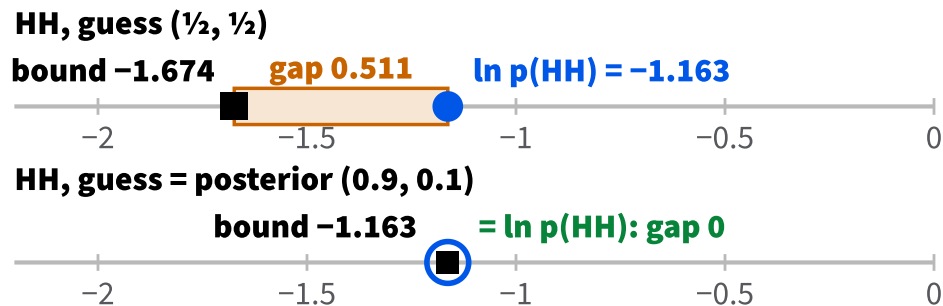
$$\begin{aligned} \ln p_\theta(x) &= \ln p_\theta(x, z) - \ln p_\theta(z | x) \\ &= \sum_z q(z) \left[\ln \frac{p_\theta(x, z)}{q(z)} + \ln \frac{q(z)}{p_\theta(z | x)} \right] \\ &= \underbrace{B(q, \theta)}_{\text{bound}} + \underbrace{D_{\text{KL}}(q \| p_\theta(\cdot | x))}_{\text{gap}} \end{aligned}$$

Lemma 1 (bound and gap). $D_{\text{KL}} \geq 0$, and $D_{\text{KL}} = 0$ exactly when $q = p_\theta(z | x)$. Hence $B(q, \theta) \leq \ln p_\theta(x)$ for every guess; with one guess $q^{(n)}$ per record, $\sum_n B(q^{(n)}, \theta) \leq \ell(\theta)$.

The posterior's definition, in logs; true for every z . HH, $z = A$: $\ln \frac{9}{32} - \ln 0.9 = \ln \frac{5}{16}$.

Average over q (weights add to 1); subtract and add $\ln q(z)$.

The second sum is the **Kullback-Leibler divergence**. HH, $q = (\frac{1}{2}, \frac{1}{2})$: $-1.674 + 0.511 = -1.163$.



$\ln p(x)$ splits into bound + gap; the gap is never negative and vanishes only at the posterior.

The gap is never negative because $\ln y \leq y - 1$

↳ To show: $D_{\text{KL}} \geq 0$. Write $y = p_{\theta}(z | x) / q(z)$, posterior $\div$ guess.

$$-D_{\text{KL}} = \sum_z q(z) \ln \frac{p_{\theta}(z | x)}{q(z)}$$

Flip the fraction, flip the sign.

$$\leq \sum_z q(z) \left(\frac{p_{\theta}(z | x)}{q(z)} - 1 \right)$$

$\ln y \leq y - 1$: $\ln$ lies below its tangent at $y = 1$.

$$= \sum_z p_{\theta}(z | x) - \sum_z q(z) = 1 - 1 = 0$$

Both add to 1.

HH, guess $(\frac{1}{2}, \frac{1}{2})$: $y_A = \frac{0.9}{0.5} = 1.8$, $y_B = \frac{0.1}{0.5} = 0.2$.

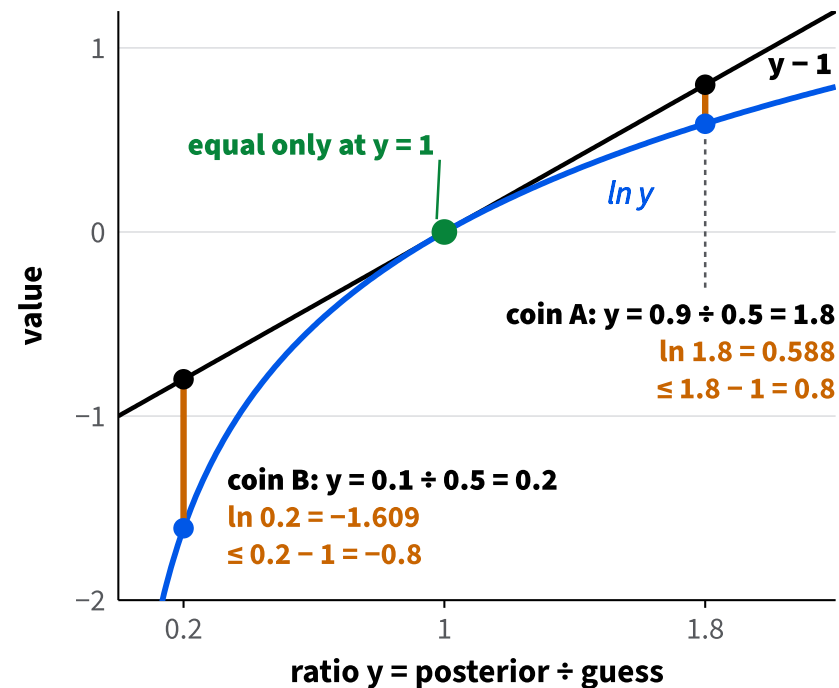
$$\frac{1}{2} \ln 1.8 + \frac{1}{2} \ln 0.2 = \frac{1}{2}(0.5878) + \frac{1}{2}(-1.6094) = -0.5108$$

$$\leq \frac{1}{2}(1.8 - 1) + \frac{1}{2}(0.2 - 1) = \frac{1}{2}(0.8) + \frac{1}{2}(-0.8) = 0.$$

So the gap is $D_{\text{KL}} = 0.511 \geq 0$.

Equality needs every $y = 1$, that is $q = \text{posterior}$. ■

$\ln y$ stays below its tangent line $y - 1$



$\ln$ lies below its tangent, so the gap is ≥ 0 : Lemma 1 holds.

Theorem 4: no EM round lowers the log-likelihood

Recall EM, round t , records $x^{(n)}$:

E-step $q_t^{(n)}(z) = p_{\theta_t}(z \mid x^{(n)})$, the posterior.

Coins at $\theta_0 = (\frac{3}{4}, \frac{1}{4})$: chance of A is 0.9, 0.5, 0.1 for HH, HT, TT.

M-step $\theta_{t+1} = \arg \max_{\theta} \sum_n \sum_z q_t^{(n)}(z) \ln p_{\theta}(x^{(n)}, z)$.

Coins: heads to A = $0.9 \cdot 2 + 0.5 \cdot 1 + 0.1 \cdot 0 = 2.3$,

flips to A = $2(0.9 + 0.5 + 0.1) = 3$, so $\theta_A = \frac{2.3}{3} = 0.767$;

heads to B = $0.1 \cdot 2 + 0.5 \cdot 1 + 0.9 \cdot 0 = 0.7$,

flips to B = $2(0.1 + 0.5 + 0.9) = 3$, so $\theta_B = \frac{0.7}{3} = 0.233$.

Theorem 4 (EM climbs; Dempster, Laird & Rubin, 1977). With Lemma 1's bound for the E-step's labels,

$$B(q_t, \theta) = \sum_n \sum_z q_t^{(n)}(z) \ln \frac{p_{\theta}(x^{(n)}, z)}{q_t^{(n)}(z)},$$

every round t satisfies

$$\ell(\theta_{t+1}) \underset{\text{floor}}{\geq} B(q_t, \theta_{t+1}) \underset{\text{climb}}{\geq} B(q_t, \theta_t) \underset{\text{touch}}{=} \ell(\theta_t).$$

In words: Read right to left: equal at θ_t , the M-step raises B , ℓ stays above B .

Predict: if the M-step only raises its objective, without maximizing it, can ℓ fall?

yes

no

Answer. No: the climb link only needs B to rise. This variant is **generalized EM**.

Touch, climb, floor: every EM round keeps or raises ℓ .

Round 1: the sum of the logs climbs at every link

From $\theta_0 = (\frac{3}{4}, \frac{1}{4})$, round 1 gives $\theta_1 = (\frac{23}{30}, \frac{7}{30})$. The bound B keeps the **old labels** $q_0 = 0.9, 0.5, 0.1$ (chance of A for HH, HT, TT).

record	$\ln p_{\theta_0}(x)$	B at θ_1	$\ln p_{\theta_1}(x)$
HH	-1.16315	-1.13739	-1.13597
HT	-1.67398	-1.72099	-1.72099
TT	-1.16315	-1.13739	-1.13597
sum	-4.00028	-3.99577	-3.99293
in the chain	$\ell(\theta_0)$	$B(q_0, \theta_1)$	$\ell(\theta_1)$

Read the sum row left to right: **each value is at least the one before**, as Theorem 4's chain says.

HH in full. Recall $\ln p_{\theta_0}(\text{HH}) = \ln \frac{5}{16} = -1.16315$.

$$\begin{aligned} B &= 0.9 \ln \frac{\frac{1}{2}(\frac{23}{30})^2}{0.9} + 0.1 \ln \frac{\frac{1}{2}(\frac{7}{30})^2}{0.1} \\ &= 0.9 \ln \frac{529}{1620} + 0.1 \ln \frac{49}{180} \\ &= 0.9(-1.1192) + 0.1(-1.3011) = -1.13739. \end{aligned}$$

$$\begin{aligned} \ln p_{\theta_1}(\text{HH}) &= \ln \left(\frac{1}{2} \left(\frac{23}{30} \right)^2 + \frac{1}{2} \left(\frac{7}{30} \right)^2 \right) \\ &= \ln \frac{289}{900} = -1.13597. \end{aligned}$$

TT mirrors HH: swap heads with tails and A with B.

Predict: HT's term falls, from -1.67398 to -1.72099. Does that break Theorem 4?

yes

no

Answer. No: the theorem is about the sum, and $-3.99293 - (-4.00028) = +0.00735$.

Summed over the records: $-4.000 \leq -3.996 \leq -3.993$, so ℓ climbed.

HT's own term falls, but the sum still rises

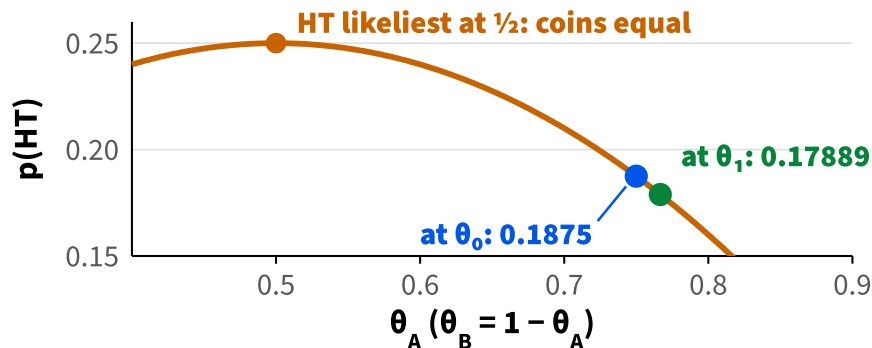
Recall round 1: $\theta_0 = (\frac{3}{4}, \frac{1}{4}) \rightarrow \theta_1 = (\frac{23}{30}, \frac{7}{30})$. HT has **two equal paths**, one per coin, so:

$$p_{\theta_0}(\text{HT}) = 2 \cdot \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{4} = 0.1875, \ln 0.1875 = -1.67398.$$

$$p_{\theta_1}(\text{HT}) = 2 \cdot \frac{1}{2} \cdot \frac{23}{30} \cdot \frac{7}{30} = \frac{161}{900} = 0.17889,$$

$$\ln \frac{161}{900} = -1.72099: \text{below} -1.67398.$$

Why: a mixed record is likeliest when the coins are near $\frac{1}{2}$; round 1 pushed them apart, to 0.767 and 0.233, to fit HH and TT.



change in each record's log-likelihood

$$(\ln p(x) \text{ at } \theta_1) - (\ln p(x) \text{ at } \theta_0)$$

$$\text{HH: } -1.13597 - (-1.16315)$$

$$+0.02718$$

$$\text{HT: } -1.72099 - (-1.67398)$$

$$-0.04701$$

$$\text{TT: the same as HH}$$

$$+0.02718$$

$$\text{sum: } 2(0.02718) - 0.04701$$

$$+0.00735$$

HT's paths stay equal, so its posterior is still 0.5: its gap is 0 and $B = \ell$ for HT.

One record may fall; Theorem 4 promises only that the sum rises.

Proof of Theorem 4: touch, climb, floor

↳ Touch and floor use Lemma 1; climb uses the M-step's definition. Coin values: round 1, summed over HH, HT, TT.

$$B(q_t, \theta_t) = \ell(\theta_t)$$

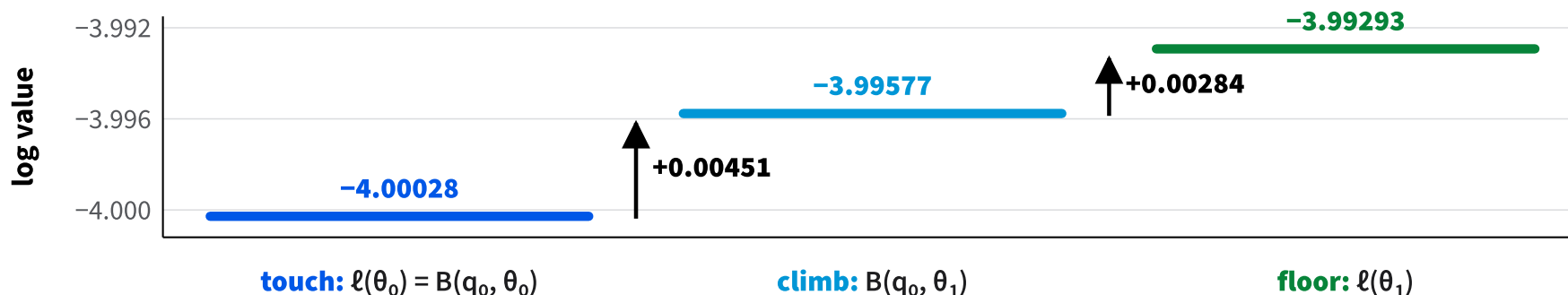
$$B(q_t, \theta_{t+1}) \geq B(q_t, \theta_t)$$

$$\ell(\theta_{t+1}) \geq B(q_t, \theta_{t+1})$$

Touch. The E-step sets q_t to the posterior at θ_t , so every gap is 0 (Lemma 1). Coins: -4.00028 on both sides.

Climb. $\ln \frac{p}{q} = \ln p - \ln q$, so $B(q_t, \theta) = \sum q_t \ln p_\theta(x, z) - \sum q_t \ln q_t =$ M-step objective – (a term with no θ). Maximizing one maximizes the other. Coins: $-3.99577 \geq -4.00028$.

Floor. Lemma 1 holds for every guess: also for the old q_t at the new θ_{t+1} . Coins: $-3.99293 \geq -3.99577$. ■



Chain the three lines: $\ell(\theta_{t+1}) \geq B(q_t, \theta_{t+1}) \geq B(q_t, \theta_t) = \ell(\theta_t)$.

Climbing is not reaching the top: the flat start never moves

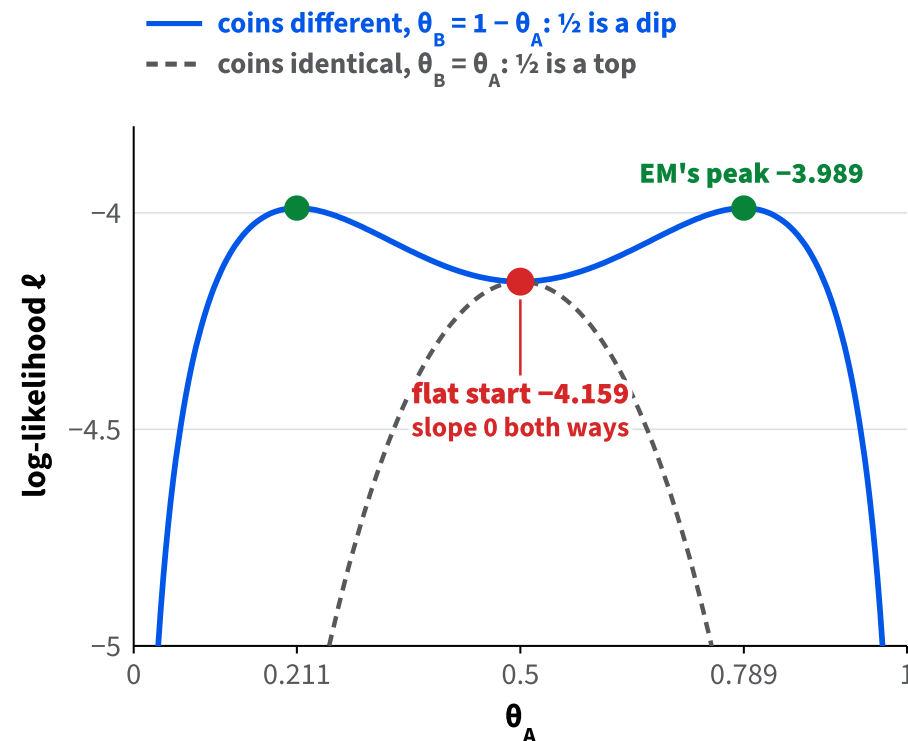
Start with identical coins, $\theta_A = \theta_B = \frac{1}{2}$. **E-step:** HH's two paths are both $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$, so the chance of A is $\frac{1/8}{1/8+1/8} = \frac{1}{2}$; the same for HT and TT.

record	chance of A	heads to A	flips to A
HH	$\frac{1}{2}$	$\frac{1}{2} \cdot 2 = 1$	$\frac{1}{2} \cdot 2 = 1$
HT	$\frac{1}{2}$	$\frac{1}{2} \cdot 1 = 0.5$	$\frac{1}{2} \cdot 2 = 1$
TT	$\frac{1}{2}$	$\frac{1}{2} \cdot 0 = 0$	$\frac{1}{2} \cdot 2 = 1$
total		1.5	3

M-step: $\theta_A = \frac{1.5}{3} = \frac{1}{2}$, and B gets the same counts: **EM returns to $\frac{1}{2}$.**

Each record has $p = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$, so $\ell = 3 \ln \frac{1}{4} = -4.159$. Recall: from $(\frac{3}{4}, \frac{1}{4})$ EM reaches $\ell = -3.989$ at $\theta_A = 0.789$ (or 0.211, the coins renamed).

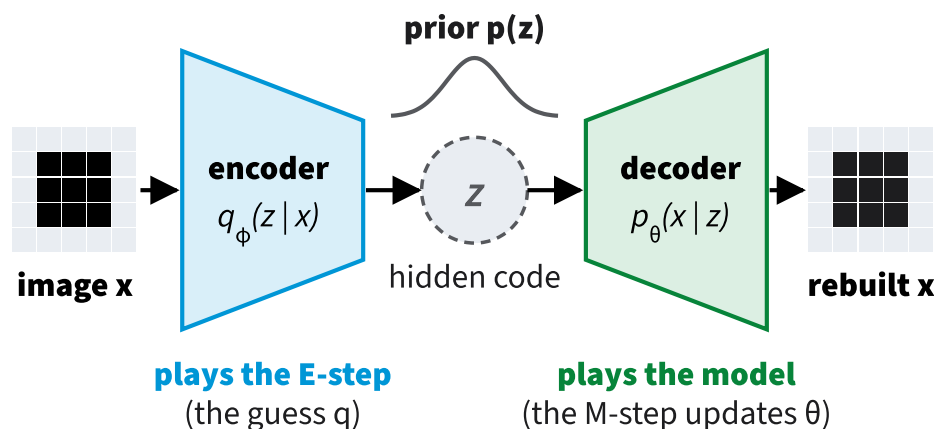
Equal chances give equal counts, so EM never leaves this **saddle**: a top along identical coins (dashed), a dip across them (blue).



Theorem 4 only forbids going down: start EM with different coins.

A variational autoencoder learns the E-step's guess

one network pass gives the guess for any image



training raises the ELBO in both θ and ϕ

Lemma 1 for one image: $\ln p_\theta(x) = \text{ELBO} + \text{gap}$

ELBO

gap

gap = $\text{KL}(\text{guess} \parallel \text{posterior}) \geq 0$

For an image x , the hidden code z is many real numbers, so the posterior cannot be computed exactly. A **variational autoencoder (VAE)** (Kingma & Welling, International Conference on Learning Representations (ICLR) 2014) trains an **encoder** with weights ϕ to output the guess $q_\phi(z | x)$.

TWO-COIN GAME

VARIATIONAL AUTOENCODER

record HH

↔

an image x

coin A or B

↔

code z , many numbers

fair pick, chance $\frac{1}{2}$

↔

prior $p(z)$, a standard bell curve

E-step: exact posterior

↔

encoder's guess q_ϕ

M-step: count

↔

gradient step on θ

bound $B(q, \theta)$

↔

evidence lower bound (ELBO)

$\text{ELBO}(\theta, \phi) = \mathbb{E}_{z \sim q_\phi} \left[\ln \frac{p(z) p_\theta(x | z)}{q_\phi(z | x)} \right]$: Lemma 1's B with path $p(z) p_\theta(x | z)$ and guess q_ϕ (the sum over z becomes an integral). **Evidence** = $p_\theta(x)$, so $\text{ELBO} \leq \ln p_\theta(x)$ by Lemma 1.

EM uses the exact posterior; a VAE learns a guess that scales to images.

An encoder that says 0.8 instead of 0.9 loses 0.0444 to the gap

Pretend a variational autoencoder's encoder looks at HH and says

$q(A) = 0.8$. Recall at $\theta = (\frac{3}{4}, \frac{1}{4})$: paths $\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$ and $\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$; exact posterior $\frac{9}{32} \div \frac{10}{32} = 0.9$. Its bound, coin by coin:

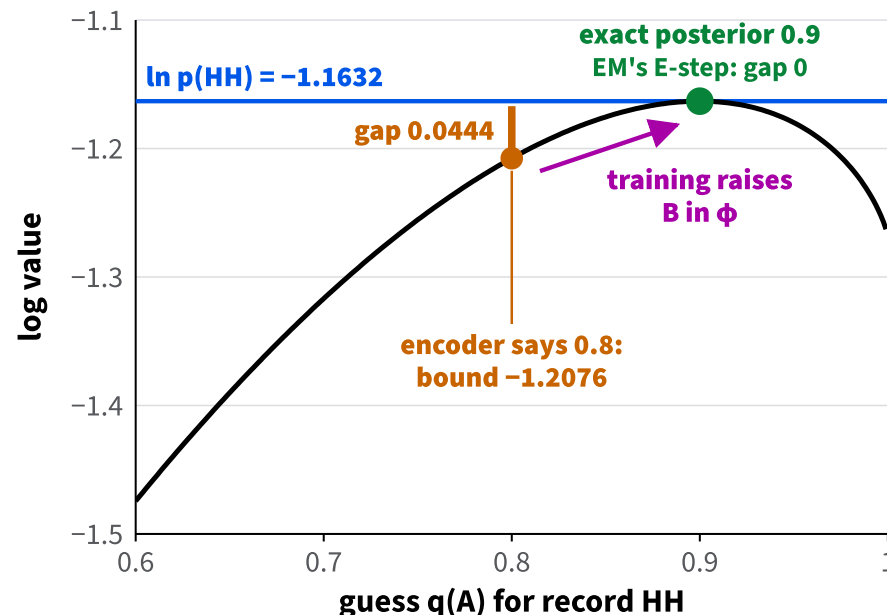
coin	guess	path	ratio = path ÷ guess	ln ratio	guess · ln
A	0.8	9/32	0.3516	-1.0454	-0.8363
B	0.2	1/32	0.1563	-1.8563	-0.3713
bound (ELBO) = sum					-1.2076

$$\text{Gap} = D_{\text{KL}}((0.8, 0.2) \parallel (0.9, 0.1)) = 0.8 \ln \frac{0.8}{0.9} + 0.2 \ln \frac{0.2}{0.1} \\ = 0.8(-0.1178) + 0.2(0.6931) = -0.0942 + 0.1386 = 0.0444.$$

Check: $-1.2076 + 0.0444 = -1.1632 = \ln \frac{5}{16} \checkmark$.

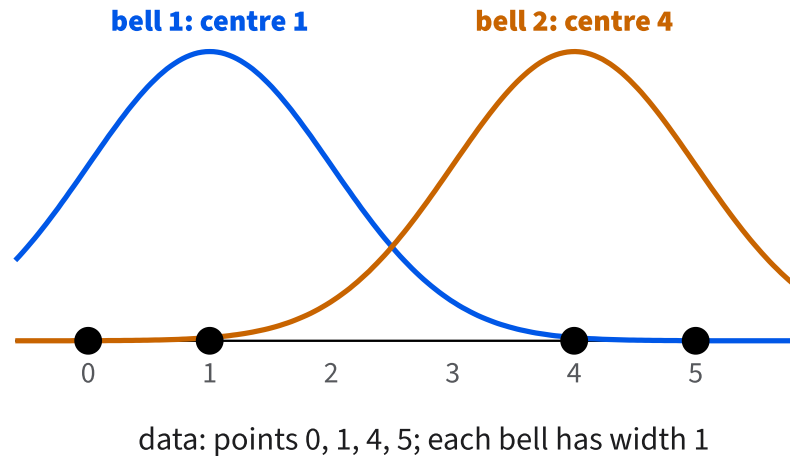
Bound = $\ln p(\text{HH}) - \text{gap}$, and $\ln p(\text{HH})$ does not depend on φ . So raising the bound in φ can only **shrink the gap**, moving the guess toward 0.9.

black curve: bound $B(q)$ on HH for each guess



The encoder's error is exactly Lemma 1's gap; training shrinks it.

Next: replace the coins by bell curves



Nothing in the proofs of Lemma 1 and Theorem 4 used coins: only an observed x , a hidden z , and a model $p_\theta(x, z)$.

The points 0, 1, 4, 5 come from two bell curves. In EM, what plays the hidden coin?

each point

each point's bell

the bell's width

Answer. The bell: hidden, picked first, then the point is drawn from it. This is a Gaussian mixture.

Same E-step, same M-step, same climb: now EM clusters points.

Same game, new records: numbers on a line instead of coin flips

Goal: find hidden groups in data with the same EM.

A friend picks **group 1 or group 2** by a fair toss, then draws a number from that group's **bell**: a normal (Gaussian) distribution with standard deviation (width) 1. We see only the numbers **0, 1, 4, 5**.

COIN GAME

coin A or B, chance $\frac{1}{2}$

record HH

θ_A , guess $\frac{3}{4}$

hidden: which coin

MIXTURE

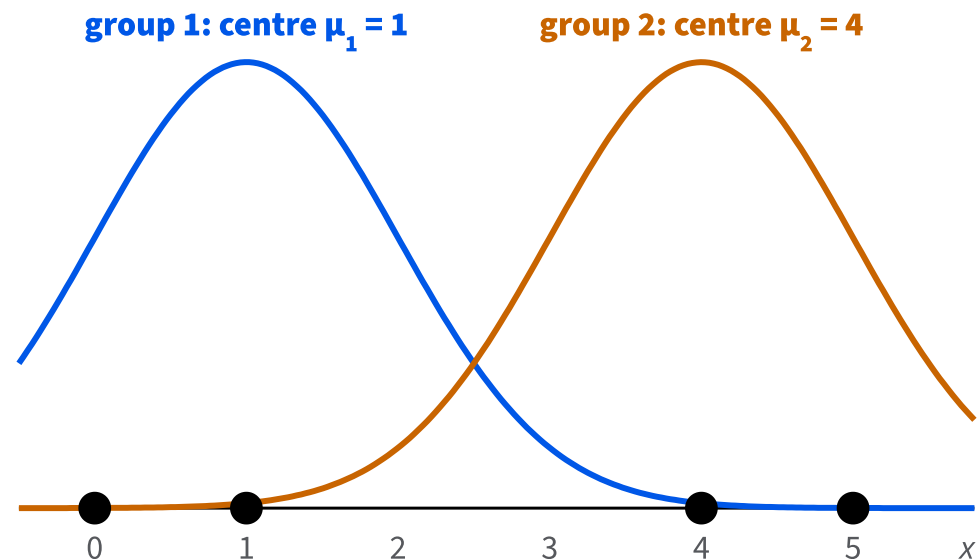
group 1 or 2, chance $\frac{1}{2}$

one number, e.g. 1

centre μ_1 , guess 1

hidden: which group

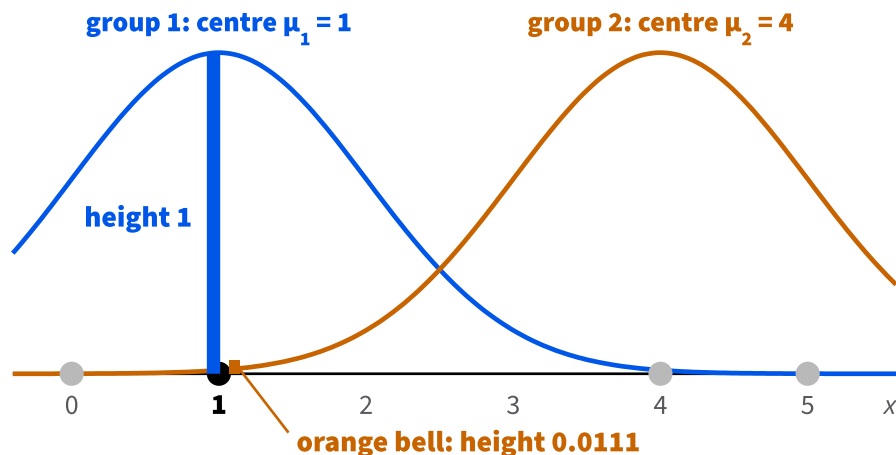
Unknowns: the centres μ_1, μ_2 . Working guess $\mu_1 = 1, \mu_2 = 4$, as $\theta_A = \frac{3}{4}, \theta_B = \frac{1}{4}$ for the coins.



data: points 0, 1, 4, 5; which group made each point is hidden
like the coins: pick a group ($\frac{1}{2}, \frac{1}{2}$), then draw a point from its bell

Coin $\leftrightarrow$ group, flips $\leftrightarrow$ a number: each EM step carries over one by one.

Step 1: a point's likelihood under a group is that bell's height



above point 1: blue bell 1, orange bell 0.0111 (almost flat)

height $e^{-(\text{distance})^2/2}$ plays the role of $p(\text{HH} | \text{coin})$

$$\mathcal{N}(x | \mu, 1) = \frac{1}{\sqrt{2\pi}} e^{-(x-\mu)^2/2}$$

The bell with centre μ , width 1. Its factor $\frac{1}{\sqrt{2\pi}} = 0.399$ is **the same for both groups**; only the ratio matters in Step 2, so we drop it and keep the height $e^{-(\text{distance})^2/2}$, **proportional to the likelihood**.

under	coins: HH	bells: point 1
A / group 1	$\frac{3}{4} \cdot \frac{3}{4} = \frac{9}{16}$	$e^{-0^2/2} = 1$
B / group 2	$\frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$	$e^{-3^2/2} = 0.0111$

Point 1 is $1 \div 0.0111 \approx 90$ **times** likelier under group 1, as HH is $\frac{9}{16} \div \frac{1}{16} = 9$ **times** likelier under coin A.

Close to a centre = tall bell = likely under that group.

Step 2 (E-step): point 1 is 0.989 group 1, as HH is 0.9 coin A

Share of a group = its path ÷ the sum of both paths, where
path = pick chance × likelihood from Step 1.

$$\text{Coins: } p(A \mid \text{HH}) = \frac{\frac{1}{2} \cdot \frac{9}{16}}{\frac{1}{2} \cdot \frac{9}{16} + \frac{1}{2} \cdot \frac{1}{16}} = \frac{9}{9 + 1} = 0.9$$

$$\text{Point 1: } r = \frac{\frac{1}{2} \cdot 1}{\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 0.0111} = \frac{1}{1.0111} = 0.9890$$

The pick chances $\frac{1}{2}$ cancel. In a mixture this share is called the **responsibility** r of the group for the point: a posterior probability.

record HH: path A vs path B



$$9 / (9 + 1) = 0.9$$

$$\text{path B: } \frac{1}{2} \cdot \frac{1}{16}$$

point 1: group 1 vs group 2



$$1 / (1 + 0.0111) = 0.9890$$

$$\text{group 2: } \frac{1}{2} \cdot 0.0111$$

share = this path ÷ (sum of both paths); the $\frac{1}{2}$ cancels

E-step = Bayes' rule on each point: the same division as for the coins.

The E-step gives every point a soft label

point	bell 1 height	bell 2 height	share r of group 1
0	$e^{-1/2} = 0.607$	$e^{-8} = 0.000335$	0.9994
1	$e^0 = 1$	$e^{-9/2} = 0.0111$	0.9890
4	$e^{-9/2} = 0.0111$	$e^0 = 1$	0.0110
5	$e^{-8} = 0.000335$	$e^{-1/2} = 0.607$	0.0006

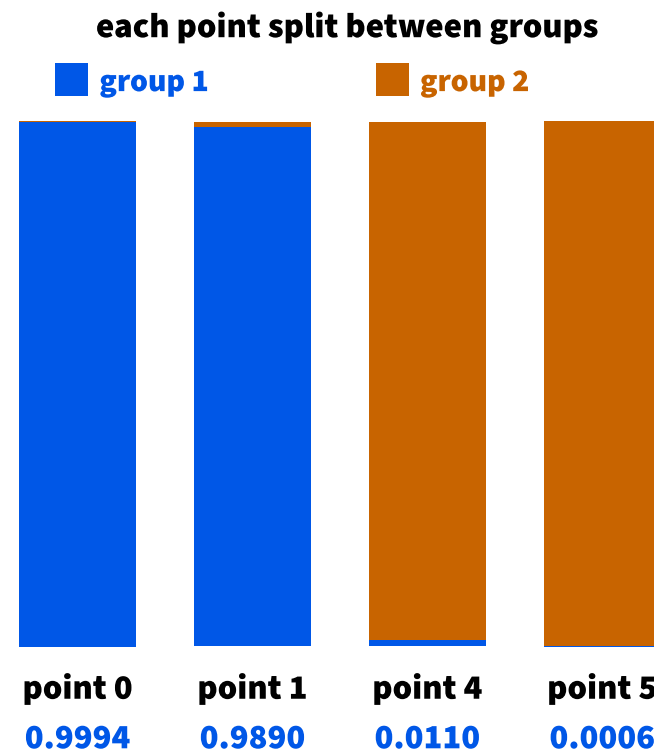
Share of group 1 = bell 1 ÷ (bell 1 + bell 2), for points 0 and 1:

$$r(0) = \frac{0.607}{0.607 + 0.000335} = 0.9994, \quad r(1) = \frac{1}{1 + 0.0111} = 0.9890$$

Points 4 and 5 mirror points 1 and 0 (the data are symmetric about 2.5):

$$r(4) = 1 - r(1) = 0.0110, \quad r(5) = 1 - r(0) = 0.0006$$

Coins, same output: soft labels 0.9, 0.5, 0.1 for HH, HT, TT.



blue part = share r of group 1

E-step output: soft labels 0.9994, 0.9890, 0.0110, 0.0006 for group 1.

Step 3 (M-step): each centre moves to a share-weighted average

Coins: θ_A = heads ÷ flips, each flip weighted by its record's share (0.9, 0.5, 0.1):

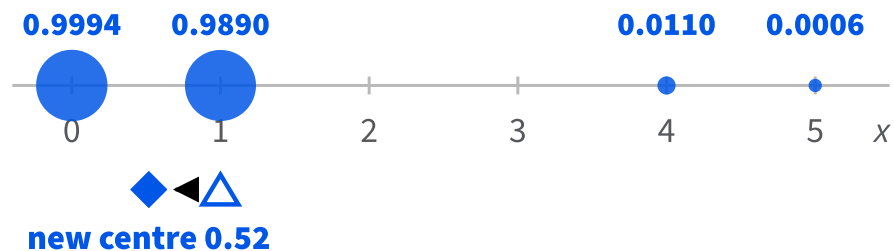
$$\theta_A = \frac{0.9 \cdot 2 + 0.5 \cdot 1 + 0.1 \cdot 0}{0.9 \cdot 2 + 0.5 \cdot 2 + 0.1 \cdot 2} = \frac{2.3}{3} = 0.767$$

point x	share r of group 1	$r \cdot x$
0	0.9994	0
1	0.9890	0.989
4	0.0110	0.044
5	0.0006	0.003
total	2.000	1.036

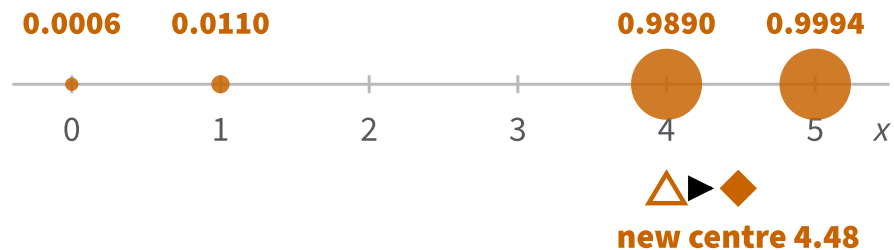
Group 1: $\mu_1 = 1.036 \div 2.000 = 0.518 \approx 0.52$.

Group 2, by the mirror $x \mapsto 5 - x$ (it swaps $0 \leftrightarrow 5$ and $1 \leftrightarrow 4$, and so swaps the two groups): $\mu_2 = 5 - 0.518 \approx 4.48$.

group 1: dot size = share r (number above)



group 2: dot size = share r (number above)



hollow: old centre · diamond: weighted average = new centre

One round: centres 1, 4 → 0.52, 4.48, each pulled to the middle of its points.

The weighted average maximizes the weighted log-likelihood

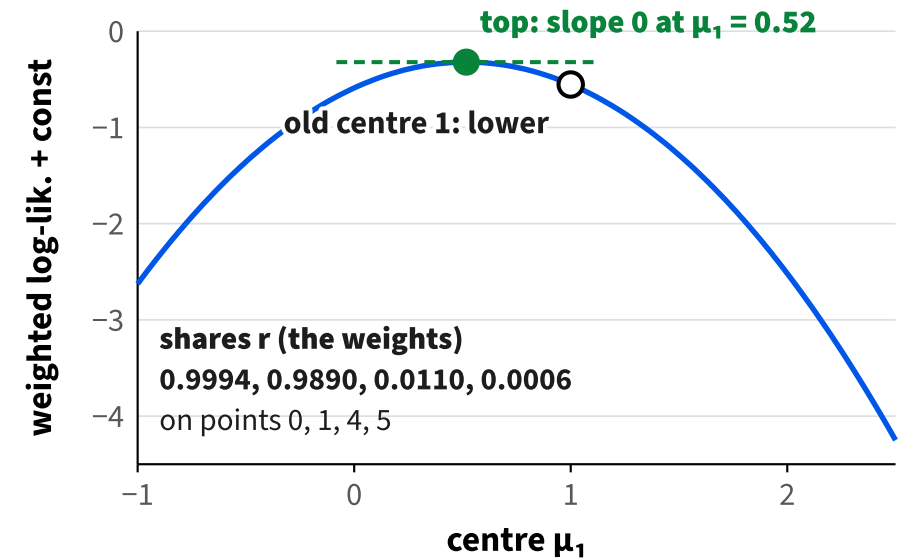
↳ Group 1's weighted log-likelihood $\ell_1(\mu) = \sum_n r_n \ln \mathcal{N}(x_n | \mu, 1)$: point n counts with its share r_n ; **EM's M-step maximizes this** (the weighted fit of Section 3). Since $\ln \mathcal{N}(x | \mu, 1) = -\frac{1}{2} \ln 2\pi - \frac{1}{2}(x - \mu)^2$:

$$\ell_1(\mu) = c - \frac{1}{2} \sum_n r_n (x_n - \mu)^2 \quad \left| \begin{array}{l} c \text{ has no } \mu \end{array} \right.$$

$$\frac{d\ell_1}{d\mu} = \sum_n r_n (x_n - \mu) = 0 \quad \left| \begin{array}{l} \text{top: slope 0} \end{array} \right.$$

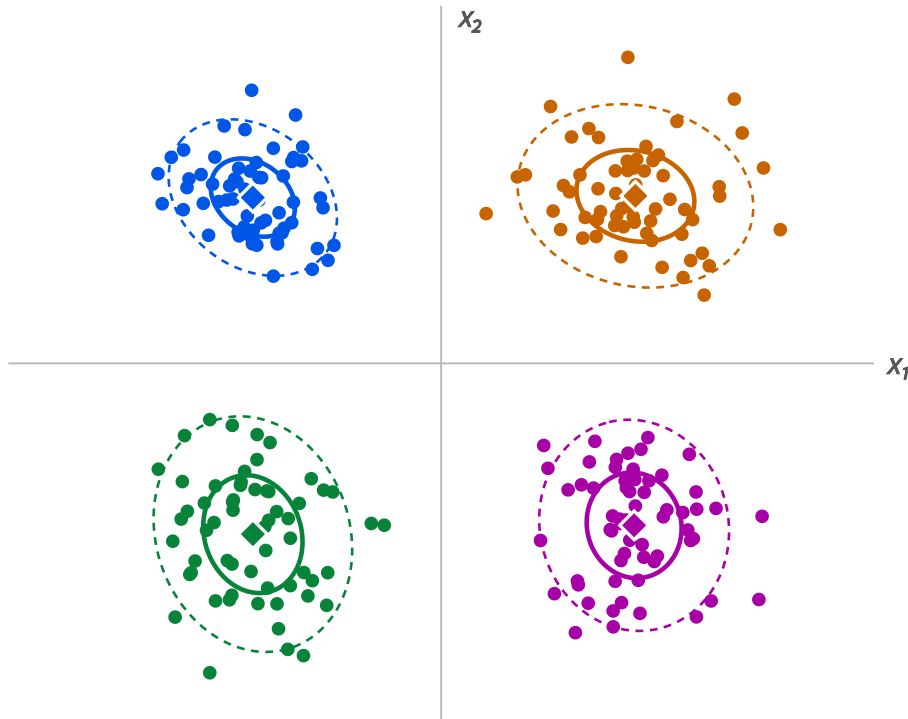
$$\mu = \frac{\sum_n r_n x_n}{\sum_n r_n} = \frac{1.036}{2.000} \approx 0.52 \quad \left| \begin{array}{l} \text{recall } \sum r x = 1.036, \\ \sum r = 2.000 \end{array} \right.$$

Coins, same steps: weighted heads $0.9 \cdot 2 + 0.5 \cdot 1 + 0.1 \cdot 0 = 2.3$, weighted tails $0.9 \cdot 0 + 0.5 \cdot 1 + 0.1 \cdot 2 = 0.7$;
 $2.3 \ln \theta + 0.7 \ln(1 - \theta)$ has slope $\frac{2.3}{\theta} - \frac{0.7}{1-\theta} = 0$ at $\theta = \frac{2.3}{3}$.



The M-step is maximum likelihood with fractional counts, for coins and bells alike.

The same three steps find four groups in 240 points



ovals: where each hill is 1 and 2 widths from its centre · diamonds: centres

round **40** · average log-likelihood per point **-2.901** · colour: the group with the largest share, paler when unsure

Points (x_1, x_2) in the plane, four groups. Each bell is now a hill whose contours are ovals.

Step 1: each hill's height at each point.

Step 2 (E): colour each point by its shares; pale = unsure.

Step 3 (M): move each centre to its share-weighted average; refit each oval from the share-weighted spread of its points.

Average log-likelihood per point **ris**es every **round**, from -4.309 to -2.901 at round 40 (Theorem 4: an EM round never lowers the likelihood).

Recolour, move, repeat: clustering is EM.

EM for a Gaussian mixture: the coin steps with bells for coins

Gaussian mixture. Pick group k with chance π_k ; draw x from the bell $\mathcal{N}(x \mid \mu_k, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu_k)^2/2\sigma^2}$ (centre μ_k , standard deviation, or width, σ), for $k = 1, \dots, K$. Data $x_1, \dots, x_N$. Here $K = 2$, $\pi_k = \frac{1}{2}$, $\sigma = 1$, $N = 4$.

$$p(x) = \sum_k \pi_k \mathcal{N}(x \mid \mu_k, \sigma^2)$$

$$r_{nk} = \frac{\pi_k \mathcal{N}(x_n \mid \mu_k, \sigma^2)}{\sum_j \pi_j \mathcal{N}(x_n \mid \mu_j, \sigma^2)}$$

$$\mu_k = \frac{\sum_n r_{nk} x_n}{\sum_n r_{nk}}, \quad \pi_k = \frac{1}{N} \sum_n r_{nk}$$

A sum over the hidden group, like
 $p(\text{HH}) = \frac{1}{2} \cdot \frac{9}{16} + \frac{1}{2} \cdot \frac{1}{16} = \frac{5}{16}$.

E-step: group k 's share of point n .
 Point 1: 0.989 (coins: 0.9).

M-step: weighted average; a group's weight = its average share. Group 1:
 $\mu_1 = 1.036 \div 2.000 \approx 0.52$,
 $\pi_1 = 2.000 \div 4 = 0.5$.

On our data: one round moves (μ_1, μ_2) from $(1, 4)$ to $(0.52, 4.48)$; the weights stay $(0.5, 0.5)$, since the shares of each group add to 2.000.

The coins' two steps, with bell heights in place of $\theta^h(1 - \theta)^t$ (h heads, t tails).

k-means: join the nearest centre, then move to the average

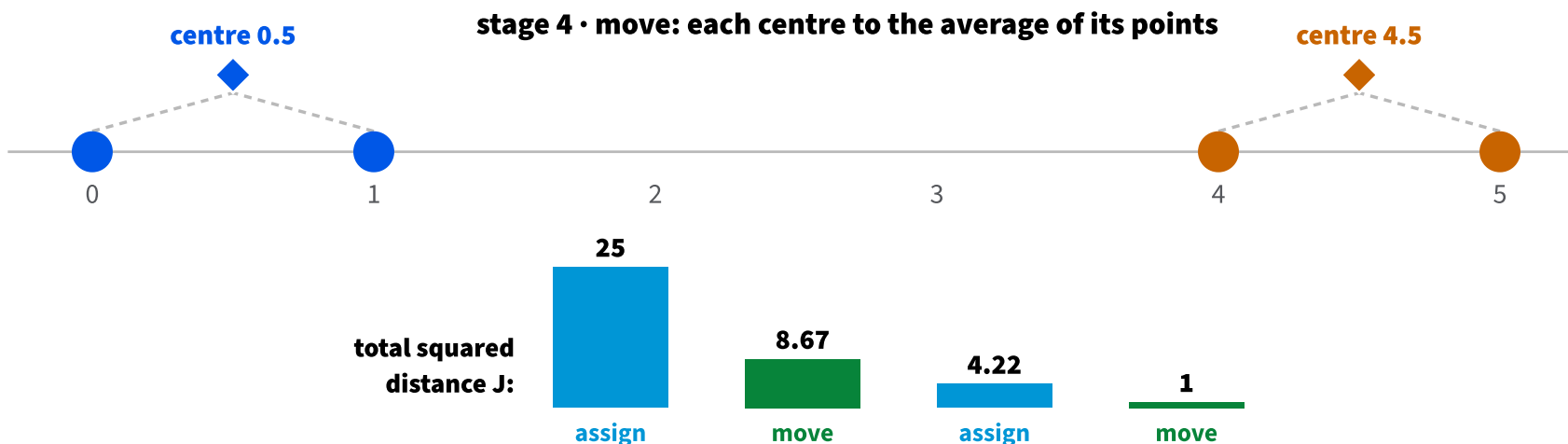
k-means (Lloyd, 1982), the most widely used clustering method, repeats two steps until nothing changes. Run it on 0, 1, 4, 5 from a **poor guess: centres 0 and 1** (in class, step through it with the button). Score $J = \text{sum of } (\text{point} - \text{its centre})^2$; smaller is tighter.

Assign

each point joins the **nearest** centre: a hard label, 1 or 0, never 0.989.

Move

each centre goes to the **plain average** of the points that joined it.



Two rounds from the poor start (0, 1): centres 0.5 and 4.5, the middles of {0, 1} and {4, 5}.

Each k-means step lowers J : $25 \rightarrow 8.67 \rightarrow 4.22 \rightarrow 1$

↳ Score of a clustering: J = sum over points of $(\text{point} - \text{its centre})^2$. Smaller J = tighter groups.

stage	centres	groups	J
assign	0, 1	$\{0\} \cdot \{1, 4, 5\}$	$0^2 + 0^2 + 3^2 + 4^2 = 25$
move	$0, \frac{1+4+5}{3} = 3.33$	$\{0\} \cdot \{1, 4, 5\}$	$0 + \frac{49}{9} + \frac{4}{9} + \frac{25}{9} = 8.67$
assign	0, 3.33	$\{0, 1\} \cdot \{4, 5\}$	$0 + 1 + \frac{4}{9} + \frac{25}{9} = 4.22$
move	$\frac{0+1}{2}, \frac{4+5}{2} = 0.5, 4.5$	$\{0, 1\} \cdot \{4, 5\}$	$4 \cdot 0.5^2 = 1$

Assign never raises J

Each point picks the centre that makes its own term smallest. Point 1:
 $1^2 = 1 < (3.33 - 1)^2 = 5.44$, so it switches.

Move never raises J

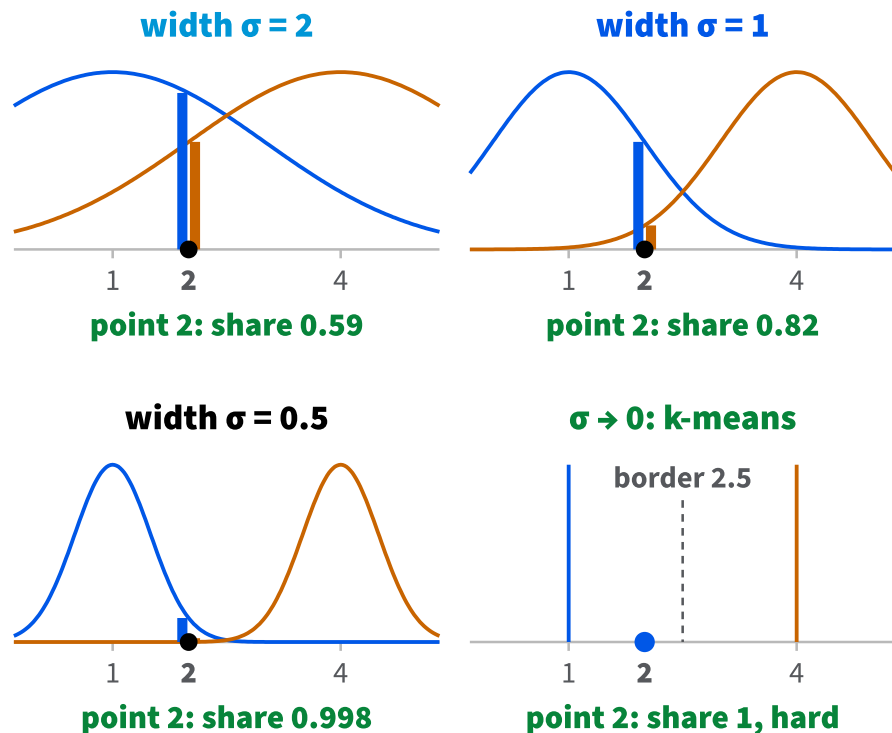
For fixed groups, $\sum (x - \mu)^2$ is smallest at the average: slope
 $-2 \sum (x - \mu) = 0$.

So it must stop

After each move, J depends only on the labelling. J strictly falls at every label change (ties keep the old label), so **no labelling repeats**; only $2^4 = 16$ labellings exist, so the labels must stop.

Like EM's log-likelihood, k-means' J only improves: a poor start still ends at a fixed point.

Narrow the bells, and EM's soft share hardens into a hard choice



A new point at **2**: distance $d = 1$ to centre 1, $d = 2$ to centre 4. With width σ , a bell's height is $e^{-d^2/2\sigma^2}$ (the shared factor $\frac{1}{\sqrt{2\pi}\sigma}$ cancels, so heights drop it).

σ	near bell ($d = 1$) $e^{-1^2/2\sigma^2}$	far bell ($d = 2$) $e^{-2^2/2\sigma^2}$	share
2	0.882	0.607	$\frac{0.882}{0.882+0.607} = 0.59$
1	0.607	0.135	$\frac{0.607}{0.607+0.135} = 0.82$
0.5	0.135	0.000335	$\frac{0.135}{0.135+0.000335} = 0.998$

Why: far $\div$ near $= e^{-(2^2-1^2)/2\sigma^2} = e^{-3/2\sigma^2} = 0.69, 0.22, 0.0025$, and share $= 1/(1 + \text{far} \div \text{near}) = 0.59, 0.82, 0.998$. As $\sigma \rightarrow 0$ the ratio $\rightarrow 0$, so **the share $\rightarrow 1$: k-means' hard label.**

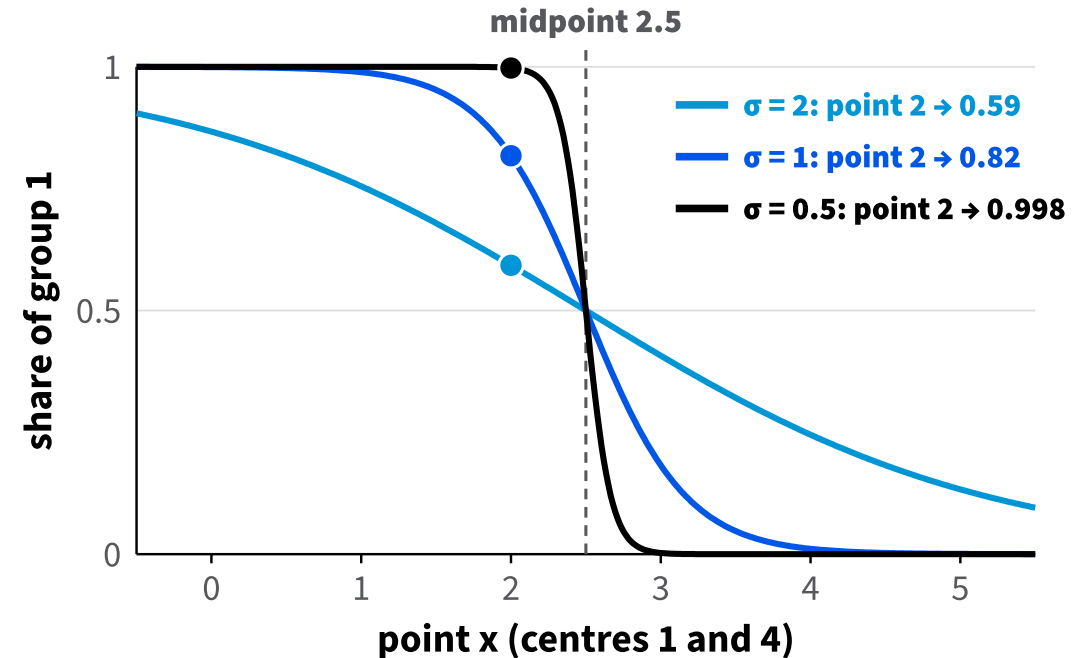
Narrower bells $\Rightarrow$ the nearer centre wins more surely $\Rightarrow$ in the limit, k-means.

Theorem 5: k-means is EM with bells of width zero

Theorem 5 (k-means as a limit). Take a Gaussian mixture with equal weights π_k and one shared width σ , **held fixed** (not re-estimated), and let $\sigma \rightarrow 0$. Then EM's E-step becomes "assign each point to its nearest centre" and its M-step becomes "move each centre to the plain average of its points": EM becomes k-means.

In words: a share is a soft vote; shrinking every bell turns it into a hard vote for the nearest centre.

On the line: at the midpoint 2.5 both distances are 1.5, so both heights are equal and **the share is $\frac{1}{2}$ for every σ** ; as $\sigma \rightarrow 0$ the curve becomes a step there, exactly **k-means' border** between centres 1 and 4.



dots: point 2; as σ shrinks the curve becomes a step at 2.5

Soft labels harden into hard ones; weighted averages become plain averages.

Proof: write the share as $1/(1 + e^{-u})$, then let σ shrink

↳ Equal weights π_k and the shared factor $\frac{1}{\sqrt{2\pi}\sigma}$ cancel in the share. Distances $d_k = |x - \mu_k|$ to centre k ; $\Delta = d_2^2 - d_1^2$.

$$r_1 = \frac{e^{-d_1^2/2\sigma^2}}{e^{-d_1^2/2\sigma^2} + e^{-d_2^2/2\sigma^2}} = \frac{1}{1 + e^{-u}}, \quad u = \frac{\Delta}{2\sigma^2}$$

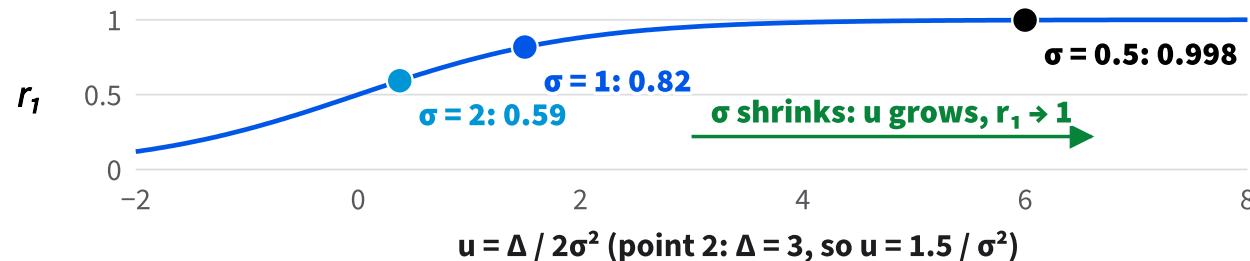
$$\Delta > 0 \Rightarrow u \rightarrow \infty \Rightarrow e^{-u} \rightarrow 0 \Rightarrow r_1 \rightarrow 1$$

$$\mu_1 = \frac{\sum_n r_{n1} x_n}{\sum_n r_{n1}} \rightarrow \frac{\sum_{n \in G_1} x_n}{|G_1|}$$

Step 2's share; divide top and bottom by the first height. Point 2: $\Delta = 2^2 - 1^2 = 3$; $\sigma = 1$ gives $\frac{1}{1+e^{-1.5}} = 0.82 = \frac{0.607}{0.607+0.135}$.

$\Delta > 0$ means $d_1 < d_2$: **centre 1 is nearer**. Then **E-step** → **nearest centre**. Point 2, $\sigma = 0.5$: $\frac{1}{1+e^{-6}} = 0.998$. If $\Delta < 0$, $r_1 \rightarrow 0$.

M-step → **plain average** over the set G_1 of points nearest centre 1 ($|G_1|$ = how many there are), since every weight is now 1 or 0. Points 0, 1: $\frac{0+1}{2} = 0.5$. ■



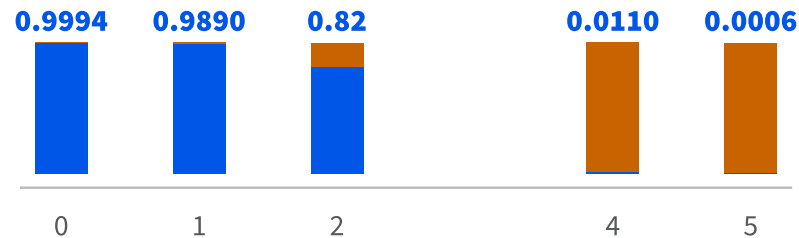
Both EM steps turn into the two k-means steps.

k-means or EM? Hard labels are fast; soft labels keep the doubt

k-means: hard labels



label: 1 or 0 · point 2 is simply "group 1"

EM ($\sigma = 1$): soft labels

blue = share of group 1 · point 2 keeps its doubt

k-means

One nearest-centre test per point: fast. Assumes round groups of equal size.

EM

Keeps the doubt (point 2: 0.82). Can also learn each group's weight (π_k = average share) and width.

Shared risk

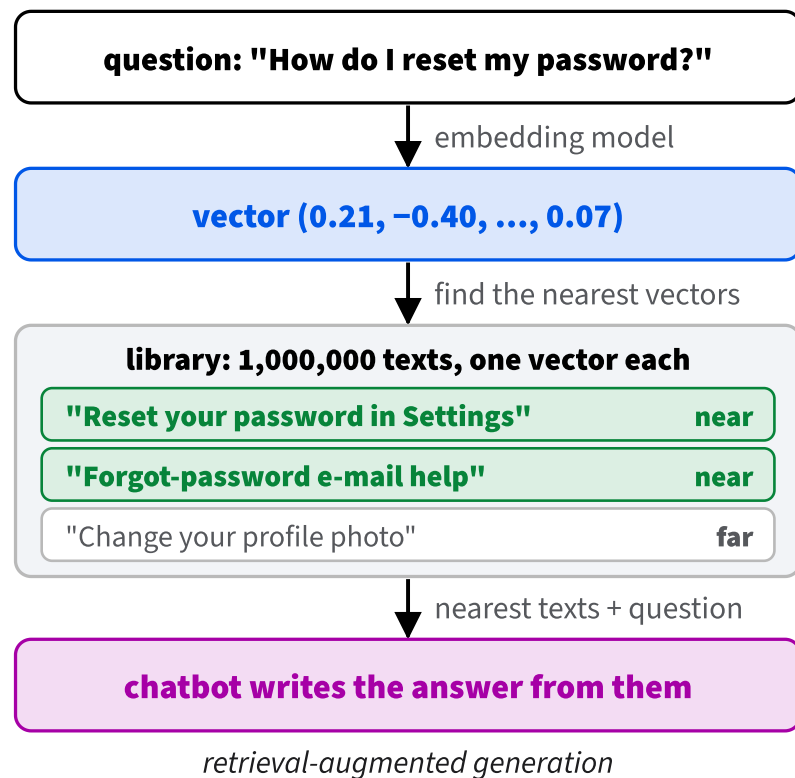
Both only improve their score (EM: log-likelihood up; k-means: J down), so a poor start can stop at a poor answer. Fix: several starts, or k-means++ (Arthur & Vassilvitskii, 2007).

Practice

Run k-means first, then EM from its centres; scikit-learn's GaussianMixture does this by default.

One family: EM is k-means that admits doubt; k-means is EM that refuses it.

Vector search: find the stored texts nearest to a question



Setting: a chatbot answers from a library of texts. Each text is stored as a **vector**, a list of numbers in which similar meanings lie close together; an **embedding model** is a network that turns a text into its vector. **Retrieval-augmented generation:** find the texts nearest the question, then answer with them.

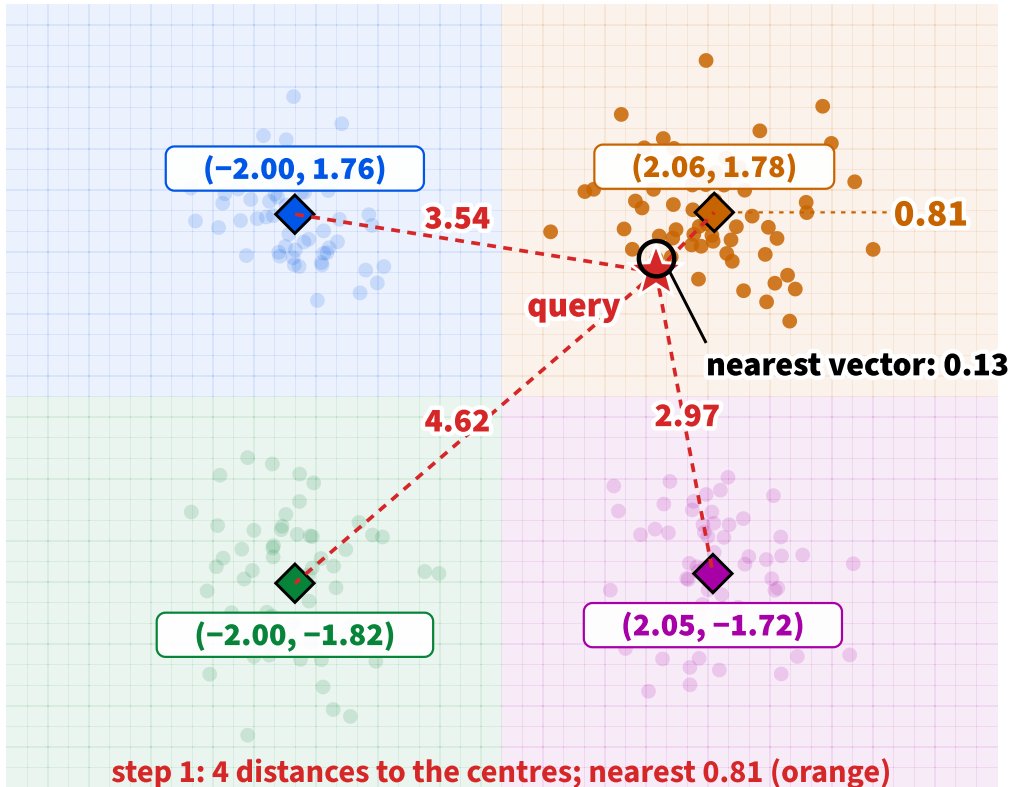
COIN GAME		MIXTURE		VECTOR SEARCH
record HH	↔	point x	↔	one text's vector
coin A or B	↔	group	↔	cell of similar texts
θ_A	↔	centre μ_k	↔	cell centre
which coin?	↔	nearest centre	↔	file a text in its cell

Cost: the slow way compares the question with every stored vector: 1,000,000 distances per question for a million texts, and many questions per second.

Finding the nearest vectors is the bottleneck; k-means' nearest-centre step cuts it.

k-means cells: 64 distance computations instead of 240

diamonds: the 4 k-means centres · shading: their cells



step 2: only the 60 vectors of that cell: $4 + 60 = 64$ of 240

Built once: the 240 points of the four-group animation are the stored vectors; k-means gives 4 centres, and each vector is filed in its nearest centre's cell, 60 per cell.

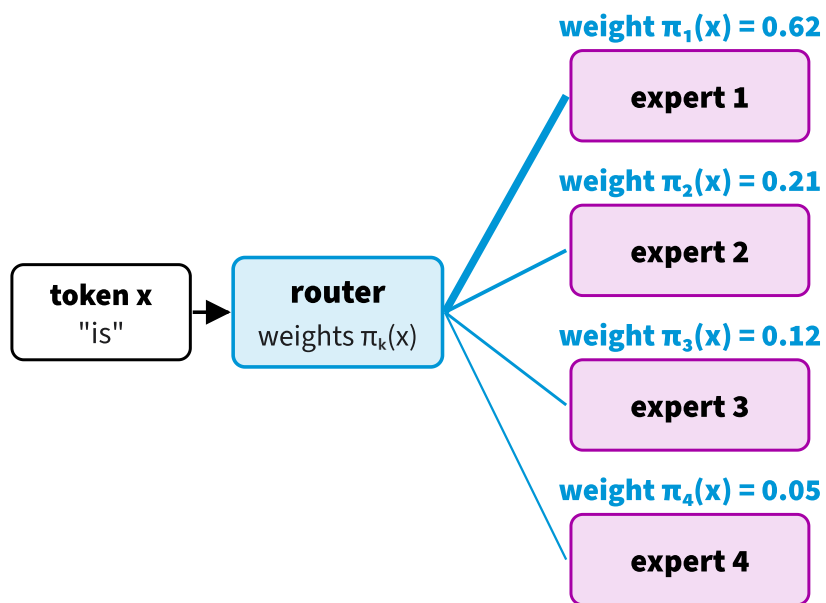
Step 1: distances from the query (1.5, 1.2) to the 4 centres (blue 3.54, **orange 0.81**, green 4.62, purple 2.97). Orange centre (2.06, 1.78): $\sqrt{0.56^2 + 0.58^2} = \sqrt{0.65} = 0.81$, the nearest.

Step 2: compare with that cell's 60 vectors only; the nearest (ringed) lies at distance 0.13. Total $4 + 60 = 64$, not 240.

At scale: 1,000,000 vectors in 1,000 cells of about 1,000: $1,000 + 1,000 = 2,000$ distances; $1,000,000 \div 2,000 = 500$ times fewer (Jégou et al., 2011). **Price:** a neighbour just across a cell border can be missed.

k-means' nearest-centre step, reused as an index: search a cell, not the library.

A mixture of experts: mixture weights that depend on the input



Setting: a language model reads text one **token** (a word or a piece of one) at a time. A layer holds several smaller networks, the **experts**. A small **router** scores each expert, and a softmax turns the scores into weights that sum to 1 (here 0.62, 0.21, 0.12, 0.05, illustrative).

$$\text{mixture: } p(x) = \sum_k \pi_k \mathcal{N}(x \mid \mu_k, \sigma^2)$$

$$\text{experts: } p(y \mid x) = \sum_k \pi_k(x) p_k(y \mid x)$$

Here x = the text read so far (the input); the predicted next token y plays the role of the mixture's point.

COIN GAME		MIXTURE		MIXTURE OF EXPERTS
coin A or B	$\Leftrightarrow$	group k	$\Leftrightarrow$	expert k
chance $\frac{1}{2}$	$\Leftrightarrow$	weight π_k	$\Leftrightarrow$	router weight $\pi_k(x)$
$p(\text{HH} \mid A)$	$\Leftrightarrow$	bell $\mathcal{N}(x \mid \mu_k, \sigma^2)$	$\Leftrightarrow$	prediction $p_k(y \mid x)$
record HH	$\Leftrightarrow$	point x	$\Leftrightarrow$	next token y

Why bother: the model holds many experts' parameters, yet each token pays only for the few experts it runs.

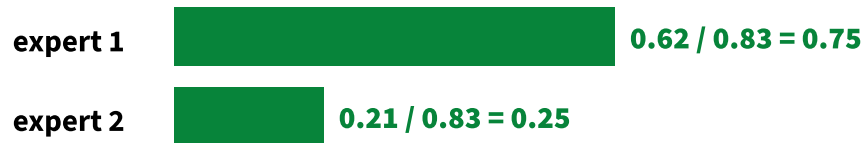
Same sum as a mixture (Jacobs et al., 1991); only the weights now depend on the token.

Top-2 routing on one token: run two experts, skip two

1 · router weights (sum 1)



2 · keep the top 2, divide each by 0.83



3 · mix the two experts' predictions

p(next token = "Paris"): expert 1: 0.9, expert 2: 0.5

$$0.75 \cdot 0.9 + 0.25 \cdot 0.5 = 0.80$$

experts 3 and 4 are never computed

Step 1: the text is "The capital of France is". For the token "is", the router gives the 4 experts weights 0.62, 0.21, 0.12, 0.05 (illustrative; they sum to 1).

Step 2: keep the two largest, $0.62 + 0.21 = 0.83$, and rescale to sum to 1: $\frac{0.62}{0.83} = 0.75$, $\frac{0.21}{0.83} = 0.25$.

Step 3: run only experts 1 and 2. Their chances that the next token is "Paris" (illustrative) are 0.9 and 0.5, so the layer outputs $0.75 \cdot 0.9 + 0.25 \cdot 0.5 = 0.80$.

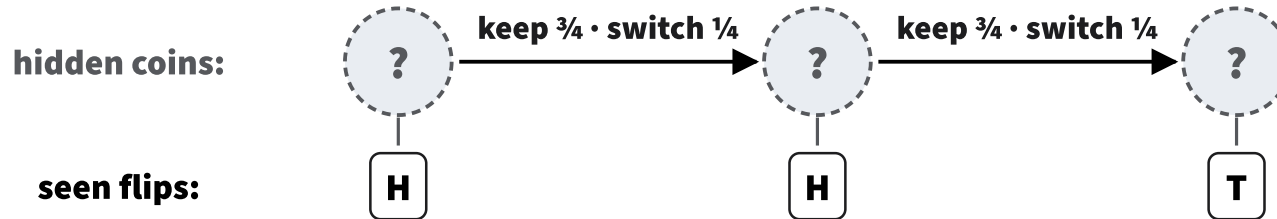
At scale: Mixtral (Jiang et al., 2024) has 8 experts per layer and runs 2 per token: $\frac{2}{8}$, a quarter of the expert computation. There each expert outputs a vector, and the layer mixes those vectors with the same kind of weights (here 0.75, 0.25).

Top-2 hardens the router's weights, as $\sigma \rightarrow 0$ hardened EM's shares into k-means.

Next: the coin may switch between flips



So far one hidden group per record. Now the friend may switch coins between flips: **one hidden coin per flip**, kept with chance $\frac{3}{4}$, switched with $\frac{1}{4}$.



Three flips, each with its own hidden coin. How many coin sequences are there?

2

6

8

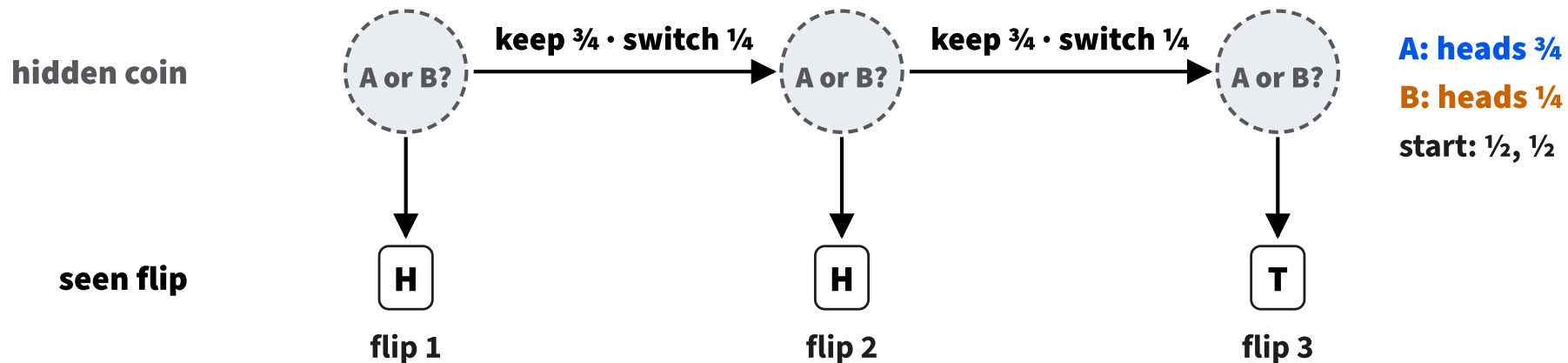
Answer. $8 = 2 \cdot 2 \cdot 2$. For 100 flips, 2^{100} : too many to list.

Hidden chains: the sum over sequences, done step by step.

If the coin can switch between flips, the coins form a hidden chain

Goal: the posterior of each flip's coin, and $p(\text{flips})$, without listing every sequence.

↳ In the two-coin game one hidden coin made both flips of a record. Now the friend may switch coins between flips, so **every flip has its own hidden coin**.

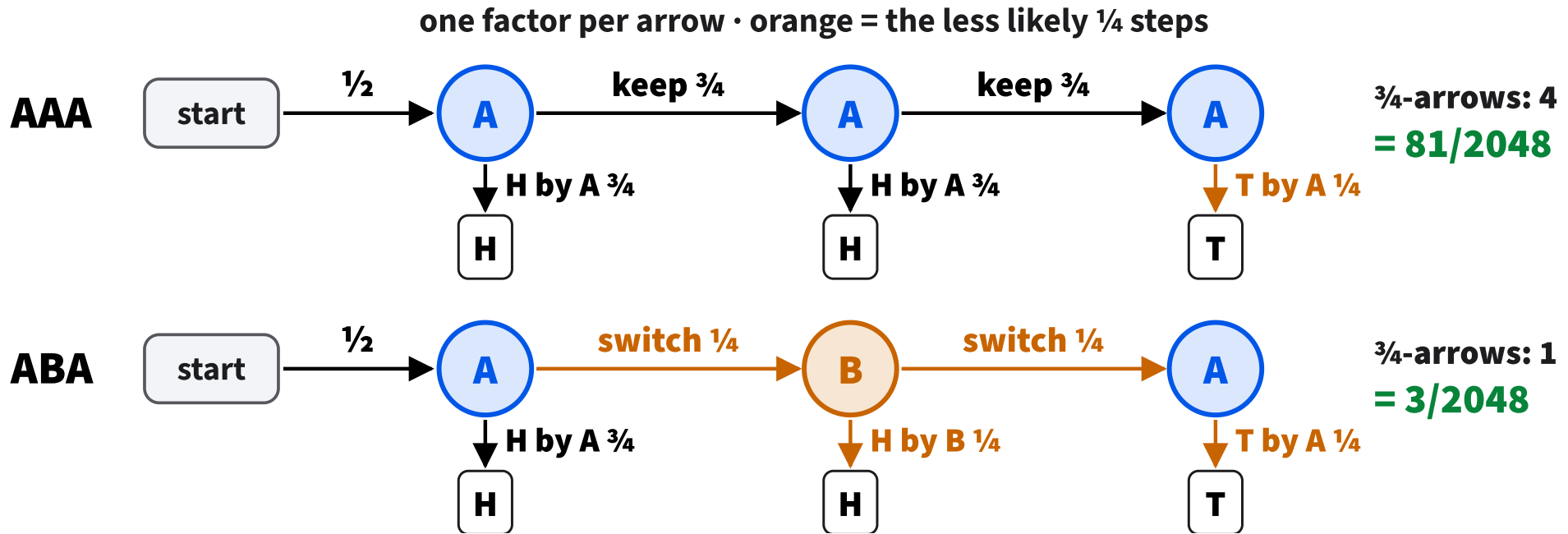


A **hidden Markov model**: each coin depends only on the coin before (Lecture 2's chain), and each flip only on its own coin. Early speech recognizers were built on it.

A hidden chain links the coins flip to flip; on H, H, T we will find flip 2's coin and $p(\text{H}, \text{H}, \text{T})$.

One coin sequence: multiply one number per arrow

↳ Rules of the switching game: the first coin is A or B with chance $\frac{1}{2}$; between flips the friend keeps the coin with chance $\frac{3}{4}$, switches with $\frac{1}{4}$; A shows H with $\frac{3}{4}$, B with $\frac{1}{4}$.



AAA: multiply its six arrows: $\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{1}{4} = \frac{3^4}{2 \cdot 4^5} = \frac{81}{2048}$.

ABA: only one arrow is $\frac{3}{4}$: $\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{3^1}{2 \cdot 4^5} = \frac{3}{2048}$.

Weight = $3^k/2048$, where k = number of $\frac{3}{4}$ -arrows ($2048 = 2 \cdot 4^5$): **AAA** $k = 4 \rightarrow 81$, **ABA** $k = 1 \rightarrow 3$.

All 8 coin sequences: their weights add up to $p(\text{HHT}) = 15/128$

↳ Each row multiplies its six factors (move = keep $\frac{3}{4}$ or switch $\frac{1}{4}$); weight $\times 2048 = 3^k$, k = number of $\frac{3}{4}$ in the row ($2048 = 2 \cdot 4^5$).

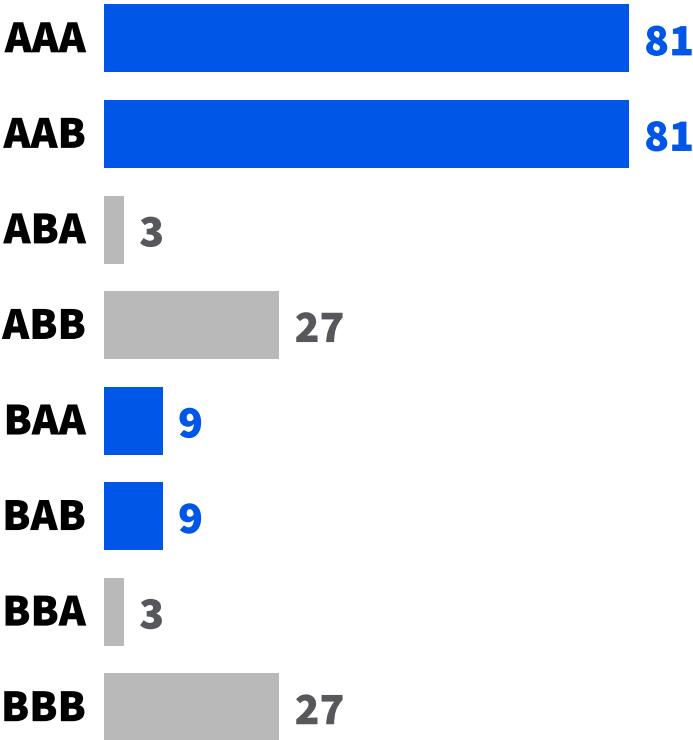
coins	first	flip 1 H	move	flip 2 H	move	flip 3 T	× 2048
AAA	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	81
AAB	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{3}{4}$	81
ABA	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	3
ABB	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	27
BAA	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	9
BAB	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{3}{4}$	9
BBA	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	3
BBB	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	27

Add all 8: $81+81+3+27+9+9+3+27 = 240$; $p(\text{HHT}) = \frac{240}{2048} = \frac{15}{128}$

A at flip 2 (Bayes: share of the weight): $\frac{81 + 81 + 9 + 9}{240} = \frac{180}{240} = \frac{3}{4}$

Correct, but n flips give 2^n rows (100 flips: about 10^{30}). We need a shortcut.

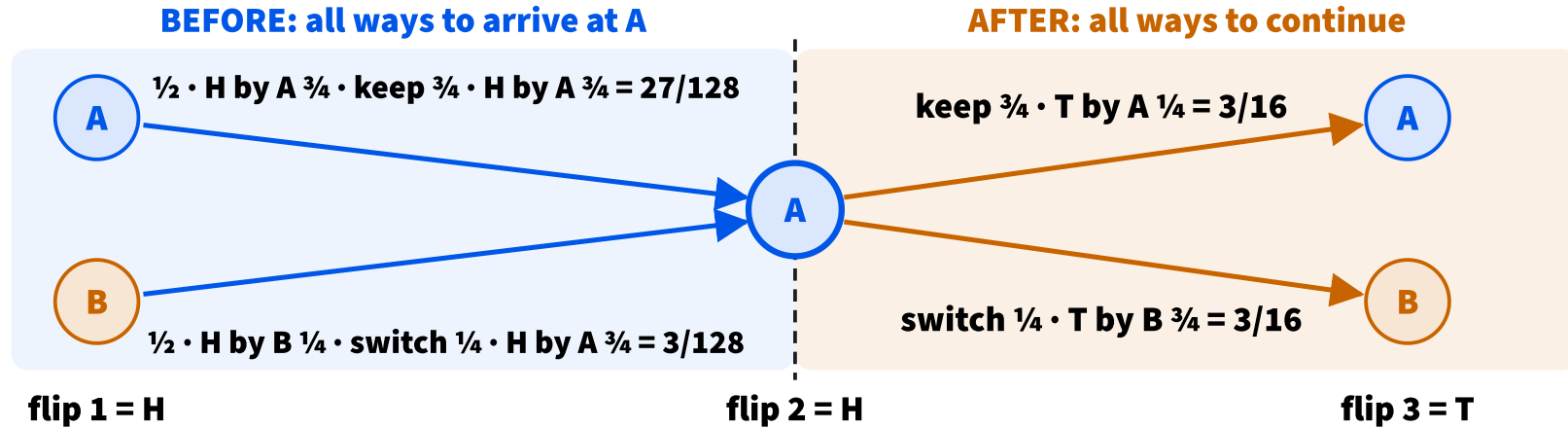
weight $\times 2048$ of each sequence



blue = coin A at flip 2

Cut every sequence at flip 2: the part before × the part after

↳ Recall the sequences with coin A at flip 2: AAA 81, AAB 81, BAA 9, BAB 9 (× 2048), 180 in all.



Why it factors: the part after flip 2 depends only on the coin at flip 2 (each coin depends only on the coin before), so **both ways in share the same two ways on**.

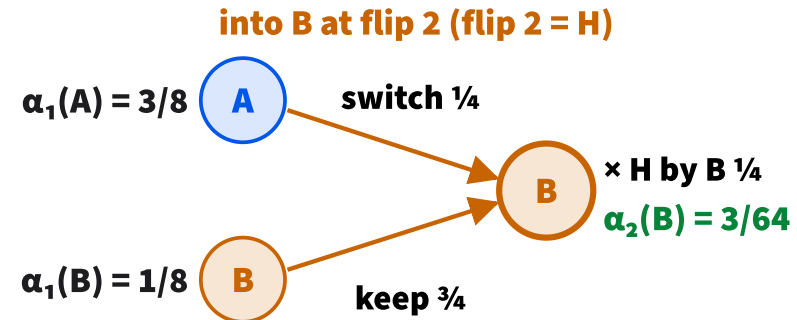
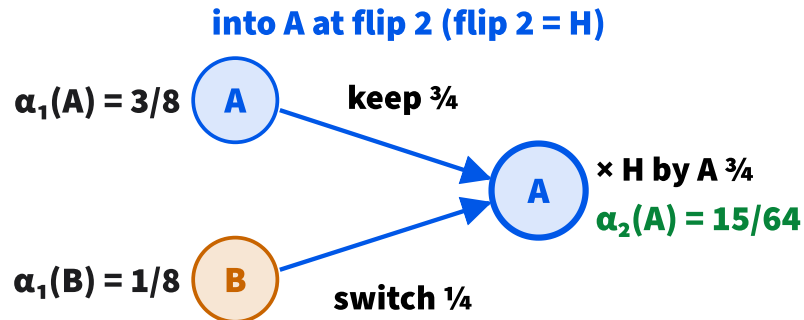
Group them (distributive law): $\underbrace{\left(\frac{27}{128} + \frac{3}{128}\right)}_{\text{all ways in}} \times \underbrace{\left(\frac{3}{16} + \frac{3}{16}\right)}_{\text{all ways on}} = \frac{30}{128} \cdot \frac{6}{16} = \frac{180}{2048} \checkmark$

Names: the first bracket is the **forward message** $\alpha_2(A) = \frac{30}{128} = \frac{15}{64}$ (flips H, H and coin A at flip 2); the second is the **backward message** $\beta_2(A) = \frac{6}{16} = \frac{3}{8}$ (flip 3 = T, given A at flip 2).

Add first, multiply once: one product of two sums replaces four products.

The forward message: add the two ways into each coin, then the flip

↳ $\alpha_t(z)$ = chance of flips 1 to t and coin z at flip t . It is built from left to right.



$$\alpha_1(A) = \underbrace{\frac{1}{2}}_{\text{first A}} \cdot \underbrace{\frac{3}{4}}_{\text{H by A}} = \frac{3}{8}, \quad \alpha_1(B) = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$$

$$\alpha_2(A) = \left(\underbrace{\frac{3}{8} \cdot \frac{3}{4}}_{\text{was A, keep}} + \underbrace{\frac{1}{8} \cdot \frac{1}{4}}_{\text{was B, switch}} \right) \cdot \underbrace{\frac{3}{4}}_{\text{H by A}} = \frac{9+1}{32} \cdot \frac{3}{4} = \frac{15}{64}$$

$$\alpha_2(B) = \left(\frac{3}{8} \cdot \frac{1}{4} + \frac{1}{8} \cdot \frac{3}{4} \right) \cdot \underbrace{\frac{1}{4}}_{\text{H by B}} = \frac{3+3}{32} \cdot \frac{1}{4} = \frac{3}{64}$$

Flip 1 = H: start chance $\times$ flip chance.

Into A: add the two ways in, then flip 2.

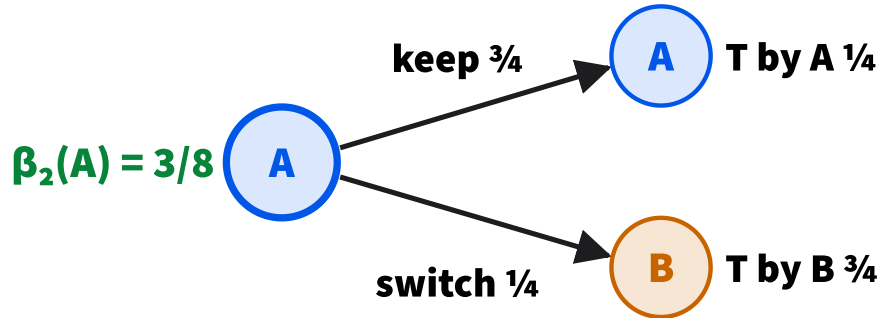
Into B: switch from A, or keep B.

Each step: (old $\alpha \times$ move, added over the old coin) $\times$ flip chance.

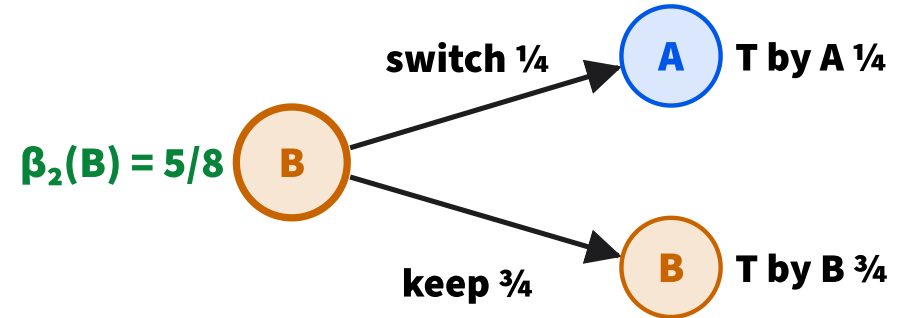
Forward × backward gives $p(\text{HHT})$ and the coin at flip 2

↳ $\beta_t(z)$ = chance of the flips after t , given coin z at flip t . It is built from right to left; after flip 3 nothing remains, so $\beta_3 = 1$.

out of A at flip 2 (flip 3 = T)



out of B at flip 2 (flip 3 = T)



$$\beta_2(A) = \underbrace{\frac{3}{4} \cdot \frac{1}{4}}_{\text{keep, T by A}} + \underbrace{\frac{1}{4} \cdot \frac{3}{4}}_{\text{switch, T by B}} = \frac{3+3}{16} = \frac{3}{8}$$

$$\beta_2(B) = \underbrace{\frac{1}{4} \cdot \frac{1}{4}}_{\text{switch, T by A}} + \underbrace{\frac{3}{4} \cdot \frac{3}{4}}_{\text{keep, T by B}} = \frac{1+9}{16} = \frac{5}{8}$$

$$\alpha_2(A)\beta_2(A) = \frac{15}{64} \cdot \frac{3}{8} = \frac{45}{512} = \frac{180}{2048}, \quad \alpha_2(B)\beta_2(B) = \frac{3}{64} \cdot \frac{5}{8} = \frac{15}{512} = \frac{60}{2048}$$

$$p(\text{HHT}) = \frac{45+15}{512} = \frac{15}{128}, \quad p(A \text{ at flip 2} \mid \text{HHT}) = \frac{45}{45+15} = \frac{3}{4}$$

Out of A at flip 2.

Out of B: the same two ways.

Before × after = 8-row list: A 81+81+9+9, B 3+27+3+27.

Add for the total; share for the posterior.

Same answers as the list of 8, with 2 numbers per flip: Lecture 2's sum-product on a chain.

Forward-backward works on any chain of n flips

↳ Notation: $\tau(z' \rightarrow z)$ = chance the coin moves from z' to z (keep $\frac{3}{4}$, switch $\frac{1}{4}$); $E(z, x)$ = chance coin z shows x (H by A $\frac{3}{4}$, H by B $\frac{1}{4}$); $\pi(z)$ = chance the first coin is z ($\frac{1}{2}$); flips $x = (x_1, \dots, x_n)$.

$\alpha_t(z)$: **forward**

chance of flips 1 to t **and** coin z at flip t

$\beta_t(z)$: **backward**

chance of flips $t+1$ to n , **given** coin z at flip t

$$\alpha_1(z) = \pi(z)E(z, x_1), \quad \alpha_t(z) = \left(\sum_{z'} \alpha_{t-1}(z') \tau(z' \rightarrow z) \right) E(z, x_t) \quad \left| \text{Forward, left to right } (\alpha_1(A) = \frac{3}{8}). \right.$$

$$\beta_n(z) = 1, \quad \beta_t(z) = \sum_{z'} \tau(z \rightarrow z') E(z', x_{t+1}) \beta_{t+1}(z') \quad \left| \text{Backward, right to left } (\beta_2(A) = \frac{3}{8}). \right.$$

$$p(x) = \sum_z \alpha_t(z) \beta_t(z), \quad p(z_t=z \mid x) = \frac{\alpha_t(z) \beta_t(z)}{p(x)} \quad \left| \text{Combine at any flip } t. \text{ Coins, } t=2: p = \frac{45}{512} + \frac{15}{512} = \frac{15}{128}; \text{ share of A} = \frac{45}{60} = \frac{3}{4}. \right.$$

Each flip, each of K coins adds K ways in: about $n \cdot K^2$ steps, not K^n . 100 flips, 2 coins: $100 \cdot 2^2 = 400$, not $2^{100} \approx 10^{30}$.

To learn the switching game, run the same EM as for two coins

↳ So far the rules were given. Now we see only the flips and must **learn** them: the heads chances θ_A, θ_B and the keep chance; the start chance stays $\frac{1}{2}$.

	two-coin game	switching game (hidden chain)
hidden	one coin per record	one coin per flip
E-step	$p(\text{coin} \mid \text{record})$ by Bayes (HH: 0.9)	$p(\text{coin at flip } t \mid \text{all flips})$ by forward-backward
extra soft label	none	$p(\text{coins at flips } t, t+1 \mid \text{all flips})$
M-step	heads to A ÷ flips to A	the same; keep chance = expected keeps ÷ moves

Why a pair label: each cell is the chance of (coin at flip 1, coin at flip 2); the diagonal (green) means the coin was kept. Both tables have coin A at flip 1 with 0.8 and at flip 2 with 0.75, yet keep 0.95 or 0.65. **Single soft labels cannot fix the keep chance.**

	kept $0.75 + 0.2 = 0.95$				kept $0.6 + 0.05 = 0.65$		
	A at 2	B at 2	total		A at 2	B at 2	total
A at 1	0.75	0.05	0.8		0.6	0.2	0.8
B at 1	0	0.2	0.2		0.15	0.05	0.2
total	0.75	0.25			0.75	0.25	

cell = chance of (coin at 1, coin at 2)
same totals (blue), different kept cells (green)

This EM for hidden chains is Baum–Welch (Baum et al., 1970), older than the name "EM".

Finish the forward and backward messages on H, H, T

↳ Current guess: heads chances $\theta_A = 3/4$, $\theta_B = 1/4$; keep $3/4$, switch $1/4$; first coin $1/2$. Recall: α_t = chance of flips 1 to t and the coin at t ; β_t = chance of the later flips, given the coin at t .

green = new

flip 1 = H

flip 2 = H

flip 3 = T

forward

$$\alpha_1(A) = 3/8 \cdot \alpha_1(B) = 1/8$$

$$\alpha_2(A) = 15/64 \cdot \alpha_2(B) = 3/64$$

$$\alpha_3(A) = 3/64 \cdot \alpha_3(B) = 9/128$$

backward

$$\beta_1(A) = 1/4 \cdot \beta_1(B) = 3/16$$

$$\beta_2(A) = 3/8 \cdot \beta_2(B) = 5/8$$

$$\beta_3(A) = 1 \cdot \beta_3(B) = 1$$

$$\alpha_3(A) = \left(\frac{15}{64} \cdot \frac{3}{4} + \frac{3}{64} \cdot \frac{1}{4} \right) \cdot \underbrace{\frac{1}{4}}_{\text{T by A}} = \frac{45+3}{256} \cdot \frac{1}{4} = \frac{3}{64}$$

$$\alpha_3(B) = \left(\frac{15}{64} \cdot \frac{1}{4} + \frac{3}{64} \cdot \frac{3}{4} \right) \cdot \underbrace{\frac{3}{4}}_{\text{T by B}} = \frac{15+9}{256} \cdot \frac{3}{4} = \frac{9}{128}$$

$$\beta_1(A) = \underbrace{\frac{3}{4} \cdot \frac{3}{4}}_{\text{keep, H by A}} \cdot \frac{3}{8} + \underbrace{\frac{1}{4} \cdot \frac{1}{4}}_{\text{switch, H by B}} \cdot \frac{5}{8} = \frac{27+5}{128} = \frac{1}{4}$$

$$\beta_1(B) = \underbrace{\frac{1}{4} \cdot \frac{3}{4}}_{\text{switch, H by A}} \cdot \frac{3}{8} + \underbrace{\frac{3}{4} \cdot \frac{1}{4}}_{\text{keep, H by B}} \cdot \frac{5}{8} = \frac{9+15}{128} = \frac{3}{16}$$

Into A: keep or switch, then T.

Into B: the same two ways.

Out of A: move, H, then β_2 .

Out of B: the same two ways.

Four short sums complete all twelve messages: no list of 8 sequences needed.

At each flip, forward × backward gives each coin's share

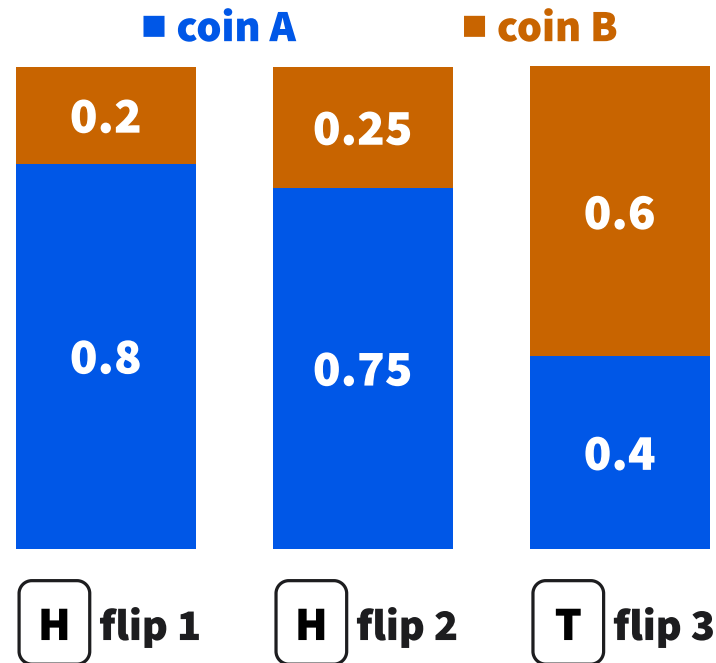
↳ $\alpha_t(z)$ (flips 1 to t and coin z at t) × $\beta_t(z)$ (later flips, given z at t) = weight of all sequences with coin z at flip t . In 2048ths:

	flip 1 = H	flip 2 = H	flip 3 = T
$\alpha_t(A)\beta_t(A)$	$\frac{3}{8} \cdot \frac{1}{4} = \frac{192}{2048}$	$\frac{15}{64} \cdot \frac{3}{8} = \frac{180}{2048}$	$\frac{3}{64} \cdot 1 = \frac{96}{2048}$
$\alpha_t(B)\beta_t(B)$	$\frac{1}{8} \cdot \frac{3}{16} = \frac{48}{2048}$	$\frac{3}{64} \cdot \frac{5}{8} = \frac{60}{2048}$	$\frac{9}{128} \cdot 1 = \frac{144}{2048}$
sum × 2048	240	240	240
A's share	$\frac{192}{240} = 0.8$	$\frac{180}{240} = 0.75$	$\frac{96}{240} = 0.4$

Check: every column adds to $\frac{240}{2048} = \frac{15}{128} = p(\text{HHT})$, the total of the 8-row list: every sequence has some coin at each flip.

A's share is the **soft label** $\gamma_t(A) = p(\text{coin A at flip } t \mid \text{H, H, T})$. **Flip 3 is a tail**, so B (tails $\frac{3}{4}$) takes over there.

chance of each coin, given H, H, T



Soft labels for coin A at flips 1, 2, 3: 0.8, 0.75, 0.4. The M-step counts with them.

M-step: new heads chances = credited heads ÷ credited flips

↳ Recall $\gamma_t(A)$ = chance coin A made flip t , given H, H, T: 0.8, 0.75, 0.4; coin B gets the rest, $1 - \gamma_t(A)$. As in the two-coin M-step, each flip is credited to each coin by its soft label.

flip	1 = H	2 = H	3 = T	sum
flips to A	0.8	0.75	0.4	1.95
heads to A	0.8	0.75	0	1.55
flips to B	0.2	0.25	0.6	1.05
heads to B	0.2	0.25	0	0.45

coin A: its share of each flip

H	H	T
0.8	0.75	0.4

1.55 heads ÷ 1.95 flips = 0.795

coin B: its share of each flip

H	H	T
0.2	0.25	0.6

0.45 heads ÷ 1.05 flips = 0.429

solid = heads, light = tails

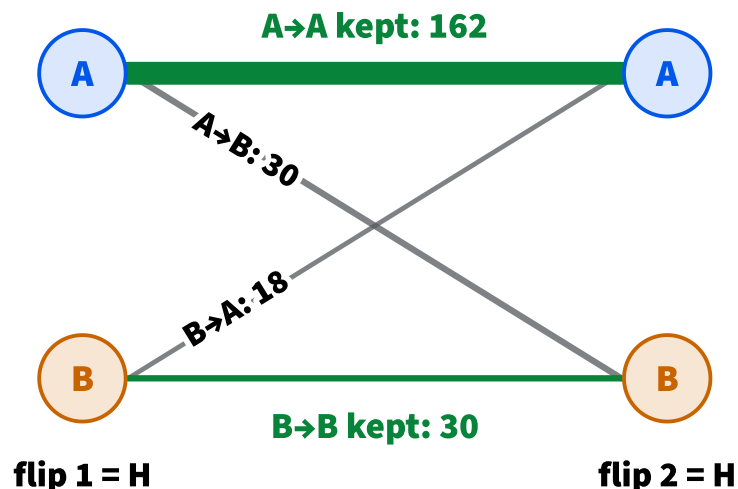
Theorem 1 with soft counts (heads ÷ flips), θ_A, θ_B = heads chances of A, B:

$\theta_A = \frac{1.55}{1.95} = 0.795, \quad \theta_B = \frac{0.45}{1.05} = 0.429$

θ_A : 0.75 → 0.795, θ_B : 0.25 → 0.429. The keep chance needs pairs of coins.

Kept between flips 1 and 2? Weigh the four coin pairs

pair weights $\times 2048$ (thicker = heavier)



↳ Pair weight = product of the four factors in a row (move: keep 3/4, switch 1/4). Recall α_1 = chance of flip 1 and the first coin (A 3/8, B 1/8); β_2 = chance of flip 3, given the second coin (A 3/8, B 5/8).

pair	α_1	move	H by 2nd	β_2	product	$\times 2048$
A → A	3/8	3/4	3/4	3/8	81/1024	162
A → B	3/8	1/4	1/4	5/8	15/1024	30
B → A	1/8	1/4	3/4	3/8	9/1024	18
B → B	1/8	3/4	1/4	5/8	15/1024	30
total		$= p(\text{HHT})$			120/1024	240

Pair soft label $\xi_1(a, b) = \text{pair weight} \div p(\text{HHT}) = \text{chance of coin } a \text{ at flip 1 and } b \text{ at flip 2, given H, H, T.}$

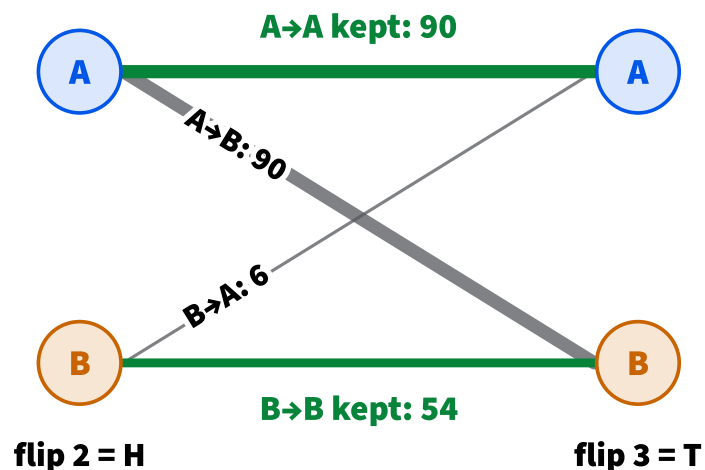
Kept between flips 1 and 2: $\xi_1(A, A) + \xi_1(B, B) = \frac{162+30}{240} = 0.8$

Check: each pair adds two rows of the 8-row list, e.g. $A \rightarrow A = AAA + AAB = 81 + 81 = 162 \checkmark$.

Between flips 1 and 2 the coin was kept with chance 0.8.

Between flips 2 and 3, then the new keep chance

pair weights $\times 2048$ (thicker = heavier)



Same rule with flip 3 = T. Recall α_2 = chance of flips 1–2 and the pair's first coin, at flip 2 (A 15/64, B 3/64); after flip 3 nothing remains, so $\beta_3 = 1$.

pair	α_2	move	T by 2nd	β_3	product	$\times 2048$
A $\rightarrow$ A	15/64	3/4	1/4	1	45/1024	90
A $\rightarrow$ B	15/64	1/4	3/4	1	45/1024	90
B $\rightarrow$ A	3/64	1/4	1/4	1	3/1024	6
B $\rightarrow$ B	3/64	3/4	3/4	1	27/1024	54
total		$= p(\text{HHT})$			120/1024	240

$\xi_2(a, b)$ = chance of coin a at flip 2 and b at flip 3, given H, H, T. **Kept:** $\xi_2(A, A) + \xi_2(B, B) = \frac{90+54}{240} = 0.6$; the tail at flip 3 makes a switch to B likelier.

New keep chance = expected keeps $\div$ moves: kept 0.8 times (flips 1–2, recalled) + 0.6 times (flips 2–3) = 1.4 of 2 moves, so $\frac{1.4}{2} = 0.7$.

New guess $\theta_A = 0.795$, $\theta_B = 0.429$, **keep 0.7: the same forward pass gives** $p(\text{HHT}) = 0.139 > \frac{15}{128} = 0.117$.

Baum–Welch = forward–backward soft labels + counting

↳ Recall: $\alpha_t(z)$ = chance of flips 1.. t and coin z at flip t ; $\beta_t(z)$ = chance of flips $t+1$.. n , given coin z at t ; $\tau(a \rightarrow b)$ = chance the coin moves from a to b ; $E(b, x)$ = chance coin b shows x ; θ_A = heads chance of coin A .

Theorem 6 (Baum–Welch; Baum, Petrie, Soules & Weiss, 1970). For a hidden Markov model with flips $x_1, \dots, x_n$, one EM round is the E-step and M-step below, and it never lowers $p(x)$. Here both coins share one keep chance and the start chance stays $\frac{1}{2}$; with a separate chance per move, $\tau(a \rightarrow b) = \sum_{t < n} \xi_t(a, b) / \sum_{t < n} \gamma_t(a)$.

$$\gamma_t(z) = \frac{\alpha_t(z)\beta_t(z)}{p(x)}$$

$$\xi_t(a, b) = \frac{\alpha_t(a) \tau(a \rightarrow b) E(b, x_{t+1}) \beta_{t+1}(b)}{p(x)}$$

$$\theta_A = \frac{\sum_{t=1}^n \gamma_t(A) [x_t=H]}{\sum_{t=1}^n \gamma_t(A)}, \quad \text{keep} = \frac{\sum_{t=1}^{n-1} (\xi_t(A, A) + \xi_t(B, B))}{n-1}$$

E-step, one coin: chance coin z made flip t , given all flips. Coins: $\gamma_2(A) = \frac{180}{240} = 0.75$.

E-step, pairs: chance of coin a at t and b at $t+1$. Coins: $\xi_1(A, A) = \frac{162}{240} = 0.675$.

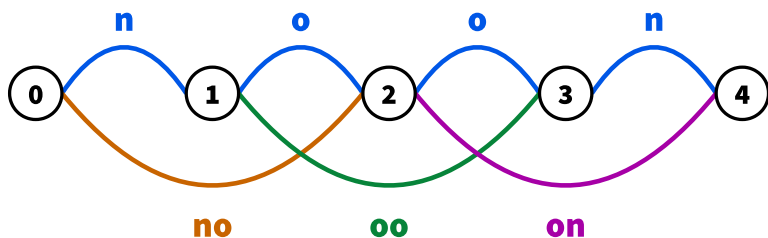
M-step: count and divide; $[x_t=H] = 1$ for heads, 0 for tails. Coins: $\frac{1.55}{1.95} = 0.795$, $\frac{0.8+0.6}{2} = 0.7$.

Proof. The expected log-likelihood of coins and flips splits into $\gamma \cdot \ln(\text{heads chance})$ and $\xi \cdot \ln(\text{keep or switch chance})$ terms; counting maximizes each (Theorem 1); Theorem 4 (EM climbs) gives that $p(x)$ never falls.

Every line is a step we just did by hand on H, H, T (θ_B : the same with B).

Tokenizers learn where to cut words by EM over hidden cuts

every path from 0 to 4 is one cut of “noon”



node k = boundary after k letters

posterior of each cut = chance $\times 625 \div 41$

n o o n	1/41
n o on	5/41
n oo n	5/41
no o n	5/41
no on	25/41

A language model reads text in pieces (**tokens**). No one labels where to cut "noon"; the **Unigram tokenizer** learns it by EM. Pieces (1–2 letters only): n, o, no, on, oo, each $1/5$ at first.

COIN CHAIN

hidden coin sequence

flips H, H, T

heads chance

keep / switch chance

soft label γ_t

forward message α_t

TOKENIZER

hidden cut, e.g. no|on

letters n, o, o, n

piece probability

none: pieces drawn independently

posterior of a cut, given the letters

α_k = chance of all cuts of the first k letters

e.g. $\alpha_2 = \text{chance}(n|o) + \text{chance}(no) = \frac{1}{25} + \frac{1}{5} = \frac{6}{25}$.

A cut's chance = product of its pieces: 4 pieces $1/625$, 3 pieces $5/625$, 2 pieces $25/625$; all 5 cuts: $(1+5+5+5+25)/625 = 41/625$. Speech recognition (CTC) runs the same sums over audio–letter alignments.

T. Kudo, “Subword regularization,” Association for Computational Linguistics (ACL) 2018; used in SentencePiece.

Cuts are hidden like coins; forward–backward adds over all cuts without listing them.

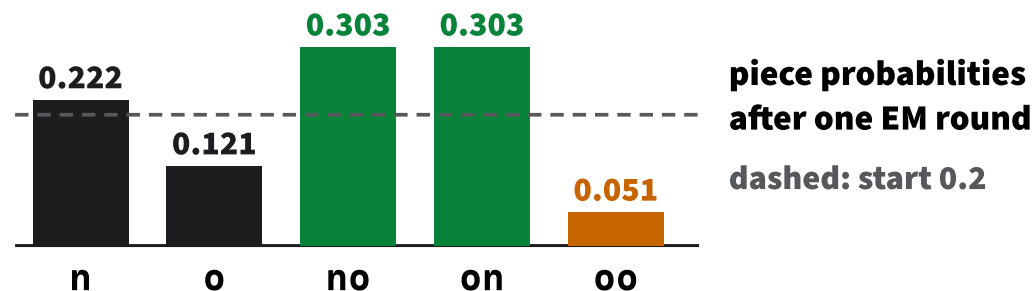
One EM round on "noon": count each piece, weighted by its cut

↳ Start: all 5 pieces at $1/5$, so a cut of k pieces has chance $(1/5)^k$ ($\times 625$ in the table). Each cell = uses of the piece in the cut $\times$ the cut's chance $\times 625$.

cut	chance $\times 625$	n	o	no	on	oo
n o o n	1	2	2	0	0	0
n o on	5	5	5	0	5	0
n oo n	5	10	0	0	0	5
no o n	5	5	5	5	0	0
no on	25	0	0	25	25	0
total	41	22	12	30	30	5
new: $\div 99$		0.222	0.121	0.303	0.303	0.051

E-step: a cut's posterior = chance $\div$ total = (chance $\times 625$) $\div$ 41, so a piece's expected uses = its column total $\div$ 41.

M-step: piece probability = its uses $\div$ all uses ($22 + 12 + 30 + 30 + 5 = 99$), like heads $\div$ flips; the 41 cancels.



Pieces of the likely cut no|on gain; oo, used only by an unlikely cut, falls from 0.2 to 0.051.

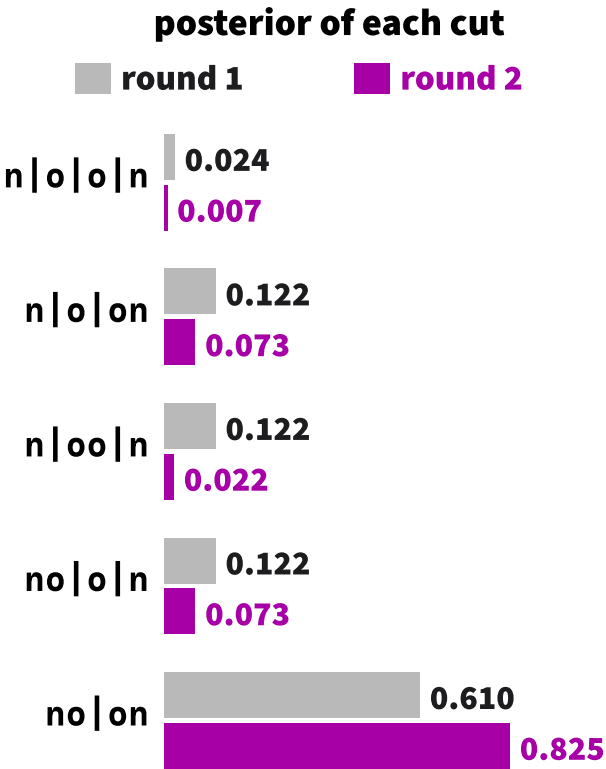
Round 2: the cut no|on rises from 0.610 to 0.825

↳ Same E-step with the round-1 probabilities n 22/99, o 12/99, no = on 30/99, oo 5/99: chance of a cut = product of its pieces; posterior = chance ÷ total.

cut	chance	× 1000	posterior, round 1	posterior, round 2
n o o n	$\frac{22}{99} \cdot \frac{12}{99} \cdot \frac{12}{99} \cdot \frac{22}{99}$	0.73	1/41 = 0.024	0.007
n o on	$\frac{22}{99} \cdot \frac{12}{99} \cdot \frac{30}{99}$	8.16	5/41 = 0.122	0.073
n oo n	$\frac{22}{99} \cdot \frac{5}{99} \cdot \frac{22}{99}$	2.49	5/41 = 0.122	0.022
no o n	$\frac{30}{99} \cdot \frac{12}{99} \cdot \frac{22}{99}$	8.16	5/41 = 0.122	0.073
no on	$\frac{30}{99} \cdot \frac{30}{99}$	91.83	25/41 = 0.610	0.825
total		111.37	1	1

Read one row: no|on has chance $\frac{30}{99} \cdot \frac{30}{99} = \frac{91.83}{1000}$; posterior = $\frac{91.83}{111.37} = 0.825$.

Every other cut **loses** posterior: pieces of no|on gained in round 1.



The likely cut gets likelier: no|on 0.610 → 0.825 after one more E-step.

Round 2 M-step: no and on rise to 0.41, oo falls to 0.010

↳ Posteriors from the round-2 E-step (column 2). Each cell = uses of the piece in the cut × the cut's posterior; a column total = the piece's expected uses.

cut	posterior	n	o	no	on	oo
n o o n	0.007	$2 \cdot 0.007 = 0.014$	0.014	0	0	0
n o on	0.073	0.073	0.073	0	0.073	0
n oo n	0.022	$2 \cdot 0.022 = 0.044$	0	0	0	0.022
no o n	0.073	0.073	0.073	0.073	0	0
no on	0.825	0	0	0.825	0.825	0
total	1	0.204	0.160	0.898	0.898	0.022

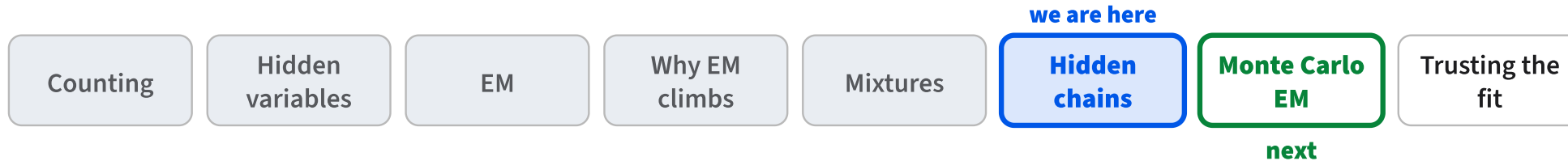
M-step (uses ÷ all uses): all uses = $0.204 + 0.160 + 0.898 + 0.898 + 0.022 = 2.182$;

$$\text{no} = \text{on} = \frac{0.898}{2.182} = 0.41, \text{oo} = \frac{0.022}{2.182} = 0.010$$

Why no|on wins: a cut of 2 pieces multiplies fewer numbers below 1 than a cut of 4, and EM feeds back: likely cut → its pieces gain → likelier cut. The tokenizer cuts "noon" as **no|on** (2 tokens, not 4) and drops pieces near 0, like oo.

no|on 0.610 → 0.825 → 0.967 (one more round, same steps); oo 0.2 → 0.051 → 0.010.

Next: when even the shortcut is too big



Forward-backward works because each coin touches only its neighbours. A hidden **chain of thought** (100 words a language model writes before its answer, each from a 50,000-word vocabulary) has $50,000^{100} \approx 10^{470}$ possible values, and each word depends on all earlier ones: **no small messages**.

How can we still get a soft label, such as HH's $p(A \mid \text{HH}) = 0.9$, without adding up every way?

guess 0.5

replay the game at random, count

skip the record

Answer. Replay and count: Monte Carlo, as in Lecture 4.

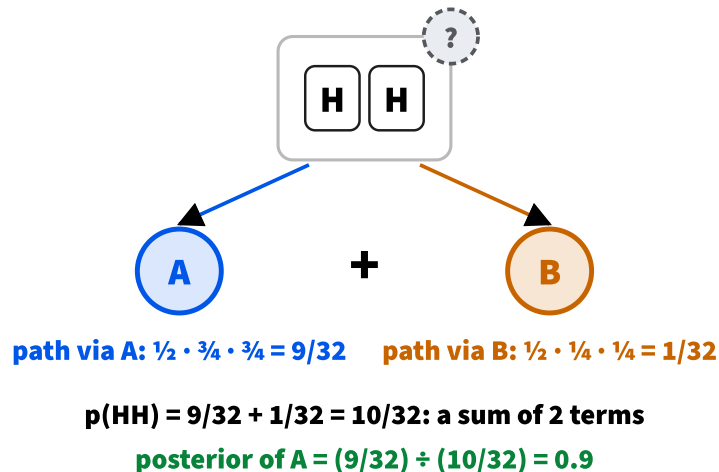
Replace the exact E-step by samples.

The E-step's sum can be impossible, yet running the model forward is cheap

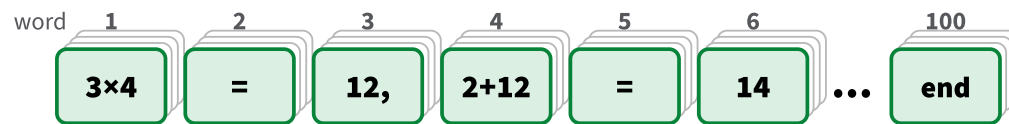
Goal: run EM when the hidden variable has far too many values to add up.

↳ Recall: the E-step needs the **posterior** $p_{\theta}(z | x) = p_{\theta}(x, z) / \sum_{z'} p_{\theta}(x, z')$ of the hidden z given a record x such as HH, at the guess $\theta_A = \frac{3}{4}, \theta_B = \frac{1}{4}$. The sum has **one term per value** of z .

Coins: the hidden z has 2 values



Reasoning: the hidden z is a 100-word text



behind each word: a stack of 50,000 possible words

Sum over every z : $50,000 \times 50,000 \times \dots$ (100 times) $\approx 10^{470}$ terms

since $50,000 \approx 10^{4.7}$, the product is $(10^{4.7})^{100} = 10^{470}$ ($4.7 \times 100 = 470$)
more terms than atoms in the visible universe (about 10^{80})

Run forward: pick one word at a time, left to right

100 easy steps; keep only the runs that reach the right answer, and the kept runs are posterior draws

Adding up every z can be hopeless; running forward and keeping the matches is cheap. Monte Carlo EM uses such draws.

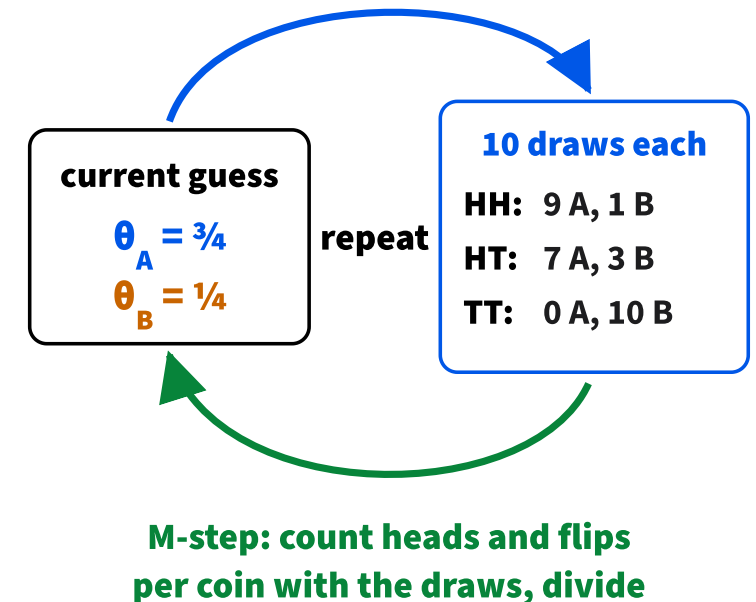
Monte Carlo EM keeps the M-step and replaces each exact share by a fraction of draws

↳ Recall at $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$: HH's path via A is $\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = 9/32$, via B $\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = 1/32$.

step	exact EM	Monte Carlo EM
E-step for HH	share of A: $\frac{9/32}{9/32 + 1/32} = 0.9$ needs the sum $p(\text{HH})$	draw the coin 10 times from the posterior; A came up 9 times: 9/10
M-step	count heads and flips with the shares, divide	the same , with the fractions of draws
repeat	at the new guess	at the new guess: draw afresh

Two questions follow. **How** can we draw from the posterior without its sum? And **how much noise** do the draws add to the new guess?

E-step: draw each record's coin from the posterior, 10 times



Exact shares → fractions of draws. Everything else in EM stays.

Replay the game and keep the runs that match the record: the kept runs follow the posterior

↳ To draw the coin for HH **without** computing $p(\text{HH})$: replay the game at the guess $\theta_A = \frac{3}{4}, \theta_B = \frac{1}{4}$ (fair toss for a coin, flip it twice) and **keep** a run only if it shows HH. Shown: 32 replays at their expected counts; a fair toss sends $32 \cdot \frac{1}{2} = 16$ runs to each coin.

picked A
16 runs



A shows HH with chance $\frac{3}{4} \cdot \frac{3}{4} = 9/16 \rightarrow 9$ of A's 16 runs are kept

picked B
16 runs



B shows HH with chance $\frac{1}{4} \cdot \frac{1}{4} = 1/16 \rightarrow 1$ of B's 16 runs is kept

kept: 9 + 1 = 10 runs; A made 9 of them $\rightarrow$ share $9/10 = 0.9$ = the posterior of A given HH

No sum was computed: we only ran the game forward and compared. This is **rejection sampling**. **Its cost:** $32 - 10 = 22$ of the 32 runs are thrown away. The keep rate $10/32$ is exactly $p(\text{HH})$, the sum we never computed, so **rare records waste most runs**.

Kept runs show A : B = 9 : 1, so each kept run's coin is a posterior draw: A with chance 0.9.

Ten draws per record give noisy fractions; more draws come closer to the posterior

↳ Each dot is one draw of a record's coin from its posterior at $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$ (recall: A with chance 0.9 for HH, 0.5 for HT, 0.1 for TT).

record	draws of its coin (blue = A, orange = B)	fraction of A's	posterior of A
HH		9 of 10 = 0.9	0.9
HT		7 of 10 = 0.7	0.5
TT		0 of 10 = 0	0.1

fractions 0.9, 0.7, 0 against posteriors 0.9, 0.5, 0.1: HT is 0.2 off

More draws: with 1000 draws per record the fractions are **0.908, 0.504, 0.107**, each within 0.01 of 0.9, 0.5, 0.1 (with 10 draws, HT was 0.2 off).

Ten draws give fractions near the posteriors but not equal (HT 0.7 against 0.5); 1000 draws land within 0.01.

One Monte Carlo EM round: credit heads and flips by the drawn fractions, then divide

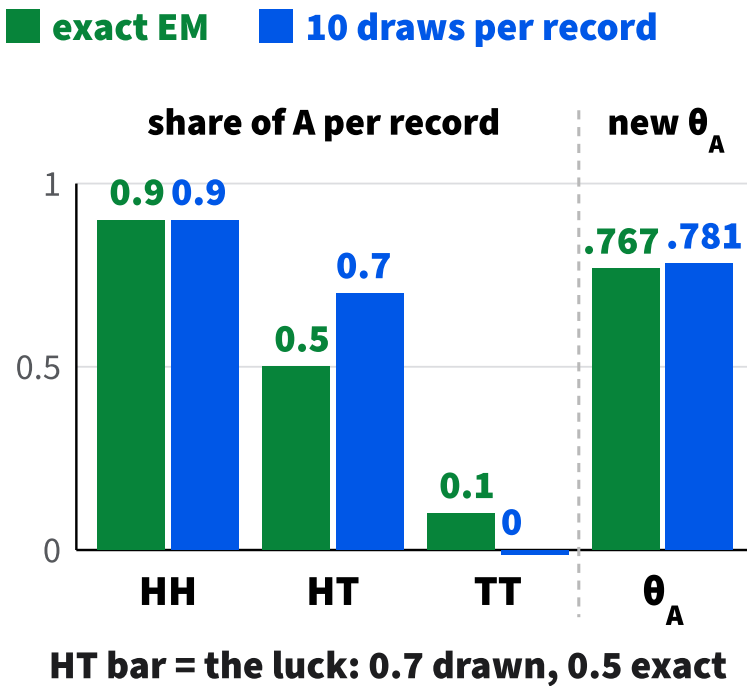
↳ Guess $\theta_A = \frac{3}{4}, \theta_B = \frac{1}{4}$; 10 draws per record gave A 9 times for HH, 7 for HT, 0 for TT. Each record's heads and flips go to A in proportion to its fraction of A's; B gets the rest.

record	fraction of A's	heads to A	flips to A
HH	$9/10 = 0.9$	$0.9 \cdot 2 = 1.8$	$0.9 \cdot 2 = 1.8$
HT	$7/10 = 0.7$	$0.7 \cdot 1 = 0.7$	$0.7 \cdot 2 = 1.4$
TT	$0/10 = 0$	$0 \cdot 0 = 0$	$0 \cdot 2 = 0$
total		2.5	3.2

B gets the rest (fractions $1 - 0.9 = 0.1, 1 - 0.7 = 0.3, 1 - 0 = 1$):
heads $0.1 \cdot 2 + 0.3 \cdot 1 + 1 \cdot 0 = 0.5$; flips $0.1 \cdot 2 + 0.3 \cdot 2 + 1 \cdot 2 = 2.8$

M-step (heads $\div$ flips): $\theta_A = \frac{2.5}{3.2} = 0.781, \theta_B = \frac{0.5}{2.8} = 0.179$.

Exact EM (shares 0.9, 0.5, 0.1): heads $0.9 \cdot 2 + 0.5 \cdot 1 + 0.1 \cdot 0 = 2.3$, flips $2(0.9 + 0.5 + 0.1) = 3.0$, so $\theta_A = 2.3/3.0 = 0.767$.



Then draw afresh at (0.781, 0.179) and repeat: EM with "share" replaced by "fraction of draws".

Monte Carlo EM: draw m hidden values per record, then fit as if they were told

Definition (Monte Carlo EM; G. C. G. Wei and M. A. Tanner, J. Amer. Statist. Assoc., 1990). Records $x^{(1)}, \dots, x^{(N)}$; start from θ_0 ; choose m , the number of draws per record. For $t = 0, 1, 2, \dots$:

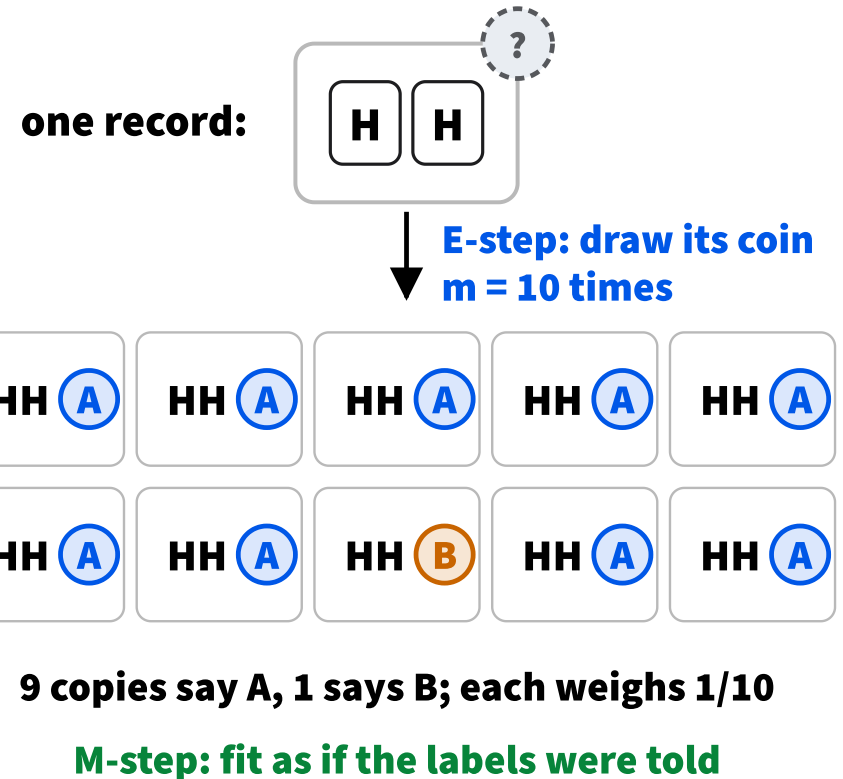
E-step: for each record n , draw m hidden values (for example by replay-and-keep):

$$z^{(n,1)}, \dots, z^{(n,m)} \sim p_{\theta_t}(z \mid x^{(n)})$$

$z^{(n,j)}$ = the j -th draw for record n .

M-step: maximum likelihood with each draw as a told label of weight $\frac{1}{m}$:

$$\theta_{t+1} = \arg \max_{\theta} \sum_{n=1}^N \frac{1}{m} \sum_{j=1}^m \ln p_{\theta}(x^{(n)}, z^{(n,j)})$$



Each draw weighs $1/m$; as $m \rightarrow \infty$ the fractions become the posteriors: EM again.

EM's M-step maximizes an expectation; Monte Carlo replaces it by an average over draws

↳ Recall Section 3 (EM = expectation–maximization): $q_t^{(n)}(z) = p_{\theta_t}(z \mid x^{(n)})$ is record n 's posterior (its soft label) at the guess θ_t .

$$\sum_n \sum_z q_t^{(n)}(z) \ln p_{\theta}(x^{(n)}, z)$$

$$= \sum_n \mathbb{E}_{z \sim q_t^{(n)}} [\ln p_{\theta}(x^{(n)}, z)]$$

$$\mathbb{E}_{z \sim q} [g(z)] \approx \frac{1}{m} \sum_{j=1}^m g(z^{(j)}), \quad z^{(j)} \sim q$$

$$\approx \sum_n \frac{1}{m} \sum_{j=1}^m \ln p_{\theta}(x^{(n)}, z^{(n,j)})$$

EM's M-step objective.

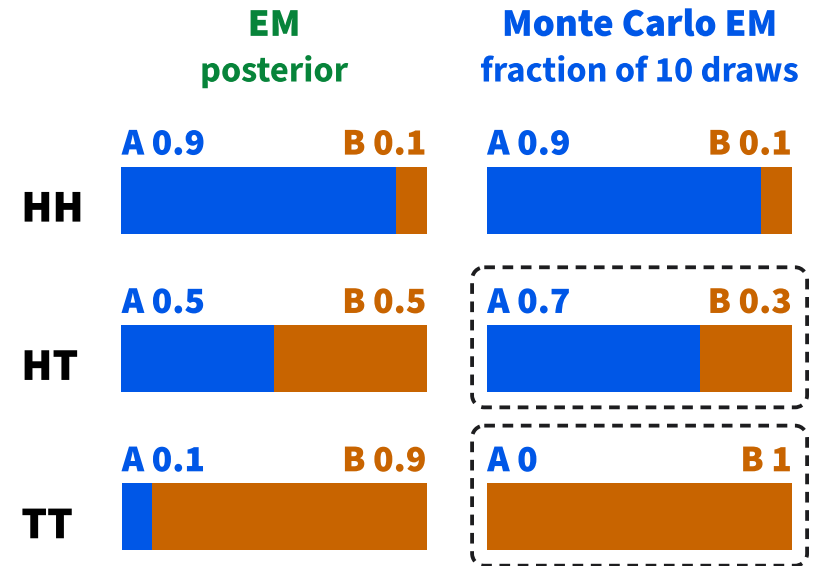
An **expectation** over the posterior.

Monte Carlo: average over m draws, with $g(z) = \ln p_{\theta}(x^{(n)}, z)$.

Monte Carlo EM's M-step objective.

On the coins ($m = 10$): for HH, $0.9 \ln p_{\theta}(\text{HH}, A) + 0.1 \ln p_{\theta}(\text{HH}, B)$ becomes $\frac{9}{10} \ln p_{\theta}(\text{HH}, A) + \frac{1}{10} \ln p_{\theta}(\text{HH}, B)$; for HT, the weights 0.5, 0.5 become **0.7, 0.3**.

weight on $\ln p_{\theta}(x, A)$ and $\ln p_{\theta}(x, B)$



same terms; only the weights move, by the luck of the draws (dashed boxes)

Same log-probability terms, new weights: each posterior share becomes a fraction of draws.

Each draw becomes a told label, so the M-step is Section 1's maximum likelihood

↳ Recall Section 1 (Theorem 1): when the friend tells every coin $z^{(n)}$, maximum likelihood maximizes $\sum_n \ln p_\theta(x^{(n)}, z^{(n)})$, and the answer is **count and divide**.

Monte Carlo EM's M-step objective ($m = 10$; EM = expectation-maximization):

$$\sum_n \frac{1}{10} \sum_{j=1}^{10} \ln p_\theta(x^{(n)}, z^{(n,j)})$$

Read it as a sum over 3 records $\times$ 10 draws = **30 pairs** (record $x^{(n)}$, drawn coin $z^{(n,j)}$), each with weight $\frac{1}{10}$.

That is the told-label log-likelihood of the **pretend data**: Section 1's objective, nothing new.

pretend data: 3 records $\times$ 10 draws
= 30 labelled copies, weight 1/10 each

HH 

9 copies (HH, A), 1 copy (HH, B)

HT 

7 copies (HT, A), 3 copies (HT, B)

TT 

0 copies (TT, A), 10 copies (TT, B)

as if a friend had told these 30 coins

The M-step averages the log-probability over draws in place of an expectation, so it is count and divide on the pretend data.

Collect the θ_A terms: 2.5 heads and 0.7 tails, so $\theta_A = 0.781$

$$\ln p_\theta(\text{HH}, A) = \ln \frac{1}{2} + 2 \ln \theta_A$$

$$\ln p_\theta(\text{HT}, A) = \ln \frac{1}{2} + 1 \ln \theta_A + 1 \ln(1 - \theta_A)$$

drop: every $\ln \frac{1}{2}$ and every B copy

$$\text{heads: } \frac{1}{10} (9 \cdot 2 + 7 \cdot 1) \ln \theta_A$$

$$\text{tails: } \frac{1}{10} (7 \cdot 1) \ln(1 - \theta_A)$$

$$\text{sum} = 2.5 \ln \theta_A + 0.7 \ln(1 - \theta_A)$$

$$\frac{2.5}{\theta_A} - \frac{0.7}{1 - \theta_A} = 0 \Rightarrow \theta_A = \frac{2.5}{3.2} = 0.781$$

Coin B, from its copies (1 HH, 3 HT, 10 TT): heads $(1 \cdot 2 + 3 \cdot 1)/10 = 0.5$, tails $(3 \cdot 1 + 10 \cdot 2)/10 = 2.3$, so $\theta_B = 0.5/(0.5 + 2.3) = 0.179$.

Exact EM (recall): $2.3 \ln \theta_A + 0.7 \ln(1 - \theta_A)$, peak 0.767; only the heads count differs, by HT's luck $(0.7 - 0.5) \cdot 1 = 0.2$.

Path $\frac{1}{2} \cdot \theta_A \cdot \theta_A$.

One head, one tail.

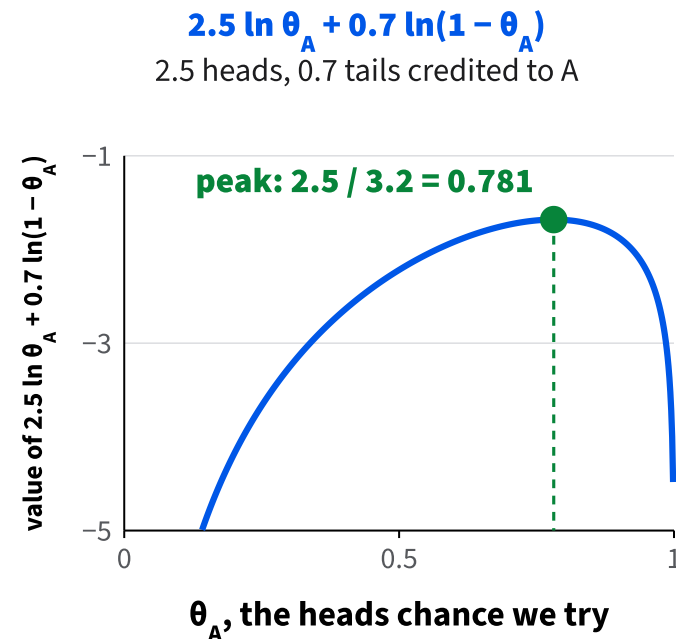
No θ_A in them: they cannot move its peak.

9 copies (HH, A) $\times$ 2 heads; 7 copies (HT, A) $\times$ 1 head.

7 copies (HT, A) $\times$ 1 tail.

A's heads and tails.

Slope zero: $2.5(1 - \theta_A) = 0.7 \theta_A \Rightarrow \theta_A = \frac{2.5}{2.5+0.7}$.



The arg max is count and divide on the pretend data: $2.5/3.2 = 0.781$ and $0.5/2.8 = 0.179$.

Theorem 7: the noise in a fraction of m draws shrinks like $1/\sqrt{m}$

Theorem 7 (Monte Carlo error). Draw the coin m times, independently, from its posterior; q = posterior of A. The fraction of A's has mean q and standard deviation $\sqrt{q(1-q)/m}$: **four times the draws, half the noise.**

$$\mathbb{E}[I_j] = 1 \cdot q + 0 \cdot (1 - q) = q$$

$$\text{Var } I_j = \mathbb{E}[I_j^2] - q^2 = q - q^2 = q(1 - q)$$

$$\mathbb{E}\left[\frac{1}{m} \sum_j I_j\right] = \frac{1}{m} \cdot m q = q$$

$$\text{Var}\left[\frac{1}{m} \sum_j I_j\right] = \frac{1}{m^2} \cdot m q(1 - q) = \frac{q(1-q)}{m}$$

$$\text{HT, } m = 10: \sqrt{\frac{1}{2} \cdot \frac{1}{2} / 10} = 0.158$$

The **new guess inherits the noise**: the hand round's $\theta_A = 2.5/3.2 = 0.7813$ is 0.015 off exact EM's $2.3/3.0 = 0.7667$, because HT drew 0.7, not 0.5.

$I_j = 1$ if draw j is A, else 0.

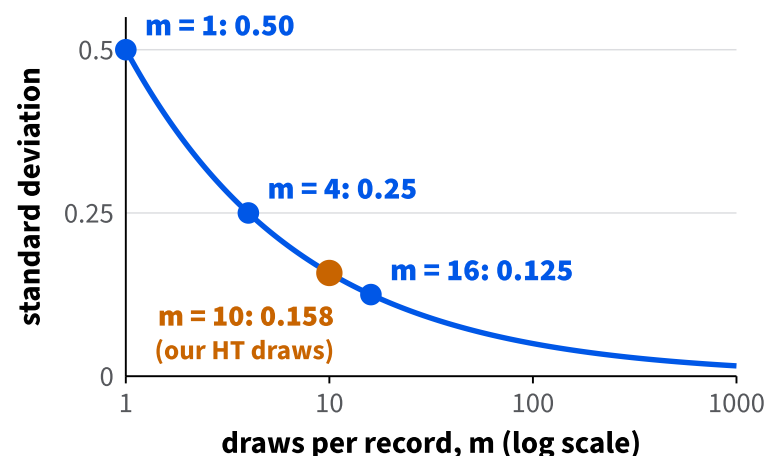
$I_j^2 = I_j$: it is 0 or 1.

Linearity of expectation.

Independent: variances add;
 $\frac{1}{m}$ comes out squared. ■

0.7 is 0.2 off: $0.2/0.158 \approx 1.3$ standard deviations.

HT, posterior $q = 1/2$: standard deviation $0.5 / \sqrt{m}$
4× more draws halves it: $0.50 \rightarrow 0.25 \rightarrow 0.125$



Noise shrinks like $1/\sqrt{m}$: pick m for the precision you need; large m gives EM.

Hard EM gives each record entirely to its likelier coin

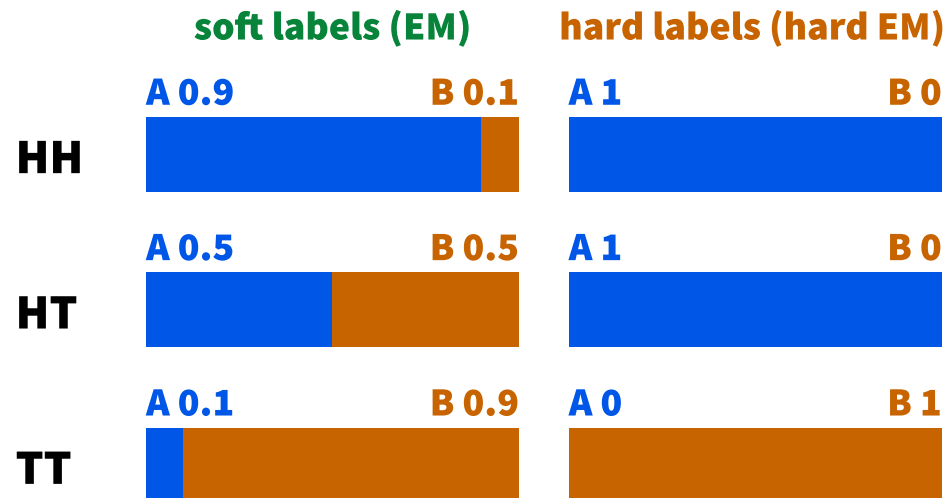
↳ **Hard EM** = one "draw" per record that is always the more likely coin, as in k-means (Section 5). Start $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$.

record	path via A	path via B	hard label
HH	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	A
HT	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{32}$	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{32}$	A (tie)
TT	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	B

M-step: A made HH and HT: 3 heads in 4 flips, $\theta_A = \frac{3}{4}$. B made TT: 0 heads in 2 flips, $\theta_B = \frac{0}{2} = 0$.

Exact EM instead uses B's shares $\frac{1/32}{10/32} = 0.1$, $\frac{3/32}{6/32} = 0.5$, $\frac{9/32}{10/32} = 0.9$ and credits B with $0.1 \cdot 2 + 0.5 \cdot 1 + 0.9 \cdot 0 = 0.7$ heads in 2 ($0.1 + 0.5 + 0.9$) = 3.0 flips: $\theta_B = \frac{0.7}{3.0} = 0.233$.

share of each record given to A / B



hard EM rounds 0.9 up to 1, 0.1 down to 0, and the tie 0.5 up to 1 (ties go to A: a fixed rule; any tie-break leaves one coin with only heads or only tails here)

Rounding the shares to 0 or 1 leaves B with only tails: a coin that never shows heads.

Hard EM then stops at once, at a fit worse than where it started

↳ Round 2 at $\theta_A = \frac{3}{4}, \theta_B = 0$: the two paths pick the labels, and their sum $p(x)$ scores the fit.

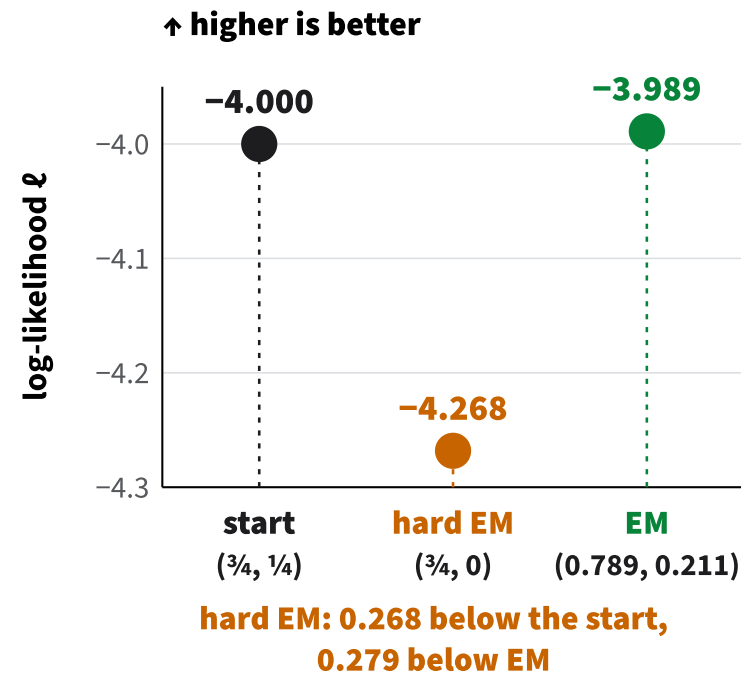
record	path via A	path via B	label	$p(x)$
HH	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{1}{2} \cdot 0 \cdot 0 = 0$	A	$\frac{9}{32}$
HT	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{32}$	$\frac{1}{2} \cdot 0 \cdot 1 = 0$	A	$\frac{3}{32}$
TT	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{1}{2} \cdot 1 \cdot 1 = \frac{16}{32}$	B	$\frac{17}{32}$

Labels A, A, B again $\Rightarrow$ same counts $\Rightarrow$ **stuck** at $(\frac{3}{4}, 0)$.

Score: $\ell = \ln \frac{9}{32} + \ln \frac{3}{32} + \ln \frac{17}{32} = \ln \frac{9 \cdot 3 \cdot 17}{32^3} = \ln \frac{459}{32768} = -4.268$.

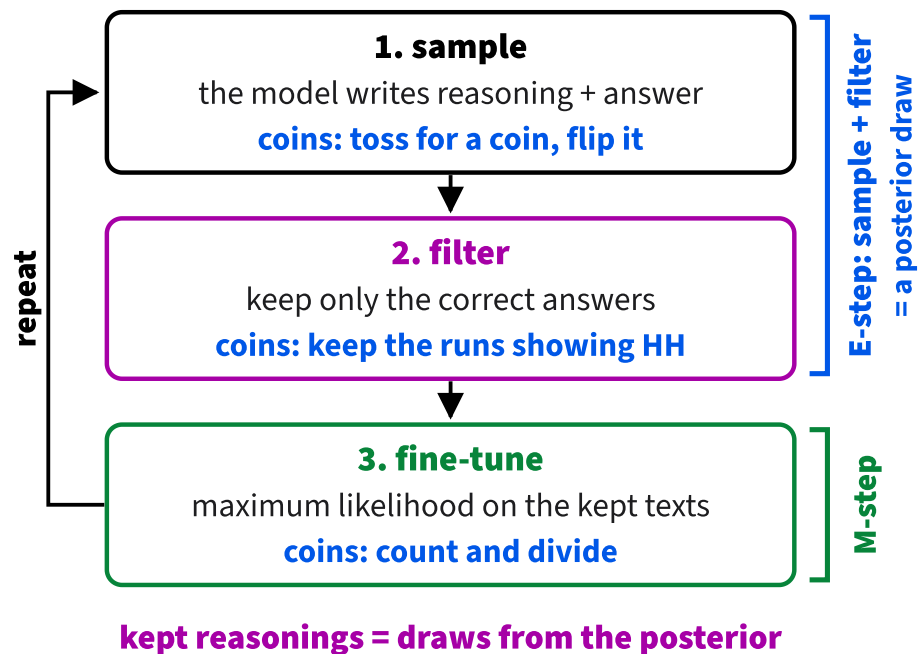
Recall: $\ell = -4.000$ at the start, -3.989 at EM's limit.

Why it can fall: Theorem 4's climb needs the exact posterior as the soft label (the bound touches ℓ only there). Hard shares of A (1, 1, 0) are not the posteriors (0.9, 0.5, 0.1), so ℓ is not protected.



Hard labels are not the posterior, so Theorem 4 no longer protects ℓ : hard EM ends below its own start.

Teaching a language model to reason from answers is the replay-and-keep game



COIN GAME

hidden coin z

record HH

θ_A, θ_B (what EM learns)

toss chance π ($\frac{1}{2}$ in our game)

posterior of the coin given HH

replay, keep HH

count and divide

The training data hold only questions and final answers; the reasoning is hidden, like the coin.

Self-Taught Reasoner, STaR (Zelikman et al., 2022) and **Reinforced Self-Training with EM, ReST-EM** (Singh et al., 2023) run this loop; ReST-EM names its two steps E and M.

REASONING MODEL

$\Leftrightarrow$ hidden reasoning trace z

$\Leftrightarrow$ correct final answer

$\Leftrightarrow$ the model's weights

$\Leftrightarrow$ how often the model writes each kind of reasoning

$\Leftrightarrow$ posterior of the reasoning given the question and the correct answer

$\Leftrightarrow$ sample, keep correct

$\Leftrightarrow$ fine-tune on kept texts

Sample, keep the correct, fine-tune: Monte Carlo EM with a reasoning as the coin.

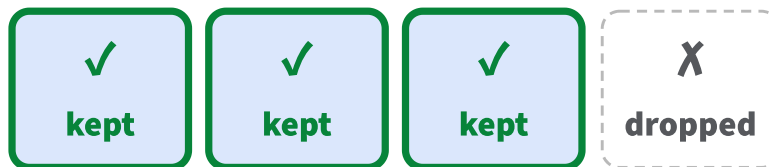
A tiny reasoner: keeping the correct answers raises careful reasoning from $\frac{1}{2}$ to $\frac{3}{4}$

↳ Toy model: it writes a **careful** reasoning with chance π (right with chance $\frac{3}{4}$) or a **hasty** one (right with chance $\frac{1}{4}$). These are coins A and B with one flip; "correct" plays heads. Start $\pi = \frac{1}{2}$. **Here EM learns the toss chance π** ; the heads chances $\frac{3}{4}, \frac{1}{4}$ stay fixed.

the model writes 8 reasonings at $\pi = \frac{1}{2}$

careful

right $\frac{3}{4}$ of the time



hasty

right $\frac{1}{4}$ of the time



kept: 3 careful + 1 hasty

fine-tune on the 4 kept texts

new $\pi = \frac{3}{4}$

coins: count and divide, 3 of 4

expected counts: $8 \cdot \frac{1}{2} = 4$ careful, $4 \cdot \frac{3}{4} = 3$ right; $8 \cdot \frac{1}{2} = 4$ hasty, $4 \cdot \frac{1}{4} = 1$ right

Why $\frac{3}{4}$: new π = share of careful among kept = $\frac{\frac{1}{2} \cdot \frac{3}{4}}{\frac{1}{2} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{1}{4}} = \frac{3}{4}$, the **posterior** of "careful" given "correct".

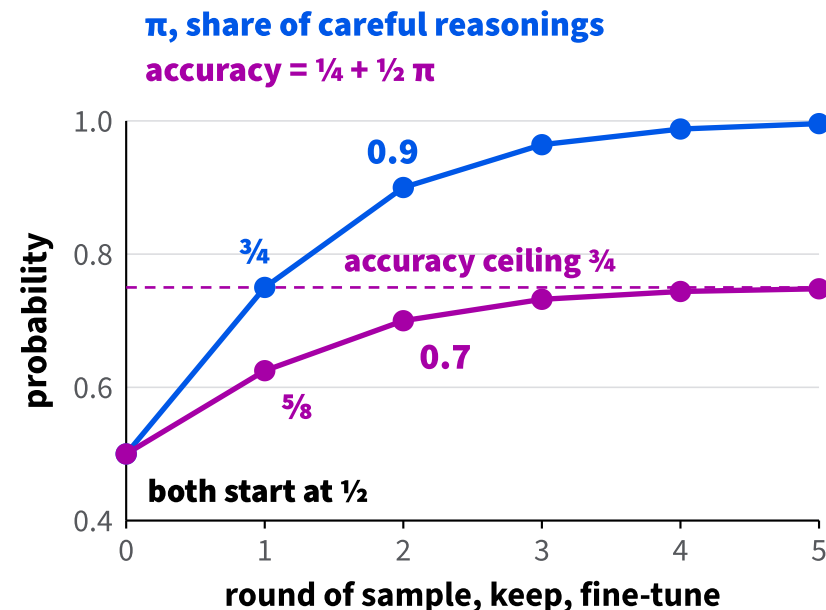
One round of STaR on the toy model = one Monte Carlo EM step for π (shown at expected counts).

Round after round, the tiny reasoner's accuracy climbs from $\frac{1}{2}$ toward $\frac{3}{4}$

↳ Each round: sample at π , keep the correct, refit $\pi' = \pi \cdot \frac{3}{4} / (\pi \cdot \frac{3}{4} + (1 - \pi) \cdot \frac{1}{4})$, the share of careful among kept. Every kept record says "correct", so its likelihood is $p(\text{correct}) = \pi \cdot \frac{3}{4} + (1 - \pi) \cdot \frac{1}{4} = \frac{1}{4} + \frac{1}{2}\pi$: **the accuracy**.

round	π , share of careful	accuracy $\frac{1}{4} + \frac{1}{2}\pi$
0	$\frac{1}{2}$ (start)	$\frac{1}{2} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{2}$
1	$\frac{\frac{1}{2} \cdot \frac{3}{4}}{\frac{1}{2} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{1}{4}} = \frac{3}{4}$	$\frac{3}{4} \cdot \frac{3}{4} + \frac{1}{4} \cdot \frac{1}{4} = \frac{10}{16} = \frac{5}{8}$
2	$\frac{\frac{3}{4} \cdot \frac{3}{4}}{\frac{3}{4} \cdot \frac{3}{4} + \frac{1}{4} \cdot \frac{1}{4}} = \frac{9}{10}$	$0.9 \cdot \frac{3}{4} + 0.1 \cdot \frac{1}{4} = 0.675 + 0.025 = 0.7$

No human-written reasoning was used, only answers. Since $\pi \leq 1$, accuracy $\leq \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$: self-training only re-weights reasonings the model **already writes**.



Here EM's likelihood is the accuracy: $\frac{1}{2} \rightarrow \frac{5}{8} \rightarrow 0.7 \rightarrow \dots$, capped at $\frac{3}{4}$ by what the model writes.

Next: the same records, three different answers. Which do we trust?

run on HH, HT, TT	start (θ_A, θ_B)	ends at	ℓ
EM (soft labels)	$(\frac{3}{4}, \frac{1}{4})$	$(0.789, 0.211)$	-3.989 , best
hard EM	$(\frac{3}{4}, \frac{1}{4})$	$(\frac{3}{4}, 0)$, stuck	-4.268
EM , identical start	$(\frac{1}{2}, \frac{1}{2})$	$(\frac{1}{2}, \frac{1}{2})$, never moves	$3 \ln \frac{1}{4} = -4.159$

Identical start: every record (HH, HT or TT) has chance $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$, so $\ell = 3 \ln \frac{1}{4} = -4.159$. Equal coins split each record $\frac{1}{2} : \frac{1}{2}$, so each coin gets half of the 3 heads and 6 flips: $1.5/3 = \frac{1}{2}$ again.

Swap the names: $\theta_A = 0.211$, $\theta_B = 0.789$. Is the fit better, worse, or the same?

☐ the same

☐ better

☐ worse

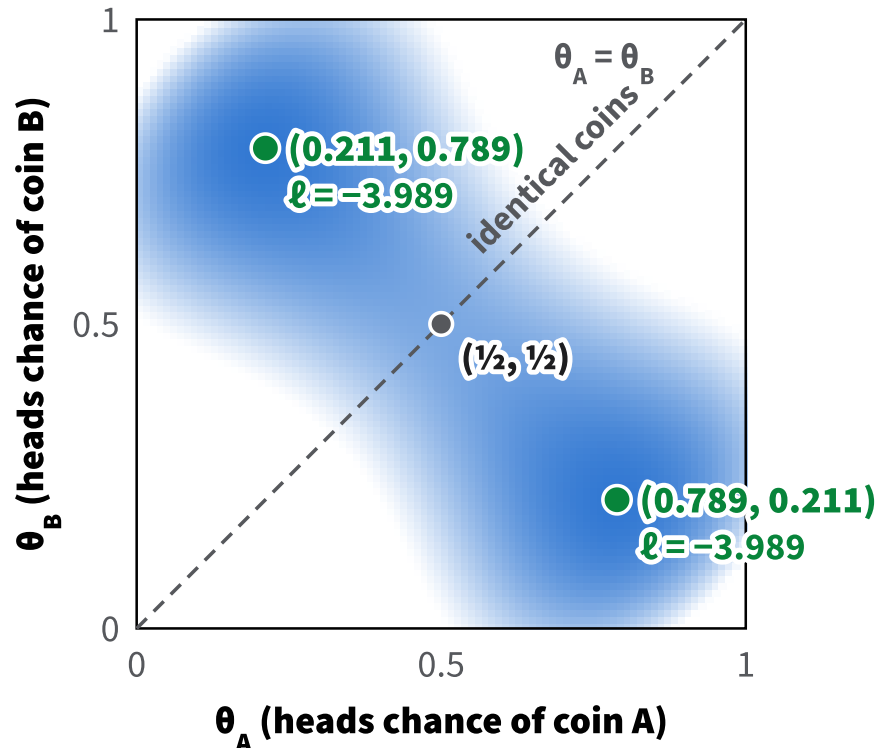
Answer. The same. $p(\text{HH}) = \frac{1}{2} \cdot 0.789^2 + \frac{1}{2} \cdot 0.211^2$ either way: the two paths just swap places, since each coin is picked with chance $\frac{1}{2}$. So every record's chance, and ℓ , is unchanged.

A fit can be stuck, or best yet not unique: which answer to trust is the next question.

The coin fit has two mirror-image peaks

ℓ : -4.59 = top - 0.6

-3.99 = top



the peaks mirror each other across the diagonal

Can a learned $\theta = (\theta_A, \theta_B)$ be trusted? Look at the log-likelihood; each record sums over the two coins:

$$\begin{aligned}\ell &= \ln p(\text{HH}) + \ln p(\text{HT}) + \ln p(\text{TT}) \\ p(\text{HH}) &= \underbrace{\frac{1}{2}\theta_A^2}_{\text{via A}} + \underbrace{\frac{1}{2}\theta_B^2}_{\text{via B}}\end{aligned}$$

Swap the names $\theta_A \leftrightarrow \theta_B$: the same two terms in the other order. Every record keeps its probability, so ℓ is a **mirror image** across the diagonal.

Peaks: recall EM's limit $\theta_A = \frac{3+\sqrt{3}}{6} = 0.789$,
 $\theta_B = 1 - \theta_A = 0.211$, so $\theta_A\theta_B = \frac{1}{6}$.

$$\begin{aligned}p(\text{HT}) &= \frac{1}{2}\theta_A \underbrace{(1 - \theta_A)}_{\theta_B} + \frac{1}{2}\theta_B \underbrace{(1 - \theta_B)}_{\theta_A} = \theta_A\theta_B = \frac{1}{6} \\ p(\text{HH}) &= \frac{1}{2}\left(\underbrace{(\theta_A + \theta_B)^2}_{1} - 2\theta_A\theta_B\right) = \frac{1}{3} = p(\text{TT}) \\ \ell &= 2 \ln \frac{1}{3} + \ln \frac{1}{6} = -3.989 \text{ at both peaks}\end{aligned}$$

Names are not facts: report "one coin 0.789, the other 0.211".

Between the peaks, $(\frac{1}{2}, \frac{1}{2})$ is a saddle that can trap EM

Between the peaks lies $(\frac{1}{2}, \frac{1}{2})$, identical coins: every record has chance $\frac{1}{4}$, so $\ell = 3 \ln \frac{1}{4} = -4.159$.

step d from $(\frac{1}{2}, \frac{1}{2})$	$p(\text{HH})$	$p(\text{HT})$	$p(\text{TT})$
along: $(\frac{1}{2}+d, \frac{1}{2}+d)$	$(\frac{1}{2}+d)^2$	$\frac{1}{4} - d^2$	$(\frac{1}{2}-d)^2$
across: $(\frac{1}{2}+d, \frac{1}{2}-d)$	$\frac{1}{4} + d^2$	$\frac{1}{4} - d^2$	$\frac{1}{4} + d^2$

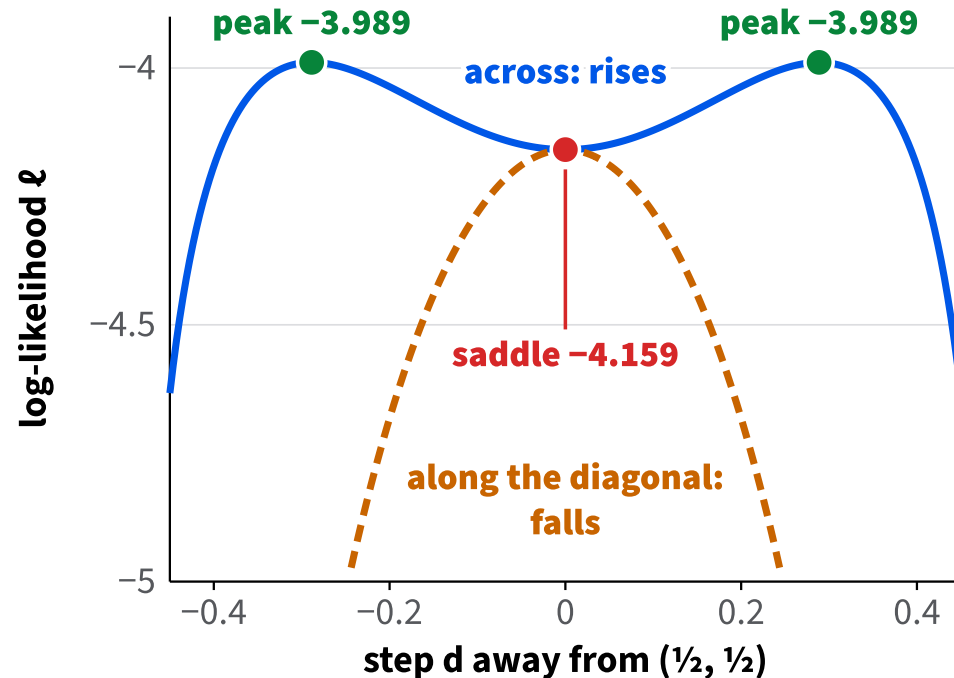
Along, both coins have $\theta = \frac{1}{2} + d$: $p(\text{HH}) = \frac{1}{2}\theta^2 + \frac{1}{2}\theta^2 = \theta^2$,
 $p(\text{HT}) = \theta(1 - \theta) = \frac{1}{4} - d^2$.

Across, the $\pm d$ terms cancel: $\frac{1}{2}(\frac{1}{2} + d)^2 + \frac{1}{2}(\frac{1}{2} - d)^2 = \frac{1}{4} + d^2$.

along: $\ell = 3 \ln(\frac{1}{4} - d^2)$, largest at $d = 0$

across: $\ell = 2 \ln(\frac{1}{4} + d^2) + \ln(\frac{1}{4} - d^2)$, smallest at $d = 0$

Highest along, lowest across: a **saddle**. Equal coins give each record posterior $\frac{1}{2}$ for A, so each coin is credited 1.5 heads in 3 flips, $\frac{1}{2}$ again: **EM stays on the diagonal and stops here.**



the same point is lowest across and highest along

The mirror peaks are one answer; the saddle is a worse fit that EM can get stuck in.

Restart EM from several guesses and keep the largest ℓ

↳ Theorem 4 promises only that ℓ never goes down, not that EM reaches the top.

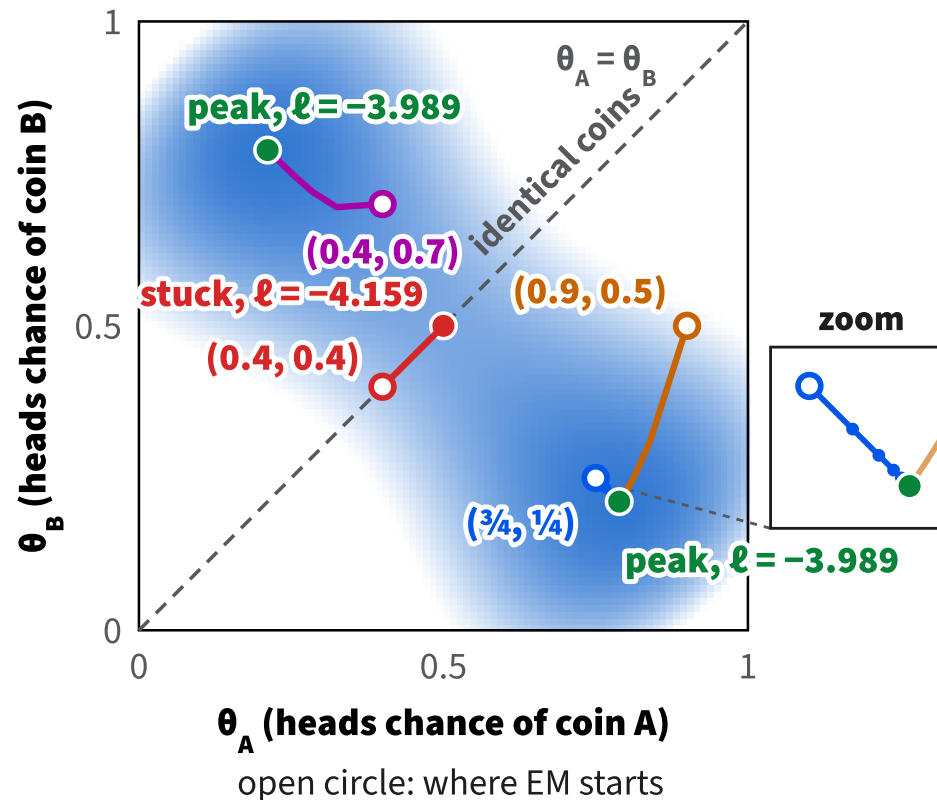
start (θ_A, θ_B)	EM ends at	ℓ
$(\frac{3}{4}, \frac{1}{4})$	(0.789, 0.211)	−3.989
(0.4, 0.7)	(0.211, 0.789)	−3.989
(0.9, 0.5)	(0.789, 0.211)	−3.989
(0.4, 0.4)	$(\frac{1}{2}, \frac{1}{2})$, stuck	−4.159

Why (0.4, 0.4) sticks: equal coins give every record the posterior $\frac{1}{2}$ for A, so each coin is credited $\frac{1}{2}(2 + 1 + 0) = 1.5$ heads in $\frac{1}{2}(2 + 2 + 2) = 3$ flips: both get $1.5/3 = \frac{1}{2}$, again and again.

Practice: run several random starts and keep the largest ℓ . Here every start off the diagonal reaches a peak with $\ell = -3.989$; in larger models the peaks differ in height.

ℓ : −4.59 = top − 0.6

−3.99 = top



Restarts escape the saddle; an equal ℓ with swapped names is the same answer.

Learn the pick chance too: 3 unknowns, but only 2 free numbers

↳ So far the friend picked A with chance $\frac{1}{2}$. Now that chance $\pi = p(A)$ is unknown too.

Unknowns: π , θ_A , θ_B : three knobs.

What the data can tell: how often a record shows 0, 1 or 2 heads. The order adds nothing: HT and TH both have chance $\theta(1 - \theta)$, with θ that coin's heads chance.

$$\begin{aligned} p(2 \text{ heads}) &= \pi \theta_A^2 + (1 - \pi) \theta_B^2 \\ p(1 \text{ head}) &= 2[\pi \theta_A(1 - \theta_A) + (1 - \pi) \theta_B(1 - \theta_B)] \\ p(0 \text{ heads}) &= \pi(1 - \theta_A)^2 + (1 - \pi)(1 - \theta_B)^2 \end{aligned}$$

Free numbers (degrees of freedom): the three always sum to 1, so any two fix the third: **2 free numbers**.

3 unknowns > 2 free numbers: some direction of the knobs leaves all three chances unchanged.

3 unknowns

$$\pi = \frac{1}{2}$$

pick chance of A

$$\theta_A = \frac{3}{4}$$

heads chance of A

$$\theta_B = \frac{1}{4}$$

heads chance of B

model
→

what the data can tell

$$\frac{5}{16}$$

p(2 heads)

$$= \frac{1}{2}(\frac{3}{4})^2 + \frac{1}{2}(\frac{1}{4})^2$$

$$\frac{6}{16}$$

p(1 head)

$$= 2(\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{4})$$

$$\frac{5}{16}$$

p(0 heads)

$$= \frac{1}{2}(\frac{1}{4})^2 + \frac{1}{2}(\frac{3}{4})^2$$

$$\text{sum check: } \frac{5}{16} + \frac{6}{16} + \frac{5}{16} = 1$$

at the working guess $\pi = \frac{1}{2}$, $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$

Three knobs, two readings: some turn of the knobs changes no reading.

Two different coin settings give exactly the same data

↳ Only 2 free numbers means different knobs can give the same readings. Write $p(h)$ for the chance of h heads.

$$p(2) = \pi\theta_A^2 + (1 - \pi)\theta_B^2, \quad p(0) = \pi(1 - \theta_A)^2 + (1 - \pi)(1 - \theta_B)^2$$

$$p(1) = 2[\pi\theta_A(1 - \theta_A) + (1 - \pi)\theta_B(1 - \theta_B)]$$

Setting 1 (EM's fit, $\pi = \frac{1}{2}, \theta_B = 1 - \theta_A$): $\theta_A(1 - \theta_A) = \theta_A\theta_B = \frac{1}{6}$,
 $\theta_A^2 + \theta_B^2 = 1 - 2 \cdot \frac{1}{6} = \frac{2}{3}$.

Setting 2: try $\pi = \frac{1}{4}$ and a coin A that **always lands heads** ($\theta_A = 1$). Then $p(0) = \frac{3}{4}(1 - \theta_B)^2 = \frac{1}{3}$ gives $(1 - \theta_B)^2 = \frac{4}{9}$, so $\theta_B = \frac{1}{3}$.

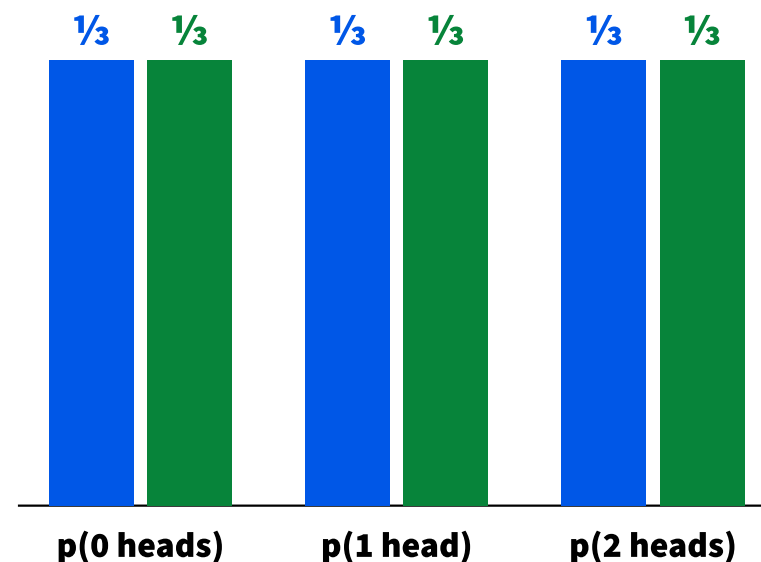
setting	$p(2)$	$p(1)$	$p(0)$
1	$\frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$	$2 \cdot \frac{1}{2}(\frac{1}{6} + \frac{1}{6}) = \frac{1}{3}$	$\frac{1}{2}(\theta_B^2 + \theta_A^2) = \frac{1}{3}$
2	$\frac{1}{4} \cdot 1 + \frac{3}{4} \cdot \frac{1}{9} = \frac{1}{3}$	$2(\frac{1}{4} \cdot 1 \cdot 0 + \frac{3}{4} \cdot \frac{1}{3} \cdot \frac{2}{3}) = \frac{1}{3}$	$\frac{1}{4} \cdot 0 + \frac{3}{4} \cdot \frac{4}{9} = \frac{1}{3}$

Each row: $\frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$. HT and TH are equally likely, so $p(\text{HT}) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$ in both, and $\ell = \ln \frac{1}{3} + \ln \frac{1}{6} + \ln \frac{1}{3} = -3.989$.

■ **setting 1:** $\pi = \frac{1}{2}, \theta_A = 0.789, \theta_B = 0.211$

■ **setting 2:** $\pi = \frac{1}{4}, \theta_A = 1, \theta_B = \frac{1}{3}$

identical bars



each setting: $\frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$

Different knobs, identical readings: both settings fit the records equally well.

Every pick chance from $\frac{1}{4}$ to $\frac{3}{4}$ has its own equal-fit pair: a ridge

↳ Setting 1 has $\pi = \frac{1}{2}$, setting 2 has $\pi = \frac{1}{4}$. What about every π in between?

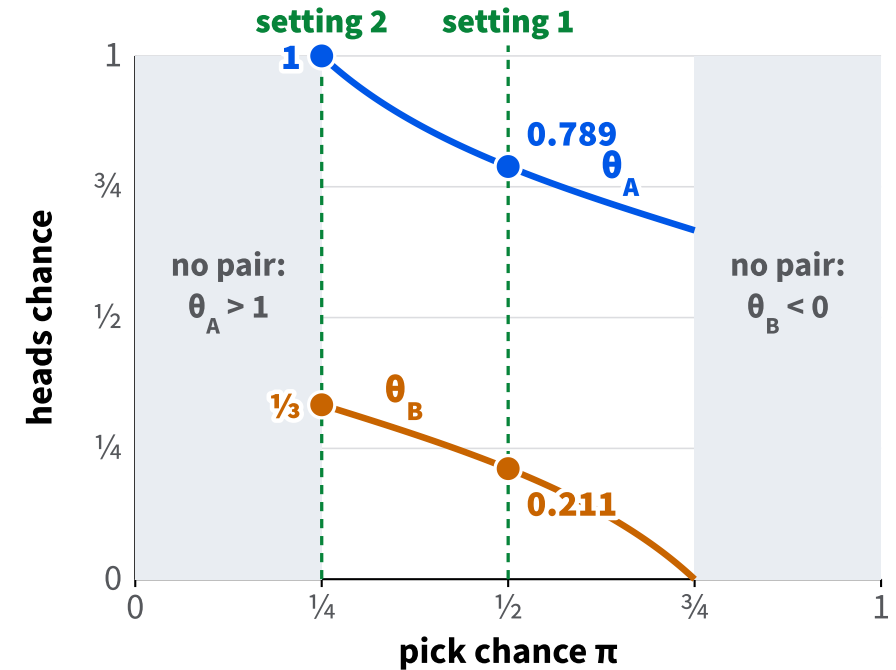
Fix any π . Matching the readings $p(2) = p(0) = \frac{1}{3}$ asks for

$$\begin{aligned}\pi \theta_A^2 + (1 - \pi) \theta_B^2 &= \frac{1}{3} \\ \pi(1 - \theta_A)^2 + (1 - \pi)(1 - \theta_B)^2 &= \frac{1}{3}\end{aligned}$$

Subtract the two: $\pi\theta_A + (1 - \pi)\theta_B = \frac{1}{2}$. Then $p(1) = 1 - \frac{1}{3} - \frac{1}{3} = \frac{1}{3}$ follows.

Try $\theta_A = \frac{1}{2} + (1 - \pi)d$, $\theta_B = \frac{1}{2} - \pi d$: the average stays $\frac{1}{2}$.
First equation: $\frac{1}{4} + \pi(1 - \pi)d^2 = \frac{1}{3}$, so $d = 1/\sqrt{12\pi(1 - \pi)}$.
 $\pi = \frac{1}{2}$: $d = 0.577$, $\theta_A = 0.789$; $\pi = \frac{1}{4}$: $d = \frac{2}{3}$, $\theta_A = 1$, $\theta_B = \frac{1}{3}$.

Each π gets its own pair (plus its mirror, with $\theta_A < \theta_B$): a **ridge** of equal fits, all with $\ell = -3.989$. Outside $\frac{1}{4}$ to $\frac{3}{4}$, $\theta_A > 1$ or $\theta_B < 0$ (shaded).



every pair: $p(0 \text{ heads}) = p(2 \text{ heads}) = \frac{1}{3}$

The data cannot choose between settings on the ridge, so no algorithm can.

More unknowns than free numbers means no unique fit

Theorem 8 (counting test). If a model has more unknowns than one record's distribution has free numbers, the parameters are **not identifiable**: near a typical setting, a whole curve of settings gives the same distribution.

In words: more unknowns than free numbers: some change of the unknowns is invisible in the data. Passing (unknowns $\leq$ free numbers) does not prove uniqueness.

$$(\pi, \theta_A, \theta_B) \mapsto (p(0), p(2))$$

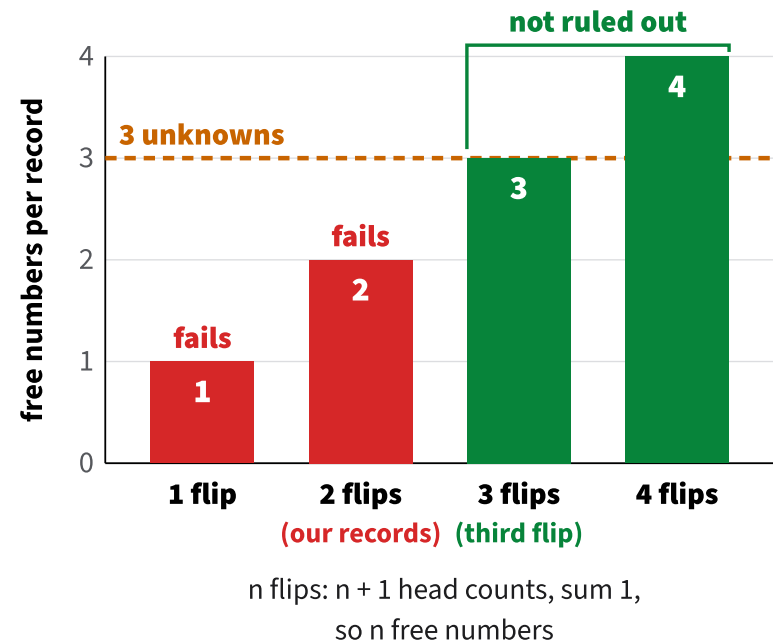
S , the slope table (the **Jacobian**): slope of each free number along each unknown, 2×3

steps along v change neither $p(0)$ nor $p(2)$

3 unknowns, 2 free numbers.

3 columns in a 2-D plane are **linearly dependent**: some step $v \neq 0$ has $Sv = 0$.

True at every nearby setting, so following v traces a curve of equal fits (constant-rank theorem). ■



Count before you fit: 3 unknowns but 2 free numbers guarantees a ridge.

On the coins, the invisible step points along the ridge

recall: $p(2) = \pi\theta_A^2 + (1 - \pi)\theta_B^2$, $p(0) = \pi(1 - \theta_A)^2 + (1 - \pi)(1 - \theta_B)^2$; the slope table S (Jacobian) holds each one's slope along π, θ_A, θ_B

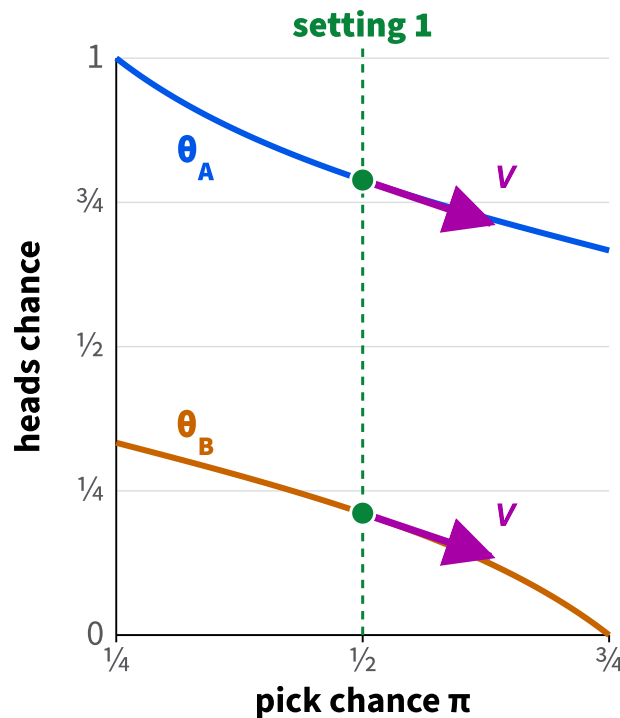
At **setting 1**, $\pi = \frac{1}{2}$, $\theta_A = \frac{3+\sqrt{3}}{6}$, $\theta_B = 1 - \theta_A$. Write $c = \theta_A - \theta_B = \frac{\sqrt{3}}{3} = 0.577$; then $\theta_A^2 - \theta_B^2 = c(\theta_A + \theta_B) = c$.

S : slope along	π	θ_A	θ_B
$p(2)$	$\theta_A^2 - \theta_B^2$	$2\pi\theta_A$	$2(1 - \pi)\theta_B$
setting 1	c	θ_A	θ_B
$p(0)$	$(1 - \theta_A)^2 - (1 - \theta_B)^2$	$-2\pi(1 - \theta_A)$	$-2(1 - \pi)(1 - \theta_B)$
setting 1	$-c$	$-\theta_B$	$-\theta_A$

$v = (1, -c, -c)$: raise π , lower both θ s.

$p(2)$: $c - c\theta_A - c\theta_B = c - c \cdot 1 = 0$; $p(0)$: $-c + c\theta_B + c\theta_A = 0$.

The ridge $\theta_A = \frac{1}{2} + (1 - \pi)d$, $\theta_B = \frac{1}{2} - \pi d$ has $d = 1/\sqrt{12\pi(1 - \pi)}$, flat at $\pi = \frac{1}{2}$ where $\pi(1 - \pi)$ peaks, so both slopes are $-d = -0.577 = -c$: **exactly** v .



v : per unit of π ,
each θ falls by $c = 0.577$
the arrows run along both curves

Fix a ridge by observing more per record, not by running EM longer.

A third flip per record tells the two settings apart

$$p(3 \text{ heads}) = \pi \theta_A^3 + (1 - \pi) \theta_B^3$$

$$\text{and } \theta_A^3 + \theta_B^3 = (\theta_A + \theta_B)^3 - 3\theta_A\theta_B(\theta_A + \theta_B)$$

$$\frac{1}{2} \left(1^3 - 3 \cdot \frac{1}{6} \cdot 1 \right) = \frac{1}{4}$$

$$\frac{1}{4} \cdot 1^3 + \frac{3}{4} \cdot \left(\frac{1}{3} \right)^3 = \frac{5}{18}$$

$$\frac{1}{4} \cdot 0^3 + \frac{3}{4} \left(\frac{2}{3} \right)^3 = \frac{2}{9}$$

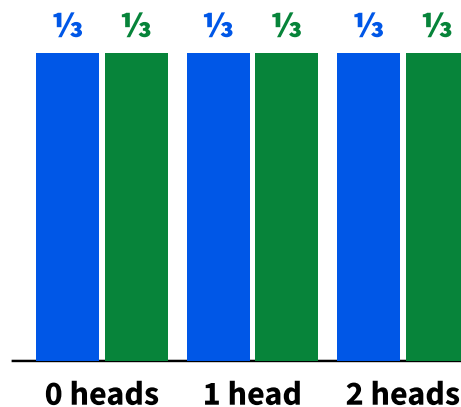
Setting 1: sum 1, product $\frac{1}{6}$.

Setting 2: **not** $\frac{1}{4}$.

Setting 2, **0 heads**; setting 1 gives $\frac{1}{4}$ again (swap H↔T, $1 - \theta_A = \theta_B$).

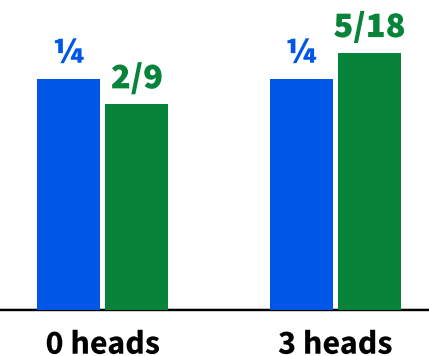
3 flips give 4 head counts summing to 1: **3 free numbers = 3 unknowns**. The counting test no longer rules out a unique fit, and indeed the coins are pinned down up to swapping names when $\theta_A \neq \theta_B$ and $0 < \pi < 1$ (Teicher, 1963; Allman, Matias & Rhodes, 2009).

2 flips per record



identical:
the data cannot choose

3 flips per record



one count that differs
is enough to tell them apart

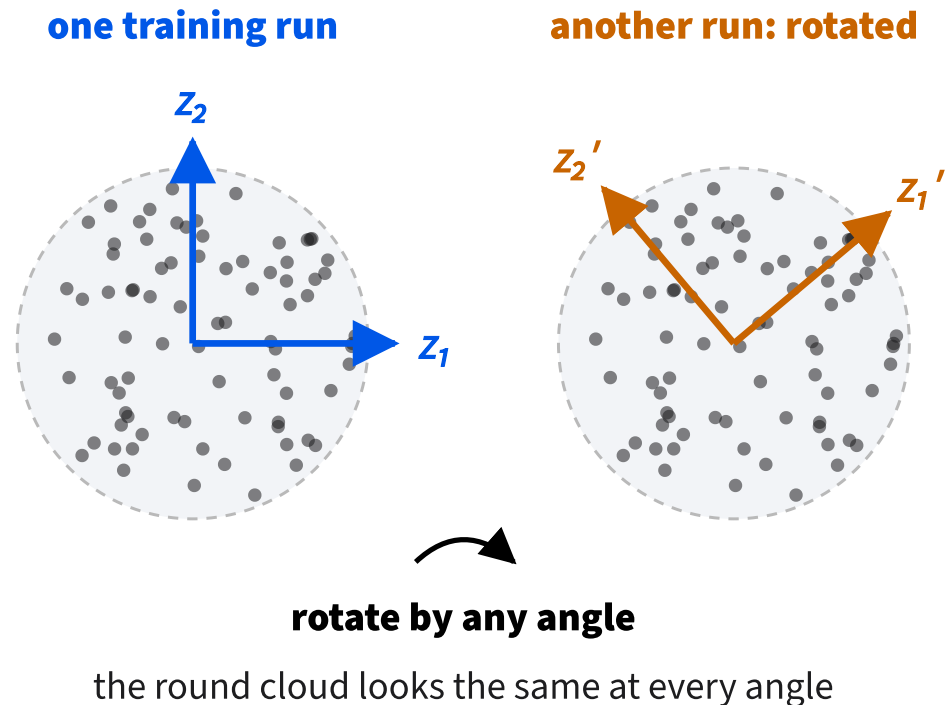
■ setting 1: $\pi = \frac{1}{2}$, $\theta_A = 0.789$, $\theta_B = 0.211$

■ setting 2: $\pi = \frac{1}{4}$, $\theta_A = 1$, $\theta_B = \frac{1}{3}$

3 flips, 1 and 2 heads differ too: $\frac{1}{4}$ vs $\frac{1}{3}$, $\frac{1}{4}$ vs $\frac{1}{6}$

One more flip per record turns the ridge back into two mirror peaks.

A learned code is a hidden coin, and its axes can be turned freely



A **variational autoencoder (VAE)** gives each image x a hidden code $z = (z_1, z_2)$; a decoder turns z into an image. We hope z_1 means pose and z_2 colour.

prior: the standard normal $p(z) = \frac{1}{2\pi} e^{-(z_1^2 + z_2^2)/2}$, which depends only on the distance from 0: **round**

COIN GAME		VAE
coin A or B	$\leftrightarrow$	code $z \in \mathbb{R}^2$
record HH	$\leftrightarrow$	image x
θ_A, θ_B	$\leftrightarrow$	decoder weights
swap A $\leftrightarrow$ B	$\leftrightarrow$	flip or permute the code axes
ridge of equal fits	$\leftrightarrow$	circle of equal fits, one per angle

Why: rotate every code by an angle and let the decoder rotate back first. The round prior gives the rotated code the same density, so every image keeps its likelihood. **So** training lands on one angle by chance.

Like the ridge, rotations give endlessly many equally good codes.

A prior that changes with an observed label pins the code axes

Tiny instance. Rotate every code by 90° , $(2, 0) \rightarrow (0, 2)$ (the decoder undoes it). Images keep their likelihood; only the prior can notice.

Round prior $\frac{1}{2\pi} e^{-(z_1^2+z_2^2)/2}$: both codes get $\frac{e^{-2}}{2\pi} = 0.0215$.
Images alone cannot fix the axes (Locatello et al., 2019).

Prior set by a label u , seen with each image: here, photo taken by day or by night.

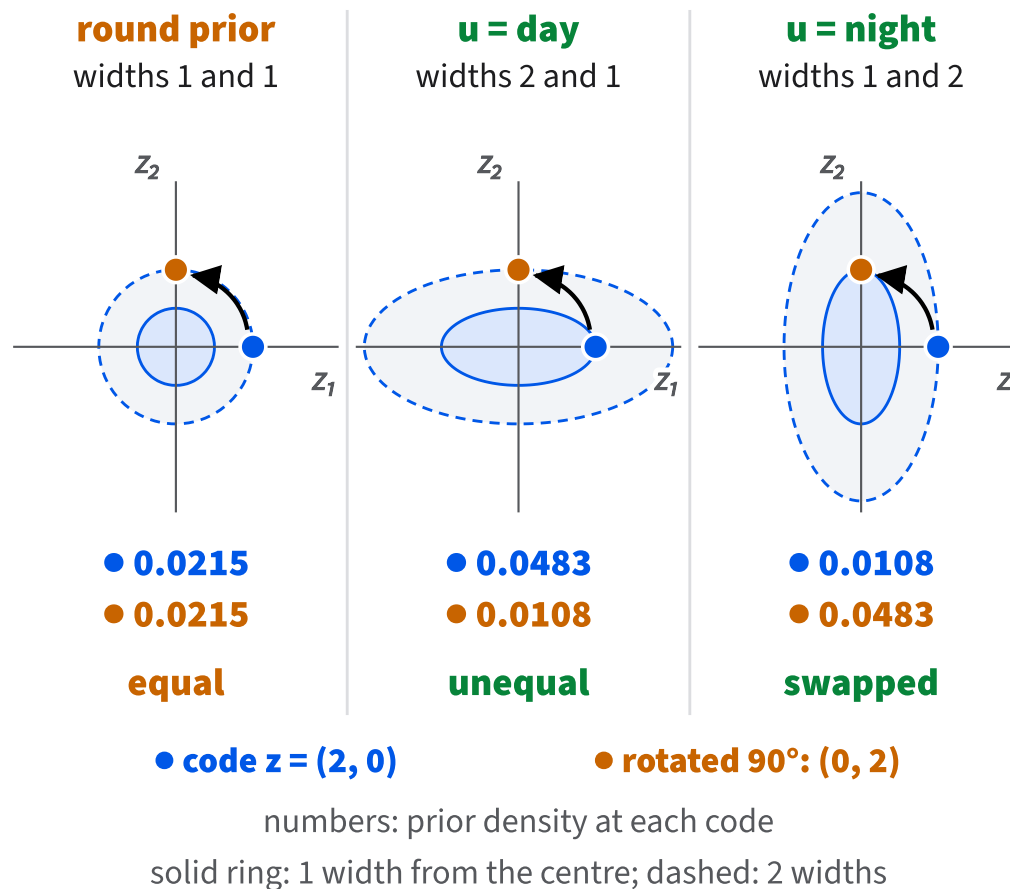
day, standard deviations (widths) 2 and 1:

$$p(z) = \frac{1}{2\pi \cdot 2 \cdot 1} e^{-(z_1^2/2^2 + z_2^2/1^2)/2}$$

$$p(2, 0) = \frac{e^{-1/2}}{4\pi} = 0.0483, p(0, 2) = \frac{e^{-2}}{4\pi} = 0.0108$$

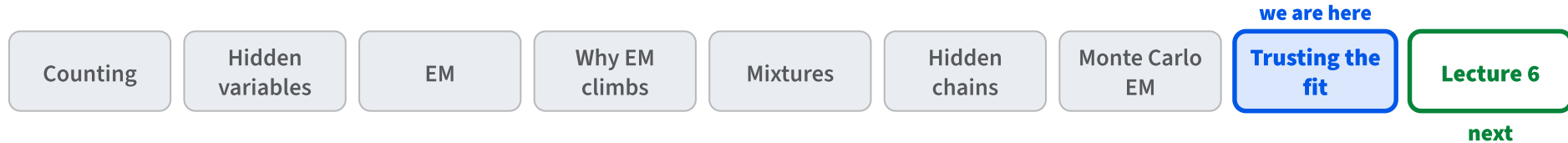
night (widths 1 and 2) swaps them

Day alone is kept by "halve z_1 , turn, double z_1 ". A linear map keeping both keeps both exponents, so their round sum $\frac{5}{4}(z_1^2 + z_2^2)$: it is a turn or mirror, and only **sign flips** keep day's stretch (Khemakhem et al., 2020).



An extra observation, like the third flip, makes the hidden code identifiable.

Next lecture: when the exact posterior costs too much



Exact E-step

The expectation step (E-step) for coins, mixtures, chains: **sum** over every hidden value.

Sampled E-step

Monte Carlo EM: **draw** hidden values instead of summing.

Lecture 6

Pick the guess q (a distribution over the hidden values) from a simple family and **maximize** Lemma 1's bound $B(q, \theta) \leq \ln p_\theta(x)$.

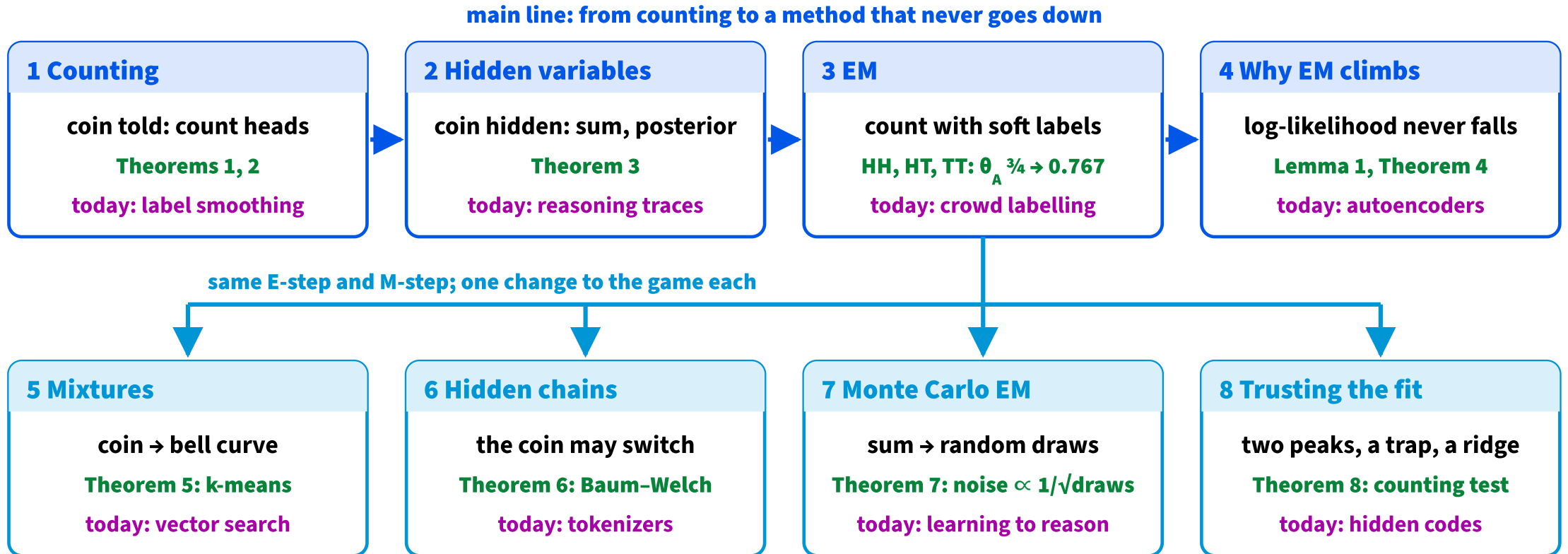
Coin game: the exact E-step sums 2 terms per record (coin A or B). With 16 hidden black-or-white pixels, how many?

32 65,536 16

Answer. $2^{16} = 65,536$: each pixel doubles the count. A 28×28 image has 784 pixels, so $2^{784} \approx 10^{236}$ terms, which **no computer can sum**. Lecture 6 approximates instead.

Today's bound $B(q, \theta) = \sum_z q(z) \ln \frac{p_\theta(x, z)}{q(z)}$ **becomes Lecture 6's main tool.**

One coin game, one method, four changes to the game



black: what changes · green: the result (proved or computed) · magenta: where it runs today

EM never lowers the log-likelihood ℓ ; Sections 5–8 change the game but keep its two steps.

Count when the coin is told; sum over it when it is hidden

Records HH, HT, TT; working guess $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$; each coin picked with chance $\frac{1}{2}$. ∂_{θ_A} = slope in θ_A .

Section	Result	On the coins
1 Counting	Theorem 1 (counting): $\hat{\theta}$ = heads $\div$ flips. Theorem 2 (Laplace's rule): uniform prior $\Rightarrow$ posterior mean (heads + 1) $\div$ (flips + 2).	Coins told. A made HH, HT: 3 heads in 4 flips, $\frac{3}{4}$. B made TT: $\frac{0}{2} = 0$; Laplace $\frac{0+1}{2+2} = \frac{1}{4}$.
2 Hidden variables	$p_{\theta}(x) = \sum_z p_{\theta}(x, z)$: a sum over the coin z . Posterior $p_{\theta}(z \mid x)$ = a path's share of the sum. Theorem 3 (Fisher's identity): slope of $\ln p_{\theta}(x)$ = posterior-weighted slopes of $\ln p_{\theta}(x, z)$, one per path.	$p(\text{HH}) = \underbrace{\frac{1}{2}\theta_A^2}_{\text{via A}} + \underbrace{\frac{1}{2}\theta_B^2}_{\text{via B}} = \frac{9}{32} + \frac{1}{32} = \frac{5}{16}.$ Posterior $p(A \mid \text{HH}) = \frac{9}{9+1} = 0.9$. $\partial_{\theta_A} \ln p(\text{HH}) = \frac{\partial_{\theta_A} (\frac{1}{2}\theta_A^2 + \frac{1}{2}\theta_B^2)}{p(\text{HH})} = \frac{\theta_A}{5/16} = \frac{3/4}{5/16} = 2.4.$ Path slopes: $\partial_{\theta_A} \ln (\frac{1}{2}\theta_A^2) = \frac{2}{\theta_A} = \frac{8}{3}$ (A), 0 (B). Posterior-weighted: $0.9 \cdot \frac{8}{3} + 0.1 \cdot 0 = 2.4$, the same slope.

Told coin: count. Hidden coin: sum the paths, and the posterior splits each record between them.

EM counts with soft labels, and θ_A climbs from 0.75 to 0.789

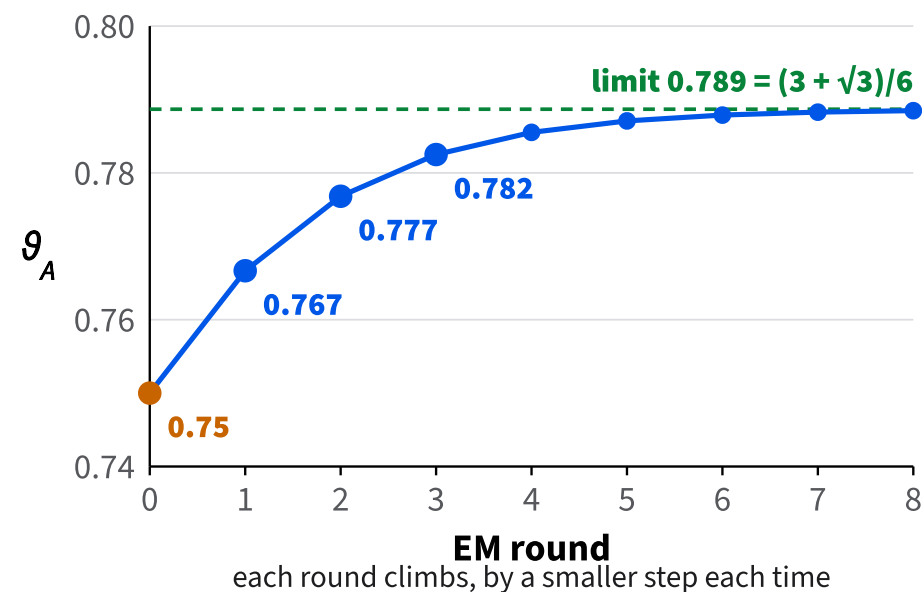
EM. E-step: soft labels $q_t = p_{\theta_t}(z \mid x)$, the posterior. **M-step:** count heads and flips with them, then divide.

record	via A	via B	soft label for A
HH	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{9}{9+1} = 0.9$
HT	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{32}$	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{32}$	$\frac{3}{3+3} = 0.5$
TT	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{1}{1+9} = 0.1$

Heads to A: $0.9 \cdot 2 + 0.5 \cdot 1 + 0.1 \cdot 0 = 2.3$.

Flips to A: $2(0.9 + 0.5 + 0.1) = 3$.

New $\theta_A = \frac{2.3}{3} = 0.767$; repeat the two steps.



E guesses the coin with the posterior, M counts with the guess; repeat until θ stops moving.

The bound touches $\ln p$ at the posterior, so EM never falls

Guess $\theta_A = \frac{3}{4}, \theta_B = \frac{1}{4}$. Recall $p(\text{HH}, A) = \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}, p(\text{HH}, B) = \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}; p(\text{HH}) = p(\text{TT}) = \frac{5}{16}, p(\text{HT}) = \frac{3}{16}$.

Log-likelihood $\ell(\theta) = \ln p(\text{HH}) + \ln p(\text{HT}) + \ln p(\text{TT})$.

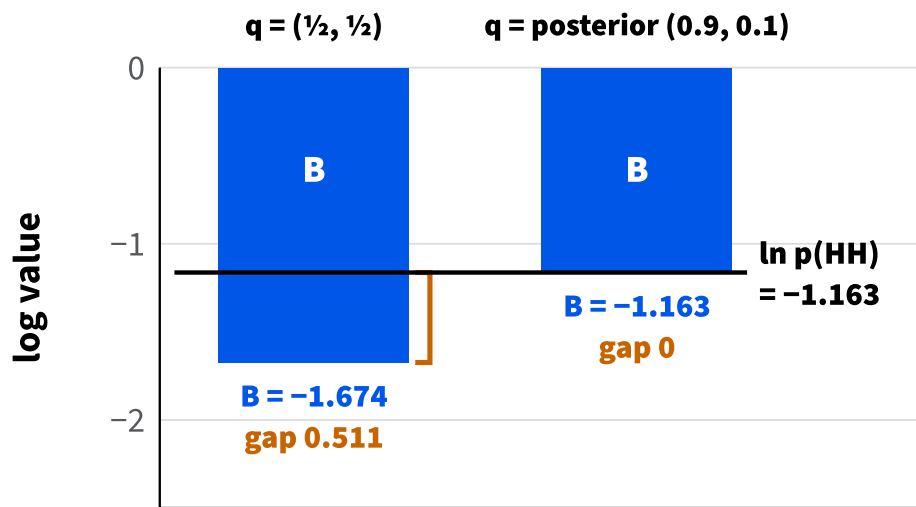
Lemma 1. $\ln p_\theta(x) = B(q, \theta) + D_{\text{KL}}(q \parallel p_\theta(z \mid x)) \geq B$, with bound $B = \sum_z q(z) \ln \frac{p_\theta(x, z)}{q(z)}$.

Theorem 4: $\ell(\theta_{t+1}) \geq \ell(\theta_t)$ in every round.

$$\begin{aligned} \text{HH with } q = (\tfrac{1}{2}, \tfrac{1}{2}): B &= \tfrac{1}{2} \ln \frac{9/32}{1/2} + \tfrac{1}{2} \ln \frac{1/32}{1/2} \\ &= \tfrac{1}{2} \ln \frac{9}{16} + \tfrac{1}{2} \ln \frac{1}{16} = \tfrac{1}{2} \ln \frac{9}{256} = \ln \frac{3}{16} = -1.674. \end{aligned}$$

Gap = $\ln \frac{5}{16} - \ln \frac{3}{16} = \ln \frac{5}{3} = 0.511$. With $q = (0.9, 0.1)$, the posterior: **gap 0**.

Round 1: $\ell(\frac{3}{4}, \frac{1}{4}) = 2 \ln \frac{5}{16} + \ln \frac{3}{16} = -4.000$; recall it moves to $(0.767, 0.233)$ with $\ell = -3.993$.



bar: the bound B · orange bracket: the gap from B up to $\ln p(\text{HH})$

The bound equals $\ln p$ at the posterior, so maximizing it can only raise ℓ .

Bells and chains: E finds each share, M counts with it

Section	Result	On the coins
5 Mixtures	E: a group's share of a point (its responsibility). M: each centre = share-weighted average. Theorem 5: width $\sigma \rightarrow 0$ gives k-means.	Points 0, 1, 4, 5; centres 1 and 4; width $\sigma = 1$. Point 1: distances $d = 0$ and 3. Heights $e^{-d^2/2}$: $e^0 = 1$, $e^{-9/2} = 0.011$. Group 1's share: $\frac{1}{1 + 0.011} = 0.989$.
6 Hidden chains	Forward $\alpha_t(z)$ = chance of flips 1..t and coin z at flip t. Backward $\beta_t(z)$ = chance of flips $t+1..n$ given coin z at flip t. Theorem 6 (Baum–Welch): EM for chains.	Flips H, H, T; keep $\frac{3}{4}$, switch $\frac{1}{4}$. Recall at flip 2, for (A, B): $\alpha_2 = \left(\frac{15}{64}, \frac{3}{64}\right)$, $\beta_2 = \left(\frac{3}{8}, \frac{5}{8}\right)$. $\alpha_2\beta_2 = \frac{15}{64} \cdot \frac{3}{8} = \frac{45}{512}$ (A), $\frac{3}{64} \cdot \frac{5}{8} = \frac{15}{512}$ (B). Sum: $p(\text{HHT}) = \frac{45 + 15}{512} = \frac{15}{128}$; A's share $\frac{45}{60} = \frac{3}{4}$.

Bell or chain, the E-step is a share of a sum; the M-step counts with the shares.

Draws shrink noise like $1/\sqrt{m}$; counting decides uniqueness

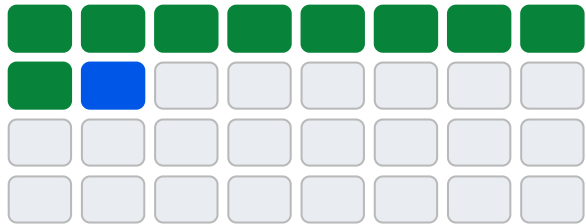
Section	Result	On the coins
7 Monte Carlo EM	E: draw the coin m times from the posterior. Theorem 7: the noise (standard deviation) of the drawn share is $\sqrt{q(1-q)/m}$, q the exact soft label.	HH: soft label $q = 0.9$, $m = 100$ draws. $\sqrt{0.9 \cdot 0.1/100} = 0.03$: drawn share 0.90 ± 0.03. Four times the draws, $m = 400$: $\sqrt{0.09/400} = 0.015$, half the noise.
8 Trusting the fit	Restart from several guesses; keep the best log-likelihood ℓ. Theorem 8 (counting test): more unknowns than free numbers $\Rightarrow$ no unique fit (a ridge).	Unknowns $\pi = p(A)$, θ_A, θ_B: 3. Two flips: $p(h)$ = chance of h heads. $p(0) + p(1) + p(2) = 1$: 2 free numbers < 3, a ridge. Three flips: $p(0) + \dots + p(3) = 1$: 3 free = 3, pinned.

More draws buy precision slowly; more observed flips decide whether the answer is unique.

Open questions: today's coin game, far larger

Wasted draws

32 reruns at $\theta_A = 3/4, \theta_B = 1/4$



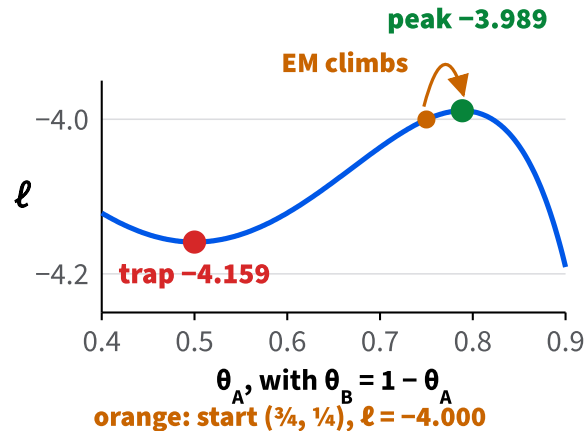
9 HH via A 1 via B 22 wasted

1 kept 999 wasted

same rerun idea: 1 correct trace in 1,000
(magenta sliver widened to be visible)

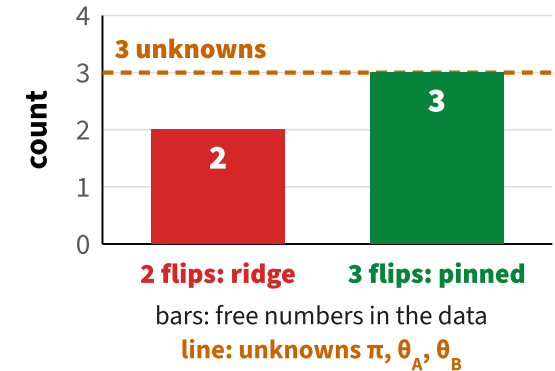
To draw the coin given HH, keep only reruns showing HH: chance $\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{10}{32}$. Rerun $\leftrightarrow$ reasoning trace; HH $\leftrightarrow$ correct answer. **Can we sample correct traces without the waste?** (Phan et al., 2023)

Trap or peak



At $(\frac{1}{2}, \frac{1}{2})$ each record has chance $\frac{1}{4}$, so $\ell = 3 \ln \frac{1}{4} = -4.159$; every soft label is $\frac{1}{2}$, so EM never moves. **When does a spectral estimate (one pass over data averages) start EM near the peak?** (Zhang et al., 2016)

Meaningful codes



Two flips: 3 head counts whose chances sum to 1, so 2 free numbers. Three flips: 4 counts, 3 free. Extra flip $\leftrightarrow$ extra observed label (e.g. time). **Which extra observations pin down a network's hidden code?** (Hyvärinen & Morioka, 2016)

Today's coin game, scaled up: wasted draws, good starts, pinning down the hidden code.

Four questions: counting, posterior, soft counts, the climb

1. A coin shows 3 heads, 0 tails. Laplace's rule estimates its heads chance as...

1

4/5

3/4

Answer. 4/5: $\frac{h+1}{h+t+2} = \frac{3+1}{3+0+2}$. Plain counting says $\frac{3}{3} = 1$: certainty from 3 flips.

2. $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$, fair pick. The record HHH came from coin A with chance...

9/10

27/28

3/4

Answer. 27/28: paths $\frac{1}{2}(\frac{3}{4})^3 = \frac{27}{128}$ and $\frac{1}{2}(\frac{1}{4})^3 = \frac{1}{128}$; A's share $\frac{27}{27+1}$.

3. Soft labels for coin A: HH 0.9, HT 0.5, TT 0.1. What θ_B does the M-step give?

0.7/3

2.3/3

0.1

Answer. 0.233. B's shares = 1 – A's: 0.1, 0.5, 0.9.
Heads: $0.1 \cdot 2 + 0.5 \cdot 1 + 0.9 \cdot 0 = 0.7$.
Flips (2 per record): $0.1 \cdot 2 + 0.5 \cdot 2 + 0.9 \cdot 2 = 3$.

4. One exact EM round takes ℓ from -5.20 to -5.31 . What happened?

a bug

normal

converged

Answer. A bug: Theorem 4 says $\ell(\theta_{t+1}) \geq \ell(\theta_t)$ in every exact round; ℓ fell by 0.11.

Bayes for the label, counting for the coin, and ℓ must never fall.

One question per section: bells, chains, draws, trust

5. Centres 1 and 4, width 1, equal weights; height $e^{-d^2/2}$ at distance d . Group 1's share of the point 3?

0.18 0.82 0.50

Answer. 0.18: distances 2 and 1, heights $e^{-4/2} = 0.135$ and $e^{-1/2} = 0.607$. Share = $\text{height}_1 \div (\text{height}_1 + \text{height}_2) = \frac{0.135}{0.135 + 0.607}$.

7. HT's soft label is $q = 0.5$. With $m = 100$ posterior draws, the noise (standard deviation) of the drawn share is...

0.05 0.03 0.0025

Answer. 0.05: Theorem 7, $\sqrt{q(1-q)/m} = \sqrt{0.5 \cdot 0.5/100} = \sqrt{0.0025}$. The noise is largest at $q = \frac{1}{2}$; HH's $q = 0.9$ gives only 0.03.

6. Chain: keep $\frac{3}{4}$, switch $\frac{1}{4}$. $\alpha_1(A) = \frac{3}{8}$, $\alpha_1(B) = \frac{1}{8}$ (chance of flip 1 and the coin). Flip 2 is T; A shows T with $\frac{1}{4}$. $\alpha_2(A)$?

5/64 15/64 10/32

Answer. 5/64: $\left(\underbrace{\frac{3}{8} \cdot \frac{3}{4}}_{\text{keep A}} + \underbrace{\frac{1}{8} \cdot \frac{1}{4}}_{\text{switch}} \right) \cdot \underbrace{\frac{1}{4}}_{\text{T by A}} = \frac{10}{32} \cdot \frac{1}{4} = \frac{5}{64}$.

8. Two EM runs end at the same ℓ but different θ . Most likely?

a bug swap or ridge not converged

Answer. Swapped names or a ridge. Swap: (0.789, 0.211) and (0.211, 0.789) both give $\ell = -3.989$. Ridge: with 2 flips, 3 unknowns > 2 free numbers, so a whole curve of $(\pi, \theta_A, \theta_B)$ fits equally.

Every answer is a rule from today applied to one new number.

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For one book on today's lecture, read Bishop, Chapter 9: mixtures, k-means and EM.

References (3 of 3): draws, reasoning, identifiability

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Every frontier paper here is expectation-maximization at scale: infer, then count.